Barcode scanning

Scan barcode on stock item: environment dossier.

A Newton 1.5 environment in which two Franka FR3 arms (or two Pandas, one FR3, or a Unitree G1 humanoid with Dex3 hands) operate a physically simulated handheld 1D/2D barcode imager of the Zebra DS2278 class to record stock: a cycle count at stockroom shelving, a checkout at a service counter with the scanner standing in its presentation cradle, and a library check-in. The scanner is a rigid tool held by fingertip friction, switched by a spring-return trigger with a microswitch or by the seat sensor of its cradle. Its images are formed by a path-traced render of the scene through its own thin-lens imager, lit by its own illumination LEDs and aim line, exposed by a monochrome sensor model with auto-exposure and noise, and decoded by zxing-cpp. **Nothing is decoded from geometry**: a read happens only when a rendered, exposed frame decodes, and every read is traced back to the printed label that produced it. This package builds and evaluates the environment. It does **not** contain an autonomous policy; interactions are shown with assisted (scripted and phantom-force) demonstrations, which are labelled as such everywhere.

Numbers in this report were measured on the final code unless a line says otherwise. Provenance labels for parameters: *datasheet*, *standard*, *search* (a source seen only as a search summary), *derived*, *estimate*, *authored*, *identified* (fitted in a bench against a published observable), *sourced* (a document read in full). Sources are listed in section 16 as [S#].

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1. The real job

The action group "scan barcode on stock item" (152 occurrences, 60 leaf groups) is this: find the target (shelf, location, item, the barcode's face), take the tool (a handheld scanner) or take the item to a fixed scanner, align, trigger, and record. The end states say "recorded", "registered", "counted", "verified": a read is half the job; the other half is that the right data ends up in the right record. The group's physics entries at 100 % are optical (photoelectric conversion, contrast modulation, field of view and standoff, perspective imaging, reflection, line of sight); mechanics (tool prehension, trigger actuation, 78 %) carries it. The environment's core is therefore a physically formed image inside a real device operated by a robot hand, not a camera with a geometric "barcode in view" test.

The tool is a general-purpose corded/cordless pistol-grip area imager, the most common retail and warehouse scanner class: a 640 × 480 monochrome sensor, 32.8° × 24.8° field of view, two 645 nm illumination LEDs and an amber aim line, a trigger, and a stand or cradle for hands-free (presentation) reading [S1, S2]. The DS2278 (cordless, charging/presentation cradle) is modelled so that no cable has to be simulated; its firmware behaviour and decode ranges are taken from the DS2208 Product Reference Guide [S2], which documents the same engine.

Three settings from the rows:

- **Stockroom cycle count** (flagship; rows on storage shelves, counting, location barcodes): steel boltless shelving with shelf-edge location labels and faced retail stock whose barcodes follow GS1 placement, so many face away from the aisle [S8–S10]. Tasks: scan a location; count a slot by scanning every unit; count by scanning one unit per item and keying the quantity; take and reseat the scanner.
- **Service counter checkout** (retail point of sale rows): the scanner stands in its cradle in hands-free mode and the robot presents each item from a customer's hand basket, then bags it; a 12.5 kg case of water is scanned with the handheld scanner without lifting it (the "bulky item" practice).
- **Library circulation desk** (library rows): returned books whose library barcode (Codabar, the library's own item label [S23]) competes with the publisher's ISBN (EAN-13), sometimes inside the front cover.

2. Scene at real scale

2.1 Physical geometry (collision, mass, friction)

Element	Geometry and physics	Source
Robots	two FR3 (or Panda) arms with Franka Hands on a wheeled pedestal cart, bases 0.50 m apart at 0.75 m, 0.40 m from the shelf face; or a Unitree G1 with Dex3 hands standing on the floor, the same cart parked at its right (section 6.5)	robot catalog, reach study
Shelving (stockroom)	Uline H-7115 class boltless bay 36 × 18 × 72 in, 5/8 in particle-board decks, 1.5 in beam pitch; posts and beams are static boxes; shelf-edge location labels on the beams	[S19]
Stock (stockroom)	cases and units per slot from the stock catalog (dimensions, masses, finishes, friction per SKU): cereal box, cans, jars, bottles, a lip balm, battery pack, shrink-wrapped pack; each a free body (boxes, or 32-facet convex hulls for cylinders; box SKUs of 5 kg or more as 8-vertex convex hulls, because MuJoCo's box-box routine threw the 12.5 kg water case from a bogus 584 mm contact at one reset. Every free box as a hull was measured and rejected: a cereal box in the seed-2 basket crept 24 µm/s against 1.0 µm/s as a box, and E7's held scanner slipped 3.3 mm against 0.9 mm. A sweep of every layout's reset looks for deep box-box contacts instead, section 11.2)	`catalog.py`
Necked containers	the 24 oz sauce jar (K04) and the 16.9 oz water bottle (K05) are not cylinders: a body + shoulder convex hull and a separate cap hull on the same body (98 % / 2 % of the mass). Jar: 85 mm × 160 mm glass (sourced 63 mm lug finish; body size an estimate in the sourced 24–32 oz range), 12 mm shoulder, 66 mm × 14 mm gold lug cap (estimates), paper panel 12–120 mm above the base. Bottle: 64.3 mm × 213.3 mm PET with a 70 mm label panel (sourced), 55 mm shoulder and 30 mm × 25 mm finish and cap (estimates), panel 47–117 mm above the base. The jar body (85 mm) is wider than the Franka Hand's 80 mm stroke: only its cap can be taken	[S30], [S31]
Counter	36 in (0.915 m) service counter, ADA §904 height	[S18]
Basket and tote	a 0.43 × 0.30 × 0.20 m hand basket (5 walls, 0.6 kg; rendered with slotted moulded walls, a rolled rim and two folded handles, the slots and handles visual only) and a 0.40 × 0.30 × 0.18 m bagging tote (0.9 kg), free bodies. The customer's items are dropped into the basket at random positions and yaws with the largest clearance the floor allows (30, 20, 10, then 0 mm), stacked only when the floor is full. In `checkout-counter` there is no basket: the same items stand on the counter left of the scanner, as they stand on a shelf, at random spots and turns at least 55 mm apart (`stock.COUNTER_SPREAD`; the group's checkout row 087, "merchandise resting on counter unscanned")	authored from common sizes
Library books	a hardcover (0.65 kg, 3 mm board cover, 80 g) and a paperback (0.25 kg, 0.3 mm card cover, 5.3 g) on a revolute spine joint with friction; the opened cover comes to rest on whatever is below it (joint limit 270°, the cloth hinge's fold-back)	section 3.5
Scanner	214 g within the DS2278's 175 × 66 × 90 mm envelope (H × W × D; the colliders measure 175.0 × 66.0 × 90.0 mm, checked when the scene is built): a 72 × 66 × 68 mm head and the 45 × 35 × 105 mm handle raked 15° back, its top 41 mm behind the window so that its rear bottom corner is the device's 90 mm depth; collision: head and handle boxes (the moulded housing, window tunnel, grip taper and neck are visual meshes); the trigger a separate body on a revolute pivot, its 3 g part of the 214 g. Mass split 45 % head / 55 % handle, derived from the two products' datasheet masses: the corded DS2208 (161.6 g, 1 cm shorter) with an estimated 60 % in its head, and the cordless unit's other 52.4 g (its battery) in the handle and base; the centre of mass lies 46 mm below the head's centre line and 46 mm behind the window (`Scene.scanner_com`). The seated scanner tips forward onto the socket's lip (its centre of mass falls 0.09 mm per degree of tipping)	[S1]
Cradle	151 g datasheet mass (fixed to the tray or counter: static); the CR2278-PC's shape: a teardrop base and a socket for the handle (floor, a front lip with the contacts, rear and side walls straight for 8 mm then flaring), 124 × 94 × 74 mm (datasheet 122 × 94 × 72 mm); seat sensor	[S1], [S32], section 3.4
Keypad	17 keys on prismatic joints (Cherry MX Red class force-travel) on the cart tray	[S13]

2.2 Visual dressing (no physics)



Figure 1. Stockroom overview (path-traced).



Figure 2. The hand basket as the checkout starts: items dropped with clearance; the clear, water-filled PET bottle; the necked jar.



Figure 3. Returned books at the circulation desk.

The rooms are dressed with pinned CC0 Poly Haven models and textures (`visual_assets.json`: shelving, hand truck, rolling shutter, fire extinguisher, clock, cameras, supply boxes, bin, crate, ladder and broom in the stockroom; store display units stocked with canned goods and cleaning bottles and back-room racks with cartons and jugs at the counter; four worn wooden bookcases filled with book rows, a circulation-desk sign and an architect lamp in the library), painted or plaster walls, laminate or tile floors, and ceiling troffers sized for EN 12464-1 illuminances (200 lx in the stockroom, 500 lx at the counter) [S17]. The troffers are sized by an analytic sum that ignores the room's inter-reflections, so the result was measured in the render: a white card at each probe point, seen by the imager with its LEDs off, calibrated against E1's 323 lx bench card, gives 136 lx on the stockroom shelf face at the design point and 109 lx at the counted slot's label (design 200), 459–483 lx on the counter top and in the scan zone (design 500), 516 lx on the library desk and 180 lx where the returned books lie (in the robots' shadow). The stockroom is about a third darker than designed (not corrected); the imager's own LEDs and auto-exposure dominate its frames at reading distances (E1, E24). The packaging art is generated per SKU (`packaging.py`, `artwork.py`): printed brand panels, nutrition-panel-like text and the barcodes themselves, rendered from the same symbol generator the decoder is tested against. The checkout lane has the furniture of a real lane around its scanner: a customer pole display and a card terminal on a tilted stand at the customer edge of the counter's free left end, a kick plate and an edge trim on the customer side, and a bag stand with hanging carrier bags past the counter's right end, all placed where no task's arm goes (`dressing.py`, `_checkout_hardware`). The robots' cart has four swivel casters (`_cart_casters`). Dressing is visual only and has no collision.

3. The scanner as a mechanism

3.1 Trigger

The trigger is a 3 g lever on a revolute pivot with a 25 mm arm, preloaded by a torsion spring and closing an Omron D2F-class microswitch [S12]. At the blade face, the force-travel law is

$F(x) = F0 + k\,x$, $F0 = 1.0$ N, $k = 0.45$ N/mm, $x \le 3.6$ mm (stop)

(estimates bounded by the D2F operating force ≤ 1.47 N and pretravel ≤ 0.5 mm scaled by the lever ratio, and by trigger surface lengths in scanner patents [S25]). The switch closes at 3.0 mm and opens at 2.81 mm (movement differential ≤ 0.12 mm × lever ratio 1.6), with 5 ms debounce. The stop is a MuJoCo joint limit made near-rigid (`STOP_SOLIMP`: impedance 0.99 → 0.999 within 1 mrad, time constant 2 ms), because MuJoCo's default limit softness let a 35 N squeeze drive the trigger ~1 mm through its plastic stop. The trigger is damped at $\zeta = 0.4$ (a plastic pivot and snap switch; undamped, the lever rang 0.4–3.5 mm). The trigger is operated only by contact: a robot fingertip or the gripper's squeeze on the blade (or, in assisted demonstrations, a labelled phantom fingertip force).

3.2 Keypad

Each key is a 2 g body on a prismatic joint with a Cherry MX Red force-travel law (0.30 N initial, 0.45 N at the 2.0 mm operating point, 4.0 mm total travel, release at 1.8 mm; $\zeta = 0.3$) [S13]; the keycaps have 1 mm clearance to the case at the stop so that the joint limit is the only stop (a keycap resting on the case at its stop made a limit cycle). The host app receives a key when the key's switch closes.

3.3 Hands and the scanner

The scanner is taken by the Franka Hand's force-mode grasp, modelled as the hand's drive behaves: a speed-controlled drive whose push is limited to the commanded force (≤ 35 N per finger in this controller profile). Each finger follows a reference slewed towards its target at the hand's 50 mm/s through its own servo spring (3500 N/m), the push capped at the grasp force: in free air a finger moves exactly as in position mode; blocked, its push builds up with the lag (175 N/s at full speed) to the grasp force and stays there, and the reference never runs further ahead than that push (no wind-up), so a finger that loses its object does not snap shut. A simpler model that switches each finger to a constant force once it lags its target by 0.5 mm fails: that is the servo's ordinary tracking lag at full speed ($kd \cdot v/kp = 0.57$ mm), so the squeeze begins in free air, about 10 mm before contact, and throws the fingers at the item at 0.8 m/s, 16× the hand's speed, and the impact trips the arm's collision reflex at grasps. The two fingers are one mechanism, as on the real hand (one drive, both fingers mirrored): the hand's URDF declares finger 2 a mimic of finger 1, which Newton's importer turns into a MuJoCo joint equality constraint, and the environment makes that constraint as stiff as the hard stops (2 ms, `world.py`). At MuJoCo's default 20 ms softness the 3 500 N/m finger servos overpower it (driven to different targets the fingers sit 10 mm apart, and an off-centre grasp carries about 8 N of constraint force between them). With the stiff constraint the hand closes symmetrically (fingers driven to 5 and 35 mm stay within 0.1 mm of each other), and an off-centre object is pushed to the middle by the first finger to touch it until the second arrives. The grip force is capped per finger by the profile's effort limit, 35 N on the FR3 and 20 N on the Panda: summed over the two fingers 70 N, the figure the catalog profile takes for the Franka Hand's continuous 70 N; if that datasheet figure is the force each jaw applies, the simulated hand squeezes at half of it. The policy's `inward_effort_n` is the two fingers' pushes summed (at most 70 N on the FR3, 40 N on the Panda; more is refused before anything runs, section 13). The fingertip pads are rubber boxes whose contact time constant is anchored at 5 ms for the FR3 finger (label_application) and scaled with finger mass; MuJoCo geom priority makes the pad's parameters govern pad contacts (section 6). The pads resist spin about a contact normal (torsional friction 1 mm, the free bodies' coefficient, i.e. a 1.9 mm contact patch at the pad's μ 0.8): with point contacts only, a scanner turned 9° in a 35 N pinch by a legal whip keeps sliding in the hand for 30 s (4.0 mm, still 0.1 mm/s at the end); with it the slide stops within a second (section 11.2). The finger joints are made rigid at their limits (the Franka Hand's lead screw does not open past 40 mm when an object is forced between the fingers). Under arm-on-arm and arm-on-world impacts at full joint speed the coupling and the stops hold to 2.7 and 3.5 mm (158 FR3/Panda impact cases, section 6.1). A line search cut short at such an impact can slide both fingers sideways past a stop (up to 104 mm with a budget of 20 evaluations); this is the solver, not the coupling, so fingers parted by more than 10 mm stop the episode as a numerical failure (section 6.3).

3.4 Cradle and seat sensor

The DS2278 stands in a CR2278-PC presentation cradle (72 × 94 × 122 mm, 151 g [S1], fastened [S2]). Its Quick Reference Guide (Zebra MN-002917-03 [S32], pp. 3 and 5, figures viewed) shows a low teardrop base with the handle's pocket at its raised rear and the charging contacts on the pocket's front edge; the scanner is inserted "straight down into the cradle top" and "naturally rotates forward to engage the … contacts", its head overhanging the front with nothing under it. Modelled (`scene.py`, `Scene.build_cradle`):

- **Base**: the teardrop as one convex hull, the round end (94 mm) around the pocket and a 60 mm nose running forward under the head, 124 × 94 mm and 30 mm tall; a pedestal hull from the socket's floor down into it.

- **Socket for the handle**: a floor; a front lip 35 mm high (the contacts); rear and side walls that stand straight for their first 8 mm, 1.5 mm from the handle, then flare out (rear 30°, sides 15°, a moulded pocket's lead-in), the rear wall's top 74 mm above the base's foot (datasheet 72 mm). The visual is the union of exactly these colliders, edges rounded: nothing drawn without a collider, nothing colliding that is not drawn.
- **The seated pose is the rest.** The head's weight turns the scanner forward until its front face rests on the lip's top edge, its bottom on the floor and against the rear wall; that lean follows from the contact geometry (`Scene.socket_lean`: $2a\,(1-\cos\theta) + c - L\tan\theta = 0$ for the handle's half depth a, clearance c and lip height L; $\theta = 2.5°$), and the socket is built turned back by it, so the authored seated pose (window 12° down) is where the scanner rests: after the reset it is 0.024 mm and 0.009° from that pose, moves 0.0013 mm in 4 s, and is carried by the floor (2.6 N), the lip (1.6 N) and the rear wall (1.1 N). It is the bottom of a well: turning further forward lifts the handle up the straight rear wall and raises the scanner's centre of mass by 7.9 mm per degree; from upright on the floor it falls 0.30 mm per degree as it tips forward onto the lip.
- **Putting it back.** Robots set the scanner in upright (`Scene.insertion_pose`: the handle parallel to the socket's walls, in the middle of the 1.5 mm fore-aft clearance), lower it to 10 mm above the floor and let go; it drops onto the floor and tips forward onto the lip, as the guide describes. Lifting it out, they move along the socket's axis, which takes the handle's face off the lip. The clearance leaves 0.75 mm at the handle's front face, which an arm holding the scanner by its head does not reach open loop: at the counter the FR3's servos left the handle 1.6–2.0 mm towards the lip, its front edge came down on the lip's top and the scanner dropped (section 10.5). The task operator therefore closes the loop over the socket's mouth on the scanner's measured pose (`skills.over_socket`), as a policy would with its cameras.

Measured (the stockroom; each case from the seated state; ASSISTED probe forces of 0.1 s on the head's centre, under which the scanner leaves at about 0.14 and 0.35 m/s, and probe pose writes for the releases, followed by 3 s hands off):

Socket (lip / straight walls)	Rest lean	Rest: from the seated pose; moved in 4 s	Knocks of 0.3 N, three directions	Knocks of 0.75 N: sideways / backwards / forwards	Releases: upright 10 and 20 mm up; at the seated lean 5 and 10 mm up	G1 side grip, fingers the cradle blocks (hand at the seat / +15 / +25 mm)
35 / 8 mm (shipped)	2.5°	0.024 mm, 0.009°; 0.0013 mm	all seated (peaks ≤ 0.04 mm)	out (178 mm away) / seated (peak 9.4 mm) / seated	all four seated, ≤ 0.049 mm from the pose	middle / none / none
30 / 8 mm	3.0°	0.033 mm; 0.0004 mm	all seated	out / seated / seated	three seated; let go at the lean 10 mm up it tipped out onto the floor	the same
25 / 8 mm	3.6°	0.048 mm; 0.0002 mm	all seated	out / seated / seated	all seated, the two lean releases 1.1 and 1.6 mm off the pose	the same
30 / 5 mm	3.0°	0.033 mm; 0.0004 mm	all seated	out / seated / seated	all seated (≤ 0.23 mm)	the same
35 / 12 mm	2.5°	0.024 mm; 0.0011 mm	all seated	out / seated / seated	all seated (≤ 0.049 mm)	the same

Every release seats in every socket but the 30 / 8 mm one, and the two 35 mm lips put the scanner closest to its pose (≤ 0.049 mm); the shipped 8 mm straight part matches the 12 mm one. A sideways shove of 0.75 N for 0.1 s (the scanner leaves at 0.35 m/s) knocks the scanner out of every socket: sideways, only the 18 mm side walls and the lip's edge hold it.

E7 (section 9.2) takes the scanner out by its head, holds it and puts it back: seated at the end.

The seat sensor (a hall-effect stand sensor on the real device [S2]; authored stand-in) reports "seated" when the handle bottom is within 3 mm and 8° of the seated pose, with at least two scanner–cradle contacts, for 0.2 s. The firmware switches to hands-free (DOCKED) mode when seated.

Why this shape. Simpler cradles fail. A straight cup 40 mm deep around the handle (1.5 mm clearance) lets the scanner rest 4.5° forward of its authored pose, holds it wedged 9 mm off its seat when a patron's book knocks it, and jams it in every G1 lift (215–258 N on the walls, 0 of 9 lifts out), and its 40 mm front wall blocks the G1's middle finger. A head rest on a post under the head, which the real cradle does not have, is what the G1's index and middle fingers run into. A socket with a 15 mm lip lets the scanner tip out over it at the reset. Walls sloping back 5° from the floor make the well shallow (0.85 mm per degree), and a scanner let go 20 mm above its seat hangs 6 mm up between the lip and the rear wall.

3.5 Book covers

A library book is its text block (with back cover) as one free body and its front cover as a link on a revolute spine joint: hardcover board 3 mm and 80 g (greyboard ~700 kg/m³ plus cloth, K12's dimensions), hinge friction 0.02 N m; paperback card 0.3 mm (10–12 pt cover stock) at 280 g/m² (5.3 g), hinge friction 0.002 N m (estimates). Library item labels are Codabar (14 digits, "A…B" start/stop) printed on label stock [S23], on the back cover or on the inside of the front cover.

4. The imaging chain

4.1 Optics

The imager is a thin-lens camera at the entrance pupil, set back `PUPIL_SETBACK_M` = 50 mm behind the window, with pupil diameter A = 0.41 mm and focus distance 232 mm from the pupil, 182 mm before the window. Both come from a decoder-independent fit: the environment's own blur law (defocus disc and diffraction, below) evaluated at every published near and far limit of PRG Table 3-3 [S2], in module widths, and the pair (A, s) that makes the worst of them least. The fit's focus lies within 12 mm of the 194 mm from the engine front that two other Zebra engines publish (SE4710 [S3], SE2707); neither is the DS2208's engine. A lens fitted with zxing's measured blur thresholds inside the fit (A = 0.5 mm focused at 130 mm, 80 mm before the window) lets the optics carry part of the decoder's shortfall: at the datasheet's 1-D far limits that lens blurs 2.5–4.2 module widths (UPC-A 3.7, Code 39 10 mil 4.2), while this fit blurs 1.9–2.6 everywhere (UPC-A 2.2), at the price of slightly more blur at the near limits (UPC-A 0.95 against 0.82 module widths). At the task distances (hands-free items at about 120 mm, handheld aim at 150 mm) the two lenses blur alike (1.6–1.9 mrad). For a lens of focal length f focused at s, a point at distance d is imaged as a blur circle of diameter on the sensor

$$c = A\, f\, |d - s| \,/\, (d\,(s - f)),$$

and the depth of field and the limiting pixel density come from this blur against the 640 px width over the 32.8° field (IFOV = 2 tan(16.4°)/640 = 0.92 mrad per pixel). The render worker applies the thin-lens model through RTX's camera (f-number = 0.1 f/A in its units). Diffraction is added in the sensor model as a Gaussian of angular $\sigma = 0.42\,\lambda/A$ at 645 nm (the renderer has none).

4.2 Illumination and aim

Two 645 nm LEDs 10 mm either side of the lens are sphere lights (1 mm radius) shaped by an IES profile: an elliptical Gaussian with full widths at half maximum 39.3° × 28.3°. The DS2208's illumination pattern is unpublished, and its engine is not the SE4710: the SE4710's published pattern (50.3° × 32.0° [S3]) is 1.20 × 1.14 times that engine's 42° × 28° field of view, and the same ratios applied to the DS2208's 32.8° × 24.8° field give the pattern used (the SE4710's pattern unscaled is 1.53 × 1.29 times this imager's field and would light the frame edges more evenly than an engine matched to its field). The LED intensity is set so that the on-axis signal is the render calibration's 10:1 LED-to-room ratio on white paper at 150 mm (measured: red-channel ratio 10.80). A hard 25° cone is not enough: it leaves the frame's corners unlit and a label in the left third of the frame under a 5x illumination step, so that label never decodes. Measured on white paper: section 9.5. The aim line is an amber (624 nm) IES line source with a 32.8° fan and 0.3° thickness, crossing the optical axis at 150 mm; the firmware turns it off during each exposure, as the SE4710 does [S3]. In hands-free mode, exposure is capped 5.8x shorter (0.22 ms against 1.27 ms, both derived from the published motion tolerances [S1]) and the LEDs are driven 5.8x brighter for the same signal.

4.3 Sensor

Each frame the renderer returns linear radiance (HdrColor, RTX's fixed tonemap exposure, calibrated in `render_calibration`), weighted to a monochrome response (0.35 R + 0.45 G + 0.20 B, estimate). The sensor model (`optics.py`) converts radiance to photoelectrons, $N_e \sim \text{Poisson}(K\, t_e\, L)$ with a calibrated constant K (`SENSOR_K`: white paper under LED + 200 lx at 150 mm gives DN 128 at 0.4 ms, gain 1), adds 8 e^- read noise and 1 % PRNU, then applies the analog gain and quantises to 8 bits, $DN = \text{round}(255\, g\, N_e / 6000)$ (6000 e^- full well); an auto-exposure loop runs from each frame to the next (exposure 20 µs … the mode's cap, then gain 1–8). The gain is applied after the Poisson draw, so the shot noise grows as g, not $\sqrt{g}$ (applied before it, the frames at the hands-free gains of 6–7 are 1.9 times too clean: 3.10 DN against 5.92 DN at t_e 0.22 ms, gain 6.47). A test checks the frame noise against the photon count at gain 1 and 6.47 (within 5 %). The loop meters differently in the two modes, and E24 (section 9.3b) is why:

- **Handheld** (the operator aims the aim line at the code): the mean of the frame's central third is brought to 110 counts. Nothing else.
- **Hands-free** (docked; items are swept anywhere through the field): the same centre target, capped so the brightest 0.5 % of the frame's 8 × 8-pixel block means stays at or below 230 counts. The cap is predictive (the sensor is linear, so the next frame's highlight is the current one scaled by the exposure change), and a clipped highlight halves the exposure.

Without the cap a lip balm swept past the docked scanner off the metering window clips white, bars and all, and is never read (E21). The cap is predictive and runs in hands-free mode only, because a cap that runs in both modes and acts only on a highlight already over 230 counts has two faults. In handheld mode it darkens the aimed code under the LEDs' near-range hot band and moves every near limit out, e.g. Code 39 5 mil from 15 to 29 mm. And the exposure hunts: a dim centre's demand overshoots into clipping, halves back, and repeats in a ×8, four-frame cycle.

4.4 Motion blur

The exposure integrates motion: the fastest image-space speed of any registered label or body in view, relative to the imager, over the exposure (from the recorded 1 kHz poses) sets a blur length in pixels; above 0.5 px the

frame is rendered as up to 16 sub-exposures at interpolated poses and averaged. Blur computed from the scanner's own motion alone would be zero for a docked scanner while the item moves.

4.5 Render sampling

A scanner frame is a path-traced render, and its Monte Carlo noise is part of the image the decoder sees. RTX clamps the samples per frame to 24 unless `clampSpp` is set, and a second render sub-frame of an unchanged scene adds no samples. The render worker turns the clamp off, so the requested count is rendered. The count comes from a convergence measurement: a white card under the LEDs, its relative noise in a 120 × 90 px region against a 4 096-sample reference (whose own noise is 0.15–0.16 %):

Samples per frame	61 mm	121 mm
24 (RTX's default clamp)	6.70 %	6.02 %
48	3.46 %	3.09 %
128	1.27 %	1.14 %
256	0.74 %	0.67 %
512	0.32 %	0.31 %
1 024	0.13 %	0.15 %

The criterion: Monte Carlo noise at most a third of the sensor's own noise. The sensor's relative noise at the auto-exposure target of 110 DN is √(255 g / (110 × 6000)): 2.0 % at gain 1 and 5.1 % at gain 6.5. Each frame's sample count is therefore chosen from the gain the auto-exposure set for it, spp = ⌈128 × 1.27 % / (σ_sensor(g) / 3)⌉, kept between 64 and 256 (`params.RENDER_MC_REF`, `RENDER_NOISE_FRACTION`, `RENDER_SPP_RANGE`): 249 at gain 1, 176 at 2, 125 at 4, 98 at 6.5 and 88 at 8. The bench experiments (E1, E1c, E2–E5, E24) render at 256. The outcome is converged: E1 decoded on radiance rendered at 256 and at 512 samples reads to the same limits within 2 mm for every 1-D row and 6 mm for Data Matrix's far limit (the Code 128 7.5 mil misread of section 9.3 at both, the PDF417 one at 256 only; both sets decoded without the two misread rules of section 5.1, so that the comparison isolates the render).

Cost, and the imager's path depth. A scanner frame costs its samples times the cost of one sample, and inside the closed rooms a sample's paths bounced up to 6 times (the worker's setting for every camera). Measured on each preset's exported scene with the backend's own render call, every frame after a 1 mm nudge of an item so that no converged buffer is reused (on a shared GPU): 62, 61 and 73 ms per sample at depth 6 (counter, library, stockroom), 52, 45 and 36 at depth 4, 36, 32 and 30 at depth 3. (The bench scene costs about 4 ms per sample.) Scanner frames are rendered at depth 3 (`params.RENDER_IMAGER_MAX_BOUNCES`; policy and diagnostic cameras keep 6), 1.7–2.5 times cheaper. What it changes, from 17 recorded task frames re-rendered at depths 6, 4 and 3 from the same recorded poses, exposed with the recorded exposure and gain and the same noise draw: every frame decodes the same at every depth; frames with a read keep 97.3–99.9 % of the depth-6 radiance over their brighter half (the handheld shelf frame loses the most: the bay's inter-reflections), empty dark frames 91.7–99.9 %. Depth 4 keeps 99.4–100 % and 97.9–100 % but saves only 1.2–2 times. (A variant that left the dressing's far background models out of the scanner frames was rejected: once the room's walls stay, it saved little, and it brightened empty frames' backgrounds by up to 18 %.) The two E22 frames of the comparison were not rendered: the worker timed out (600 s per request) when the GPU ran five render workers. Spreading the samples over motion-blur sub-frames does not change the cost. A repeated render of an unchanged scene reuses its converged buffer and costs 0.06–0.1 s. Section 12 has the runtime this costs.

Artifacts still visible at 256 samples: a 1 px bright outline along the edge of the benches' symbol quad, which sits 0.3 mm in front of the white card; task scenes print each symbol in its package or label artwork and show no outline. And isolated single-pixel fireflies in dark backgrounds.

5. Decoder and firmware

5.1 Decoder

Each exposed 8-bit frame is padded by 48 px of edge replication (zxing-cpp is blind in a band at the frame border) and decoded by zxing-cpp in six passes (three binarizers × native and 2x upsampled). Results are grouped by symbology family and location; within a group a 1-D text is reported only if at least two passes agree and it has more than twice the votes of any rival (checksum-valid misreads appear beside correct reads on blurred, pitched symbols); Data Matrix and QR, whose Reed–Solomon check is taken as conclusive, need one pass. PDF417 needs two: zxing's PDF417 decoder, which corrects with erasures, returned a wrong 22-character text from one pass on E1's bench (107 mm, one frame of at least 1 695), and requiring two passes removes it at a cost of 14 mm of far range (115 to 101 mm; datasheet 145 mm). Requiring two for Data Matrix rejects valid reads at close range (E1), and neither Data Matrix nor QR has misread. **Symbol lengths** are the DS2208's defaults [S2] (PRG ch. 12, "Set Lengths for ..."; the number of human-readable characters including check digits, for Codabar without its start and stop characters): Code 128, Code 39 1–55, Codabar 4–55, Interleaved 2 of 5 6–55; a read of another length is dropped before the consensus (`decoder.LENGTHS`). Without this rule E1c's roll sweep decodes a blurred Code 128 5 mil card 23 mm from the window as the Codabar "C82D" (two data characters) in 17 frames at four roll angles, two passes agreeing; with the rule those frames read the Code 128 or nothing. Two passes agreeing is weaker than it sounds: the passes are three thresholding methods on the same frame at two scales, and two of them can read the same scan line the same wrong way (E1, section 9.3: a Code 128 decoded as another checksum-valid text by two binarizers on one row of the upsampled frame). A 1-D read also requires both quiet zones (from the decoded symbol's own module count) inside the frame: a real imager cannot tell a symbol's end from the frame edge, and with this rule the simulated UPC near limit lands on the datasheet's 13 mm. One symbol is reported per frame: the one whose centre is nearest the frame centre (picklist off, the DS2208 default [S2]).

Any roll (E1c, section 9.3c). The DS2208 reads at any roll [S1]; zxing's 1-D readers scan rows (and columns), so a symbol rolled by more than its bar height allows is missed: on E1's bench the six passes read a 5 mil Code 128 at 3 of 36 roll angles and the library label's Codabar at 6 of 36. The decoder therefore works as an area imager does: it locates the bars and samples along their axis. When the passes report nothing, the bars' orientation is found from the frame's gradient structure (the structure tensor of Sobel gradients after a 1.5 px Gaussian, per 16 × 16 block, kept where strong and coherent, histogrammed modulo 90°), the frame is turned to level them (bicubic), and the same passes run on it; positions map back to the frame, so the quiet-zone rule and the choice among symbols see the true frame. A second rule guards against a failure of the row scan itself: across bars near 45° it decoded a clean UPC-A and a clean EAN-13 as other checksum-valid numbers, agreed by several passes. Where the bars lie 30° or more from level, the levelled frame is decoded too, and its read of the same family at the same place replaces a different one. A frame that decoded before is decoded as before unless that guard fires.

Uneven illumination (flat-field stage). A card skewed 30° to the window is lit about 1.7 times as brightly on its near side as on its far side (quiet zones at 150–180 against 87–102 counts in one row of the rendered frame), with the dark room beyond its edge in the same rows, and zxing thresholds each 1-D row as a whole: without this stage the E1c skew sweep reads only to ±25–30° where the DS2208 reads to ±65°. An imager binarizes against the local white level. The last stage of the decoder therefore divides the frame by it (a 15 px running maximum smoothed over 20 px, floored at 30 counts) and decodes again when nothing else was found. On the front end's synthetic bench, with a misread check confined to the read's neighbourhood, the stage lets six misreads through, all of its own (an EAN-13 or UPC-E read as another UPC-E at 1.1 px per module); the check therefore looks further (below). The zxing consensus reads 2 884 of the bench's 8 120 frames (2 046 without the stage); the shipped decoder, the consensus and then the front end, reads 7 404, with no misread, and nothing on the bench's frames without a complete symbol. Skew: section 9.3c. `BS_DECODER_FLAT_FIELD=0` turns the stage off for comparisons.

The DS2208's decode security settings (from the PRG [S2]): Security Level 1 (default) "eliminates most misdecodes" for Code 128, UPC/EAN and Code 93, without stating the mechanism; Redundancy Level 1 (default) requires two reads of Codabar ≤ 8 characters, MSI ≤ 4, D2of5 and I2of5 ≤ 8; I 2 of 5 Security Level 1 requires two reads. None of the environment's symbols falls under Redundancy Level 1 (library Codabar is 16 characters; ITF-14 is 14 digits).

The misread check. The within-frame consensus alone does not eliminate misdecodes. On the front end's synthetic bench (8 120 frames of seven 1-D symbologies at 0.6–3 px per module with optical blur, shot and read noise and distractors, and 600 frames holding no complete symbol) it returned a checksum-valid wrong UPC-E in 11 frames at 1.1–1.5 px per module (in one of them the wrong text was an EAN-13 distractor's), every pass agreeing: UPC-E carries its check digit only in the parity pattern of its six digits. Each 1-D read is therefore checked against a physical model of its bars (`decoder.veto_1d`): the sub-pixel front end of section 9.3 fits an edge model (Gaussian blur, pixel aperture, the symbol's module grid, ink spread) to the read's neighbourhood, gated against the noise measured in the frame, and where it reads a different text of the same family on the same symbol, the read is dropped. Where it reads nothing, the read stands, so the check adds no read and

cannot extend the range. The neighbourhood (the read's extent and 0.75 of its length plus 16 px beyond it on every side) and the same-symbol test (inside the front end's quadrilateral grown by a quarter, or within half the read's length of its centre) are authored. When the front end finds no symbol of the read's family on the same symbol in that neighbourhood, it looks at the whole frame: a UPC-E misread off the left part of an EAN-13 at 1.1 px per module can leave the EAN-13's far end at the neighbourhood's edge, where the front end reads nothing, so the misread would stand (the six flat-field misreads above; a wider neighbourhood catches five of them).
Scan-line separation. zxing accepts a 1-D symbol once two scan lines decode it to the same text and reports the first and last agreeing lines as the edges of its quadrilateral. Those two lines may be adjacent rows: in one checkout run the consensus read the marinara jar's UPC-A as the checksum-valid EAN-13 8012871001583 from rows 196 and 197 alone, which graze the tops of the bars where they stand at uneven heights on the curved label (not a glare effect: the read's rows hold no saturated pixel). A consensus read is therefore reported only if, in one of its agreeing passes, those lines lie at least 4 px apart across the bars (`decoder.SCAN_SEP_PX`, `line_sep_px`), unless the front end, which fits the whole bar height, read the same text on the symbol. On 1 430 saved imager frames of task runs, replayed, the rule removes the misread and 30 of 430 correct frame reads (7 %, on marginal frames: a can at the edge of a fast swipe, the curved jar, a Codabar at the edge of the field); every symbol those runs read is still read, two of them one frame (33 ms) later. The 4 px are authored: the misread's lines were 1 px apart, and of the 445 1-D reads in those frames 36 had lines closer than 4 px and the rest mostly tens of pixels apart.

The front end as the last 1-D stage. Where the consensus reports no 1-D symbol in a frame, the sub-pixel front end (section 9.3) decodes the whole frame, and its reads (the DS2208's length rule applied) are reported like the consensus's. zxing needs about 1.2–1.5 pixels per module and one module of blur; the front end reads to about 0.7 px per module, as the device's published ranges imply its decoder does. Its cost is small: 0.03 s median on a frame where the consensus found nothing, against the consensus's own 0.18–0.48 s and a scanner frame's 2.5–6 s of rendering. Ranges: section 9.3.

The consensus, the edge-model check and the line rule are this environment's model of Security Level 1. Measured rates: every recorded task frame (section 9.3), E1 (section 9.3), the optics bench (section 9.4) and the synthetic bench above. Against a published reference (the AIM-sponsored Ohio University test of laser scanners with reference decoders: worst-case substitution rates of 1 per 394 003 characters for UPC-A, about 1 per 33 000 twelve-digit symbols, and 1 per 2 764 151 for Code 128), the environment's records are far too few to show such a rate; they bound it (section 9.3). A misread is reported to the host application and fails the task's `no_misreads` check (section 5.2).

5.2 Attribution (privileged; the verifier's)

Every reported read is traced to a printed label: the decoded symbol's centre is back-projected from the entrance pupil at the frame's recorded poses and intersected with every registered label patch (flat or cylindrical) facing the imager; the nearest hit inside a patch is the physical source, and its payload is compared with the decode (geometry first, payload second). Two outcomes are not a clean read:

- A decode that no label on its ray can explain is a **ghost read**: the backend made it up. It is a fault that stops the episode, never an outcome.
- A decode traced to a label whose payload differs is a **misread**, what a real imager does now and then (this model's rates: sections 5.1, 9.3, 9.4). It reaches the host application like any read, the evaluator counts it (`misreads`), and every task's `no_misreads` check fails on it. A misread does not end the episode: a legal aim at a glossy label must not end a policy's episode with a simulator fault.

5.3 Firmware

The firmware state machine (`device.py`) follows the PRG [S2]:

State	Enters when	Behaviour
LIFTED_IDLE	out of the cradle, not moving for 2 s	aim off
AIM	moved (Auto Aim)	aim line on
SESSION	trigger switch closed	illumination on, frames at 30 Hz on a fixed grid, aim off during exposures, decode every frame; a read, a release or the 9.9 s decode-session timeout ends it
WAIT_RELEASE	after a read or a timeout	nothing until the trigger opens (one read per pull)
DOCKED	seat sensor true	hands-free: frames while the presentation zone changes (0.25–2.5 s sessions); the same symbol is re-read only after it has been out of the field of view 0.5 s; a different symbol after 0.1 s

Timeouts and session limits are the PRG defaults [S2 pp. 5-22]. Image-change wake-up in the cradle is an authored stand-in: a body other than the scanner moving faster than 5 mm/s within 0.4 m in front of the window, inside the field of view. A symbol left stationary in view (no frames once the session lapses) is not re-read as soon as it moves again: the PRG requires the symbol to be *out of the field of view* for the timeout, so while nothing in the zone changes, the last frame's symbols count as still in view (section 10.3).

5.4 Host application

A keyboard-wedge host (`HostApp`) collects scans and key presses into the task record (location, counts per GTIN, sale, check-ins, errors) and draws its screen, which is rendered on the cart's display and visible to the policy cameras.

6. Contact, grasp and the robots

6.1 Solver

MuJoCo through Newton's SolverMuJoCo (CPU `mj_step`), 1 kHz (0.5 ms for the G1, section 6.5), implicitfast integrator, elliptic friction cones, Newton solver with 50 iterations and up to 50 line-search evaluations per iteration (`Config.solver_ls_iterations`, MuJoCo's own default, below), `impratio` 3, and a no-slip pass of 100 iterations at a tolerance of 1e-10. MuJoCo's midphase is disabled: in the dual-Panda model it silently dropped every scanner–cup contact.

The line search must not be cut short. Three legal commands in a fresh dual-FR3 session (counter, `checkout-bulky`, seed 2: both grippers, then in-limit targets for all 14 arm joints) swing the two arms into each other at full speed. With a budget of 20 line-search evaluations, 4 ms after the impact the right hand's two fingers slide 33 mm sideways in one 1 ms tick, at 33 m/s, to 51 mm past the 40 mm stop; the same happens on the dual Panda and at arm-on-world impacts, up to 104 mm past a stop at 73 m/s, unflagged. Cause, from MuJoCo's own solver statistics at that step: the Newton solver's third line search needs 21 evaluations, a budget of 20 cuts it off, the following searches cannot recover, and the solver stops on its no-improvement test with the fingers' mimic coupling (section 3.3) carrying −2 275 N where the converged solve gives +2.1 N (Panda: 1 111 N against −5.1 N). Applied to two 29 g fingers (14 g on the Panda), that force is the 33 m/s slide. The same step re-solved with 50 evaluations converges (finger speed 0.065 m/s); a 0.1 kg finger armature, a 20 ms coupling, a pyramidal cone, `impratio` 1, no warm start or the CG/PGS solvers also avoid it at that step, but they change the model or the solver everywhere, and 0.01 kg of armature or a 5 ms coupling do not avoid it on the Panda. Replaying 158 recorded FR3/Panda cases of legal abuse (1 733 s) with 20 and with 50 evaluations: with 20, 112 cases had a step whose line search was cut off (2 590 steps), 21 drove a finger more than 5 mm past its stop (5 more than 40 mm) and 36 parted the coupled fingers more than 10 mm; with 50, no line search used more than 32 evaluations (in the first 20 islands, the ones MuJoCo keeps statistics for; the counter's scene has 22), no finger went more than 3.5 mm past a stop and no hand's fingers parted more than 2.7 mm. The 36 worst cases replayed with 200 evaluations are bit-identical to 50: 50 never bound there, in any island. Cost: the extra evaluations happen only at the steps that need them. A budget of 20 also binds during ordinary settles: from step 29 of a resting counter's settle the two budgets differ (by 4e-16 m/s at first, 0.02 mm after the 3 s settle; MuJoCo 3.11 uses the budget only as the line search's loop bound, so a search there reached 20, in an island it keeps no statistics for). The evidence records of this report were made with a budget of 20 and would come out slightly different with 50 (section 15).

The no-slip pass must converge. At MuJoCo's default no-slip tolerance the counter's hand basket and its contents creep steadily on four of five layouts (up to 13.4 µm/s, $r^2 = 1.000$, the basket itself walking across the counter), and a scanner held in a 35 N pinch turns about 7° and slips 8 mm over its first minute. Both are the same mechanism: MuJoCo's no-slip pass, which removes the slip its soft friction allows, stops at its tolerance, and MuJoCo's default (1e-6, with 10 iterations) stops it long before the slip is gone; more iterations at that tolerance change nothing. Measured on the counter's worst layout (checkout-present seed 2, 60 s hands off):

No-slip iterations, tolerance	Fastest steady creep (least squares over the run)	Largest displacement in 60 s	Physics wall/sim
10, 1e-6 (MuJoCo's tolerance)	14.07 µm/s	0.858 mm	1.37
100, 1e-6	14.04 µm/s	0.857 mm	1.35
30, 1e-10	6.12 µm/s	0.369 mm	1.55
50, 1e-10	1.96 µm/s	0.118 mm	1.70
100, 1e-10 (shipped)	**0.11 µm/s**	**0.064 mm**	2.12
10, 1e-6 at a 0.5 ms step	12.3 µm/s	0.742 mm	2.67

Shipped: 100 iterations at 1e-10. The physics then costs 1.55 times the default's at the same load, less than halving the step, which barely changes the creep. Rest on every layout and the 120 s hold: sections 11.1 and 9.2. The corner-impact energy (E19b) is unchanged by the pass: re-run on the shipped settings, every value of its table below is the same.

Friction impedance (`impratio`) and a fast corner impact (E19b). A high `impratio` helps the grip: with 10 no-slip iterations a 17.5 N pinch of the 214 g scanner creeps 0.031 mm/s at 10, 0.11 at 3 and 0.12 at 1 (E7, 30 s holds at 1 ms). But in an E17 run at 1 ms and `impratio` 10 a 430 g cereal box the hand pushed off the back of a shelf landed on its corner at 4.2 m/s, spinning at 14 rad/s, and left the floor at 6.95 m/s: in one tick (3.2 mm of first-contact penetration, 3673 N) it gained 3.2 times its energy and flew 4 m. E19b replays that impact from the box's recorded state 7 ms before its first contact (the first table is on the shipped settings; the second table was measured with the no-slip pass at 10 iterations and MuJoCo's tolerance, and the converged pass, above, removes the creeps it lists):

Step, `impratio`	energy gain	energy kept 3–30 ms after the strike	energy after 0.4 s	rebound of the lowest point
1 ms, 10	**× 3.24**	324 %	15.0 J	930 mm

Step, `impratio`	**energy gain**	**energy kept 3–30 ms after the strike**	**energy after 0.4 s**	**rebound of the lowest point**
1 ms, 3 (shipped)	none	60 %	0.15 J	28 mm
0.5 ms, 10	none	84 %	1.24 J	132 mm
0.5 ms, 3 (the G1's step)	none	60 %	0.15 J	30 mm
0.25 ms, 10	none	60 %	0.15 J	30 mm
0.25 ms, 3	none	60 %	0.15 J	30 mm

With `impratio` 3 the impact is already converged at 1 ms (it matches both 0.25 ms references); with 10 it needs 0.25 ms, four times the physics cost. Turning the no-slip pass off did not help (× 8.5). What each alternative does to the two creeps the contact model leaves (MuJoCo's regularised static friction; real contacts would hold still):

Setting	**17.5 N pinch of the scanner, 30 s still hold (E7)**	**10 g lip balm under a sideways load, 60 s rest (E20c)**	**corner impact (E19b)**	**physics cost**
1 ms, `impratio` 10	0.031 mm/s: 1.5 mm, 1.0°	1.6 µm/s	gains × 3.2	1
1 ms, `impratio` **3 (shipped)**	0.11 mm/s: 3.9 mm, 3.2°	6.1 µm/s (r^2 1.00)	converged	1
0.5 ms, `impratio` 10	0.048 mm/s: 1.7 mm, 0.8°	0.87 mm settling, then −1.1 µm/s	no gain, too bouncy (132 mm rebound)	2
0.25 ms, `impratio` 10	the head grasp failed: the fingers closed to 60 mm on the 66 mm head and the scanner slid 28 mm (not investigated: not a shipped setting)	0.04 mm, no trend (0.03 µm/s)	converged	4

The shipped setting takes the converged impact at the lowest cost. With the unconverged no-slip pass, its price is the hold's creep, 0.11 instead of 0.031 mm/s, and a faster creep of a small item under a steady sideways load; the converged pass removes both (the lip balm 0.23 µm/s, E20c; the held scanner slips 0.09 mm in a 120 s hold, E7).

MuJoCo keeps its own state. Newton's MuJoCo solver by default copies Newton's float32 joint state into MuJoCo's float64 `qpos`/`qvel` before every step. Resting bottles and cans then walked in straight lines ($r^2 = 1.0$) at up to 5.5 µm/s, 5 nm per step, far below float32's ~60 nm resolution at 1 m: the rounding of every round trip is a ratchet. With `update_data_interval = 0` (Config `solver_update_data_interval`) MuJoCo keeps its float64 state between steps and Newton's state is its float32 mirror; `World.sync_solver()` pushes a direct write (initial pose, checkpoint restore, a probe setting a pose) into MuJoCo. E20b, the stockroom, 60 s hands off after the reset settle, the same seed:

MuJoCo state before each step	**bodies walking linearly (> 0.5 µm/s, r^2 > 0.9)**	**fastest walk**	**largest displacement in 60 s**	**wall**
overwritten from Newton's float32 state (Newton's default)	16 of 53	5.51 µm/s	0.34 mm	288 s
kept (interval 0, shipped)	0 of 53	0.002 µm/s	0.16 µm	268 s

Warnings in the logs. The only Python warning in the logs is `mujoco_warp`'s "MULTICCD is enabled, but the scene contains CCD pairs without multicontact support: (CYLINDER, CYLINDER), (CYLINDER, BOX), (CYLINDER, MESH). At most 1 contact will be generated for these pairs", printed when a G1 scene is built (the G1's links include cylinders). It describes MuJoCo Warp's GPU collision pipeline: Newton builds a MuJoCo Warp model for its data mapping in both modes (`solver_mujoco.py`, `put_model`), but here stepping and collision are MuJoCo's CPU `mj_step` (`use_mujoco_cpu=True`, `use_mujoco_contacts=True`), which the warning does not concern. The render workers' warnings are Kit start-up notices (no display or audio device, deprecated extensions, texture streaming unavailable).

6.2 Contact parameters, and why

Setting	**Value**	**Why (evidence)**
Rigid contacts (every non-pad shape)	time constant 4 ms, impedance 0.95–0.99	MuJoCo's default (20 ms, 0.9–0.95) lets two arm links sink 28 mm into each other and an item 17 mm into a shelf under the arms' own torque; with 4 ms the E17 stress test meets 5.0–8.8 mm for ≥ 5 ms in six random cases, impacts of a fingertip or link 7 at arm speed (11.2); 2 ms (MuJoCo's floor at 1 kHz) goes unstable in one case (E17)
Fingertip pads	5 ms × √(finger mass / 29.1 g), anchored; geom priority 1	anchored in the label application environment; priority makes the pad govern its pairs (a 12.5 ms mixed contact lets a 17.5 N pinch creep and roll the scanner out)

Setting	Value	Why (evidence)
Friction	Coulomb, one coefficient per shape by material (`params.MU`: pads 0.8, cardboard and the tote's and bin's walls 0.40, cans 0.30, glass and film 0.35, books and the scanner 0.30, robot links 0.8–1.0); MuJoCo applies the LARGER of the two shapes' coefficients at equal priority, the pads' at theirs. The coefficients the solver applies per contact pair, read from the built models: stock on a shelf deck 0.35–0.40 (the item's own value; the decks' 0.30 never governs), stock on stock 0.35–0.40, stock and books on the counter or desk 0.35–0.50, the scanner in its cradle 0.30, on the cart tray 0.40–0.50, on the counter 0.35–0.50 and on the floor 0.60, a fingertip pad on anything 0.8 (priority, condim 3), a robot link on stock or the scanner 0.5–1.0. The values are per shape, not per material pair: explicit MuJoCo contact pairs are not built	estimates
Spin resistance	torsional friction (MuJoCo condim 4) 1 mm on every free body's contacts and on the fingertip pads' (priority: the pad's value governs; 1 mm is a 1.9 mm contact patch at the pad's μ 0.8, an estimate). On the pads: a scanner turned 9° in a 35 N pinch by a legal whip keeps sliding in the hand for 30 s with point contacts (4.0 mm, 0.1 mm/s at the end); with the term every held case of E17 stops sliding (≤ 0.001 mm/s over the last 5 s of a 15 s hold, 11.2), and E7b's torsion breakaway stays at the corner-loaded bound (0.385 N m, 9.2)	a point contact resists no spin about its normal: after a 20 s legal random abuse a 510 g bottle stood tilted on its rim against the stopped right arm and slid back and forth 0.6 × 0.4 mm at 1.75 mm/s for 30 s without decaying. Not the no-slip pass, the solver's iterations or tolerance, impratio 1–10 or a pyramidal cone (all unchanged), nor the arm's servo (held where it was: unchanged); with any torsional coefficient from 0.5 mm it stopped (0.0005–0.001 mm/s). 1 mm = (2/3) μ a for a contact patch of a ≈ 4 mm at μ 0.35–0.4 (estimate)
impratio	3	10 holds a pinch best (0.031 mm/s creep, with the unconverged no-slip pass) but lets a fast corner impact gain 3.2× its energy at 1 ms; 3 is converged at 1 ms and creeps 0.11 mm/s (section 6.1, E19b)
No-slip pass	100 iterations, tolerance 1e-10	MuJoCo's tolerance (1e-6) stops the pass early: resting stock in the counter's basket creeps up to 14 μm/s and a held scanner turns 7° in a minute; converged, the basket's stock creeps ≤ 0.3 μm/s and the held scanner slips 0.09 mm in a 120 s hold (sections 6.1, 9.2, 11)
Contact detection gap and the box-box routine	0 for boxes and planes, 2 mm for convex hulls. Box SKUs of 5 kg or more and the counter top are 8-vertex convex hulls: MuJoCo 3.11's box-box routine returned a contact hundreds of millimetres deep against the counter top with a box resting still on it, twice (the 12.5 kg water case, 584 mm, at a reset; the bagging tote's 4 mm floor, 753 mm, during E17's held-whip settle, which recurred at another tick on rerun); with the top a hull the same E17 sequence ran clean. Every free box as a hull was tried and rejected on measurement (a cereal box in the seed-2 basket crept 24 μm/s against 1.0 μm/s as a box; E7's held scanner slipped 3.3 mm and turned 3.5° against 0.9 mm and 0.7°: the convex collider's contact points on a box face are fewer and less steady than the box routine's); the counter top as a hull left the baskets' creep unchanged. The other box-box pairs (stock on shelf decks and on stock, the scanner in its socket) are checked by a reset sweep of every layout (11.2) and by E17's single-tick spike count	with Newton's default gap of 0.1 m, MuJoCo's box-box routine returns false contacts up to 0.64 m deep and hurls a 5 kg case at 3–6 m/s from a finger 9 cm away (E17); a 2 mm gap on boxes still produces an 89 mm spike between two boxes; hulls without a gap flicker and a shelf of jars creeps faster and faster

6.3 Robots, controllers and protective stops

The robots come from the shared catalog (`robots/`, `for_environment("barcode-scanning", robot)`): `dual-fr3` (default; rig `scan-fr3`), `dual-panda` (`scan-panda`) and `single-fr3` (`scan-fr3-single`, left mount), each arm with a Franka Hand and a wrist camera, and `g1` (`scan-g1`: the floating Unitree G1 with Dex3 hands, section 6.5). The joints are position servos with the catalog's gains; targets are slewed at the datasheet joint speeds. The gains are this environment's own controller profiles (`barcode_scanning` in `robots/models/fr3`, `panda` and `g1` `robot.json`; the rigs name them), copies of the label application profile for the arms and the whiteboard profile for the G1, so that a gain change made for another environment cannot change this one's robots. Their start pose is this environment's own too (pose `barcode_scanning` in each `robot.json`; a test pins it). Selection is a closed list of the four configurations, by alias or as the rig objects themselves: a setup passed as an object (a `RobotSetup`, its dict, or a rig JSON) is accepted when its catalog fingerprint is one of the four rigs' and then runs as that alias (`barcode_scanning.robot_setup.validate_setup`, the catalog's `validator`); any other setup is refused with what the environment needs (the cart's two mounts, an IK tool frame per arm, a parallel hand with fingertip pads, wrist cameras and a measured reach). Custom setups are not built (section 14, item 18). Two protective stops of the real robot are modelled:

- **Velocity reflex**: a joint faster than 1.05x its datasheet limit stops its arm (targets frozen), reported in the step info.
- **Collision reflex** (libfranka `setCollisionBehavior`; the reference setting of the libfranka examples, `examples_common.cpp setDefaultBehavior`): an arm whose *external* joint torque exceeds 10 N m (nominal) or 20

N m (while its targets are still slewing) on any joint stops. External torque is J^T f over the contact rows only (not joint limits, the gripper's own squeeze, or gravity). Each reflex records the joint, the torque and the contact that caused it. Without it, legal full-torque jams drive a fingertip 14 mm into a keycap and push a finger housing into a shelf upright at about 500 N. A stopped arm keeps its frozen targets and refuses new ones for 1 s (authored; the real robot stays stopped until an error recovery, whose duration libfranka does not publish); finger targets are still accepted, so a stopped arm can let go. Refused commands are counted.

The arms' joint limits are hard stops (time constant 2 ms, impedance 0.99–0.999): with MuJoCo's soft default a jammed joint ran 0.135 rad past its limit.

What each configuration can reach (kinematic, every arm's IK with restarts at 1 mm and 1°, seed 0 of each preset):

Configuration	Stockroom: take the scanner / aim at the target label / key 5	Counter: take the scanner / present to the cradled scanner / key 5 / over the bagging tote	Library: present / over the returns bin	Collision-free top grasps of stock where it lies
`dual-fr3`	both arms / both / both	both / both / both / right only	both / right only	none (35, 5 and 3 items)
`dual-panda`	both / right only / both	both / both / both / right only	both / right only	none
`single-fr3` (left mount)	yes / yes / yes	yes / yes / yes / **no**	yes / **no**	none

So the one-arm FR3 can take, aim and key in the stockroom and present at the counter and the library, but cannot reach the tote or the bin from the left mount: with one arm `checkout-present`, `checkout-bulky` and `library-checkin` cannot be finished as built (the item cannot be bagged or binned), and a two-hand trigger pull (E6r) is impossible. Those three pairs are refused at launch with the reason, as is `g1` with `checkout-bulky` (the case is 181 mm out of reach standing, 6.5.4): `UNREACHABLE_TASKS` in the backend, stated in `gar inspect`'s notes. The two-arm Panda's left arm cannot reach the target label's scan pose in the stockroom (its right arm can); the two-arm Panda ran the scan-location task once (E9, section 10.1).

No arm of any configuration has a collision-free top grasp of any stock item where it lies (`grasping.plan`, which grasps from above across a pair of horizontal faces at most 80 mm apart, the hand's stroke): the books, the AA pack and the glove boxes lie on their largest face, and both their horizontal widths exceed 80 mm (the almanac 156 × 235 mm); around the basket's other items and the shelved cereal the hand collides at the grasp pose itself (the basket's walls and neighbours, the shelf above). That is why every task run hands items over through a labelled phantom customer, stocker or patron.

The robot's own attempts. Through the actions a policy has, each arm of the two-arm FR3 executes, for each item where it lies (seed 0 of each preset), the planner's least and most tilted rejected candidates and a horizontal approach from the aisle (a side grasp of a shelf unit; for a book lying flat, a pinch across its thickness at its near edge): a collision-checked path to a start pose 12–40 cm back along the approach, then a guarded descent in 5 mm steps that stops at the first contact over 5 N with anything but the item or at a protective stop; where the hand reaches the grasp point it closes and lifts 10 cm. A side grasp across a unit wider than the hand's 80 mm stroke is not tried. Result: **15 physical attempts, none lifts an item; 16 more found no collision-free arm path to any start pose** (planner refusals, counted apart):

Preset	Items (the planner's reachable candidates, accepted)	Physical attempts	What stopped them	Planner refusals
counter, basket	tomato paste, spring water, marinara jar, lip balm (21–94 per arm, 0 accepted); the AA pack has no candidate (lying flat, both widths over the stroke)	10	the hand on the basket's rim, then the collision reflex (8), a 26 N contact with the basket (1), the cradled scanner 31 N on the right arm's approach (1); no item touched	4
stockroom, facing of A03-3-B/C	soup can, cereal box, tomato paste (0 accepted); the cereal box's side grasp not tried (195 mm across the fingers, both arms)	2	the side grasp meets the neighbouring unit (contact over 5 N)	10 (the beam above, the neighbours, the cart's mast)
library, desk stack	three books (0 candidates: lying flat, both widths over the stroke)	3 (the edge pinch)	the lower finger meets the desk, then the collision reflex	2

So in every preset the physical attempt fails the way the planner predicted, by contact with the basket, the neighbouring stock or the desk. Not tried: pushing an item to the edge of the basket or the desk first, tilting it against a wall, or a two-hand grasp; the phantoms stand in for those. **Where the items stand on the counter** (`checkout-counter`, section 8) the hand takes every item itself where it lies (section 10.5). Besides the two-arm FR3's task runs (10.5–10.7), the two-arm Panda and the one-arm FR3 ran the scan-location task (10.1).

Numerical-failure guard. An object faster than 20 m/s or spinning faster than 1000 rad/s stops the episode as a backend fault, like a non-finite state (`params.MAX_OBJECT_SPEED_M_S`, `MAX_OBJECT_SPIN_RAD_S`: 3× a 2 m drop's 6.3 m/s, 8× a 16 mm lip balm rolling at 1 m/s). Nothing in the evidence reaches it. The robot's links are outside

that bound (driven at its URDF joint speeds, a G1 fingertip can reach 24 m/s), but a Franka Hand whose two coupled fingers stand more than 10 mm apart stops the episode the same way (`params.MAX_FINGER_COUPLING_M`): the fingers ride one drive, the hardest legal impacts of 158 FR3/Panda abuse cases part them by at most 2.7 mm, and the unconverged solves of a cut-short line search (section 6.1) part them by more than 10 mm in 36 of those cases, up to 232 mm.

6.4 A humanoid hand on the scanner (G1 Dex3 component probe)

Scope: can a Dex3 hand, within its URDF effort limits, hold the 214 g scanner by its pistol grip and pull its trigger with the index finger? Component level: the palm is carried by a stiff labelled servo standing in for the arm (no G1 arm, no balance).

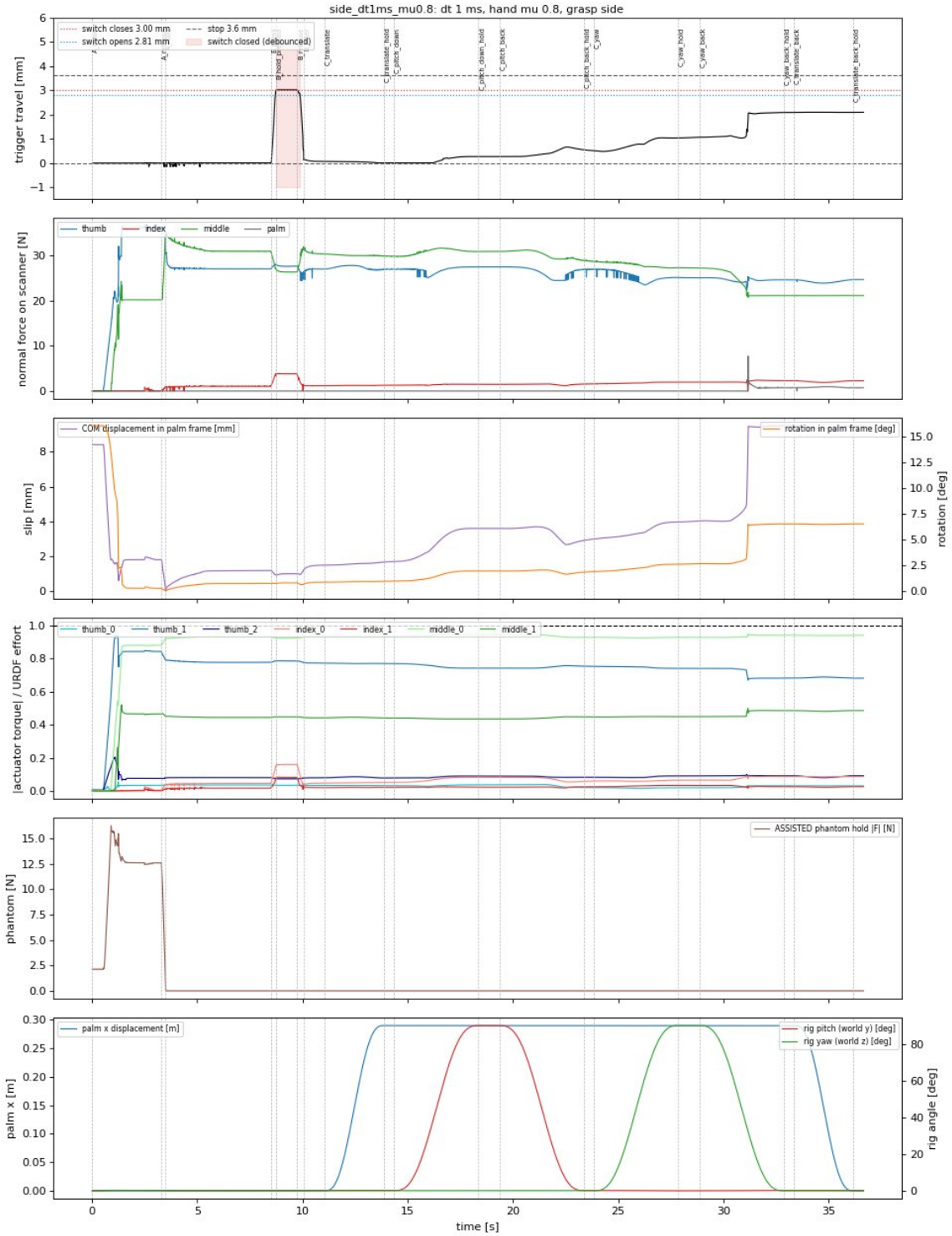


Figure 11. The Dex3 hand on the scanner at 1 ms (shipped `impratio` 3): trigger travel, finger contact forces, the scanner's shift in the hand, the finger efforts, the phantom that holds the scanner only while the fingers close, and the wrist rig's motion in experiment C (the switch closes at 3.01 mm; during the turns the trigger creeps to 2 mm without closing).

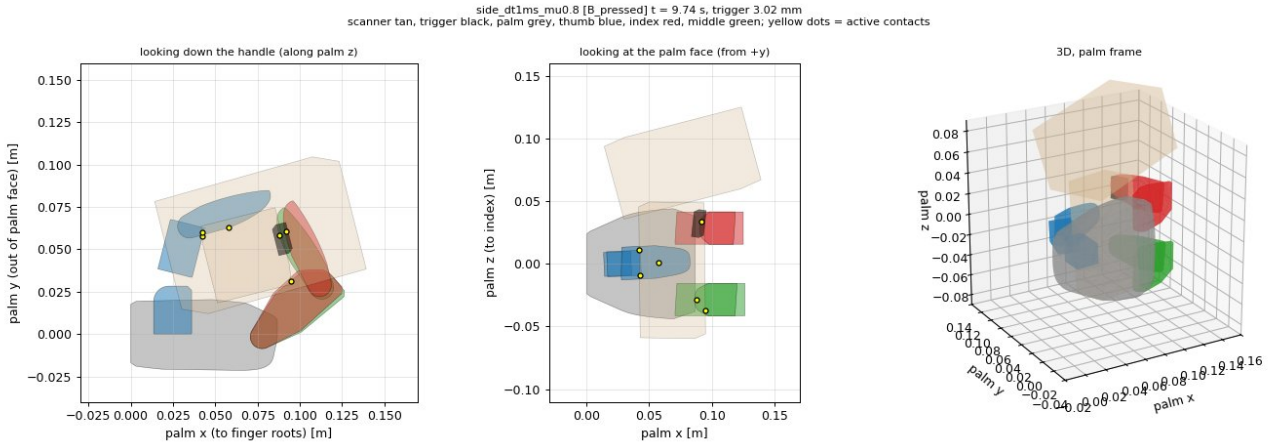


Figure 12. The collision geometry at the pull (trigger 3.0 mm): scanner tan, trigger black, palm grey, thumb blue, index red, middle finger green; yellow dots are active contacts.

The scanner is the environment's own (`Scene.scanner_and_cradle`, the trigger spring and stops of `world.py`); the hand is the right Dex3 of the pinned G1 URDF (masses from the URDF, efforts 1.4 N m and 2.45 N m for thumb_0, the catalog's whiteboard gains kp 1.8, kd 0.035, no armature: neither the URDF nor Unitree's MJCF gives one); the solver settings are the environment's. ASSISTED: a spring hold on the scanner while the fingers close, ramped to zero before anything is measured, and the palm's servo (deflection ≤ 66 µm, 0.28°). Grasp found by search: palm on the scanner's right side, thumb on the back of the handle, middle finger on the front, index finger resting on the trigger blade.

Measured (friction 0.8; shipped `impratio` 3)	**1 ms (the FR3 configurations' step)**	**0.5 ms (the G1's step)**	**0.25 ms**
5 s hold, largest shift of the scanner in the hand	1.22 mm, 0.72°	0.92 mm, 0.50°	0.96 mm, 0.55°
creep after settling	0.094 mm/s	0.043 mm/s	0.043 mm/s
index finger pulls the trigger: switch closes / reopens	3.01 / 2.76 mm	3.07 / 2.76 mm	3.08 / 2.76 mm
0.3 m move and two 90° turns while holding, largest shift	8.0 mm, 5.8°	8.6 mm, 6.3°	8.8 mm, 6.5°

(At `impratio` 10: 5 s hold 3.43 / 0.74 / 1.08 mm, creep 0.008–0.017 mm/s, the move 3.1 / 7.8 / 7.5 mm.) The index finger pressed the blade with 3.8–4.0 N using 16–17 % of its effort limit; the middle finger's proximal joint ran at up to 0.96 of its limit during the hold. Qualifications, all measured at `impratio` 10: a placement with the handle 70 mm off the palm vibrates at 1 ms (thumb force flickering 0–50 N) and not at 0.25 ms; at the manufacturer's 0.49 N m pushing torque (Unitree Dex3-1 page; the URDF's 1.4 N m is its other direction) the scanner slips 7.5 mm and turns 12.7° when released; at finger friction 0.45 (whiteboard's) instead of 0.8 (unsourced) the hold shifts 5.2 mm and 4.3°. **So a Dex3 hand can hold this scanner and pull its trigger; the G1 configuration (arm, balance, controls, evidence) is section 6.5.**

6.5 The G1 humanoid with Dex3 hands (robot `g1`)

The fourth robot configuration: a Unitree G1 (29 body joints) with two Dex3-1 hands (7 joints each), floating, standing on the floor in front of the shelf or the counter, operating the same scanner, cradle, keypad, display and stock as the FR3 configurations. `gar run barcode-scanning --robot g1 ...` selects it (catalog alias `g1`, rig `robots/rigs/scan-g1.json`). **What is demonstrated and what is not, in one line:** the G1 stands unsupported in every preset, holds the scanner by finger contact alone, aims it with its arm and waist and pulls the trigger with its own right index finger, and the rendered imager frame decodes the shelf label it was aimed at. It takes the scanner out of its cradle by itself only with a tipped lift (3 of 7 lift variants), in a grip from which its index finger cannot reach the trigger, and it cannot put the scanner back: let go at the lowest height the pistol grip clears (30 mm), the scanner falls out (below). So the read starts from a labelled phantom setup and `scan-location` end to end is not achieved (unmodified evaluator: the scanner is not seated at the end).

6.5.1 What was built

Part	**Where**	**What**
Robot	`robots/rigs/scan-g1.json`; `scene.py _g1_robot`	the catalog G1 (`g1@1`, pinned URDF `g1_29dof_with_hand_rev_1_0`, sha256 555edb93…) with this environment's own `barcode_scanning` controller profile (`robots/models/g1/robot.json`; the whiteboard gains copied unchanged, own start pose `barcode_scanning`; no retuning); floating base, no weld, no pelvis support; 34.39 kg (summed from the model); instance name `g1`, joints `g1/<urdf joint>`

Part	Where	What
Placement	`g1.py` LAYOUT	stockroom: pelvis at (0, -0.55) m facing the shelf; counter and library: (0, -0.22) m facing the counter, arms starting down at the sides (shoulders pitched 0.1 rad forward and rolled 0.3 rad out, elbows 1.3 rad; every arm and hand link at least 22.7 mm from the body, the other arm and the world, MuJoCo distance test) so the forearms do not start inside the 0.915 m counter top
Cart and tray	`scene.py` `cart_point`, `g1.py` TRAY	the FR3 cart stands where a standing operator must be (0.28 m from the shelf face) and a G1 beside the unmoved cart is 0.5–0.6 m from the cradle, so for the G1 the same cart is parked turned 90° at the robot's right (centre (0.65, -0.56) m), its tray facing the robot; on the tray the cradle keeps its FR3 position (x = +0.19 m along the tray, now the shelf end) turned so its window faces the shelf, the keypad is turned so its top row is away from the robot, the display is laid back towards the robot with its text upright. FR3/Panda scenes are as described above
Physics step	`config.py` `G1_RIGID_DT_S`	0.5 ms (`rigid_dt` resolves per robot; FR3/Panda stay at 1 ms)
Upper-body gravity feed-forward	`world.py` `_g1_gravity_feedforward`; `Config.g1_arm_gravity_ff`	model gravity torque of the 3 waist and 14 arm joints at their commanded positions, refreshed every 5 ms, clipped to the URDF efforts, added to the PD servos
Protective stops	`world.py`	none: the FR3's libfranka velocity and collision reflexes are not applied to the G1 (section 6.5.2)
Controls	`gym_anything_robotics/backends/newton_barcode_scanning.py` `_plan_g1`	named joint targets for all 43 joints (applied at once); Cartesian pose/delta for `left`/`right` through the bounded 7-joint IK (`g1.G1HandIK`, whiteboard HandIK convention: TCP 80 mm along the wrist-yaw link, 2 mm / 0.03 rad acceptance), streamed along a quintic ramp over the action; `gripper` and `keyboard` refused with ActionRejectedError before any side effect
Observation	same	proprioception: 43 joint positions and velocities, `robot_status.imu` (pelvis roll/pitch/yaw, tilt), refused-target count; cameras: `head_rgb` (policy default; URDF `d435_joint` on the torso, 69° HFOV, the whiteboard mount) and `cart_rgb` (mast camera, turns with the cart); diagnostic `g1_left`, `g1_back`
Scripted operator (ASSISTED)	the probes' G1 operator	IK waypoints and a joint-space path check (kinematic clearance, 1 cm from the world), guarded finger closing of the Dex3 component probe, fall detector (pelvis tilt > 10° stops the run)

6.5.2 Parameters and provenance

Parameter	Value	Provenance
joint gains, efforts	legs kp 800 / kd 20; waist, shoulders, elbows 120 / 6; wrists 80 / 3; Dex3 1.8 / 0.035; URDF efforts (Dex3 1.4 N m, thumb_0 2.45 N m)	`robots/models/g1/robot.json` profile `barcode_scanning` (the whiteboard profile's values copied, not re-derived)
target slew	URDF joint velocities (arms 37 rad/s, wrists 22, hand 12, thumb_0 3.14)	URDF (the World's existing slew applies the joint's `velocity_limit`)
upper-body gravity feed-forward	model torque at the commanded pose	web-sourced practice: Unitree's G1 teleoperation stack computes `pin.rnea(q_target, 0, 0)` and sends it as the motor `tau` with the arm targets (github.com/unitreerobotics/xr_teleoperate `robot_arm_ik.py`, `robot_arm.py`); measured need: without it the waist sagged 0.074 rad (4.2°) and the hand sat 3–4 cm below its kinematic pose; with it the arm/waist errors at rest are 0.4–4 mrad
protective stops	none	web-sourced: in low-level mode the G1 has no documented automatic collision or velocity reflex; the emergency stop is a manual damping mode (Unitree FSM notes, GetSoloTech unitree-g1-replay FSM_AND_SAFETY.md). Not modelled: over-current/thermal motor protection, the manual damping stop
physics step	0.5 ms	measured (6.5.3)
contact law	the scene's rigid law (4 ms, impedance 0.95–0.99) on every G1 shape; no Dex3 pad model	environment value (world.py), as in the Dex3 probe
finger friction	μ = 0.8	the environment's unsourced robot value (params.MU robot_default)
placements	table in 6.5.1	authored, checked by the reach study (6.5.4)
head camera	URDF d435_joint, 69° HFOV, zero lens offset, ideal pinhole	URDF + whiteboard mount (Intel D435 RGB FOV, as in the whiteboard environment)

6.5.3 The physics step (measured; all rows at `impratio` 10, and the rest rows with PD servos only, without the arms' gravity feed-forward)

Run	1 ms	0.5 ms
G1 hands-off rest, stockroom, PD only (no feed-forward), Newton update_data_interval 1	**fell** (at the end: pelvis 662 mm from its start, tilt 88°)	not run
same, update_data_interval 0	**fell** (at the end: pelvis 641 mm from its start, tilt 87°)	stood 30 s: pelvis 5.8 mm settle then constant, tilt 0.89°
Dex3 probe grip, qualified placement, 5 s hold slip (update_data_interval 1 / 0)	3.43 / 3.70 mm, 2.5 / 2.7°	0.74 (recorded) / 0.72 mm

Run	1 ms	0.5 ms
Dex3 probe grip at the placement that vibrated, trigger ringing before the index touches (1 / 0)	12.8 / 5.9 mm, 12 / 8 spurious switch events	not run / 3.1 mm, 0 events

Keeping MuJoCo's float64 state (update_data_interval 0) halves the ringing at the bad placement but does not remove it, does not change the qualified grip's 1 ms slip and does not prevent the 1 ms fall; the step size is the cause. (At the shipped `impratio` 3 the qualified grip slips 1.22 mm at 1 ms and 0.92 mm at 0.5 ms, section 6.4.) Cost: G1 physics alone at 0.5 ms runs at 2.4–6.2x wall/sim at rest (the spread is host load: two simulations shared the CPU for the slower runs).

6.5.4 Placement and reach (kinematic)

Arm alone = the Cartesian action's 7-joint IK; +waist = with the waist yaw/roll/pitch (a policy commands those by name).

Preset	Target	arm alone	+ waist
stockroom	pistol grip on the cradled scanner (full orientation)	yes	yes
stockroom	scan pose (window 0.15 m before the label, 20° down) at A03-3-B, A03-3-C	yes	yes
stockroom	scan pose at A03-2-A/B/C, A03-3-A	no	yes
stockroom	scan pose at A03-4-A/B/C (1.43 m)	no	**no** (52–90 mm short): the task's seed 4 slot A03-4-B is out of reach standing
stockroom	keypad keys 1, 5, 9, ENTER	right hand yes	both hands yes
counter/library	keypad keys 1, 5 (9 and ENTER within 12 mm)	right hand yes	both hands yes
counter/library	presentation zone 0.10 m before the cradled scanner's window	both hands yes	yes
counter/library	0.12 m above the basket, bagging tote, returns bin, book stack	no (28–204 mm short)	yes, both hands
counter	0.12 m above the 24-pack of water at the counter's far end (0.70 m right)	no	**no** (181 mm short): `checkout-bulky`'s handheld scan of the case is out of reach standing
counter/library	pistol grip on the counter cradle's scanner (window towards the robot)	no	no (not needed for presentation scanning)

The counter/library placement is 0.22 m from the edge with the arms down at the sides. At 0.50 m from the edge with the forearms forward every basket, tote, bin and book target is 88–420 mm out of reach even with the waist, and at 0.40 m the fingertips start pressed into the counter edge (100–210 N).

6.5.5 Evidence

(a) Hands-off rest, 30 s, every preset. No command at all after the environment's reset; PD servos plus the upper-body gravity feed-forward; no support.

Preset	pelvis drift (total)	pelvis speed, last 5 s	tilt at end (max)	feet drift	robot contacts other than feet	scanner
stockroom	2.19 mm	1.1e-5 mm/s	0.69° (0.85°)	< 0.004 mm	none	seated throughout, moved 0.002 mm
counter	0.06 mm	3.6e-6 mm/s	0.21° (0.23°)	< 0.001 mm	none	seated throughout, moved < 0.001 mm
library	0.69 mm	3.6e-6 mm/s	0.21° (0.27°)	< 0.002 mm	none	seated throughout, moved 0.0001 mm

The start pose keeps every arm and hand link 22.7 mm clear of the robot's own links: an arms-down start with the elbows at 1.5 rad and no shoulder roll puts each hand 34 mm inside its own hip links, and the robot falls before tick 0 (pelvis on the floor, tilt 102°). The environment's reset raises an error if the G1 is down after the settle (`env.py`), and a physics test checks the start pose of every preset for robot penetration and a standing reset. Stock at rest: at most 0.047 mm at the counter (the water bottle in the basket, settling while the G1 stands beside it), 0.004 mm in the library.

(b) Taking the scanner out of its cradle by the handle: achieved with a tipped lift, in a grip that cannot pull the trigger (no assistance).

- **What the cradle leaves the hand** (kinematic, the Dex3 probe's qualified pistol grip posed by the right arm's IK on the seated scanner, each finger closed alone from the preshape): with the hand at the seated scanner the thumb closes on the handle and the index on the trigger blade, and the middle finger first touches the cradle's lip (0.38 mm penetration); raised 15 mm along the handle, every finger closes on the scanner (thumb and middle on the handle, index on the head) and none touches the cradle; at 25 mm the thumb base meets the head at the preshape.
- **The approach** rises from the start, crosses above the scanner on a straight Cartesian line and comes straight down to the point 70 mm beyond the handle; the scanner does not move during it in any variant. (A

single joint-space move through two waypoints dips 12 cm between them, the thumb strikes the seated scanner and the robot tips past the operator's 10° fall limit.)

- **The lift**, from an upper-handle grip found by a kinematic sweep (palm 40 mm off the right flank, thumb high on the back strap, middle finger on the front face 60–65 mm above the handle bottom, index extended), the first 30 mm along the socket's axis and the rest as listed:

lift	scanner rise along the socket's axis	slip in the hand	lip / rear wall force max	out
straight, thumb squeeze 1.0 (of its URDF effort)	0.0 mm	11.0 mm, 11.4°	111 / 93 N	no
straight, thumb 0.6	1.0 mm	5.2 mm, 10.3°	101 / 89 N	no
straight, thumb 0.4	1.6 mm	9.2 mm, 9.7°	86 / 82 N	no
thumb 0.4, hand 4 mm towards the handle front	1.4 mm	11.3 mm, 11.4°	104 / 97 N	no
+ handle bottom tipped 8° forward	**117 mm**	10.2 mm, 10.3°	89 / 85 N	**yes**
+ tipped 15° forward	**126 mm**	13.6 mm, 9.6°	62 / 63 N	**yes**
+ tipped 8° and the hand 4 mm away from the side wall	**113 mm**	11.1 mm, 10.1°	77 / 76 N	**yes**

The closed grip already presses the handle against the lip and the rear wall (about 20 N each). Lifted along the socket's axis, the handle wedges between them like a drawer: the forces climb to 82–111 N while the hand slides 5–11 mm up the handle and turns it 10–11°, and the arm stops short of its target. Tipping the handle's bottom forward frees it: in the three tipped lifts the scanner comes out after up to 3 s of wedging, still held, 10–14 mm and about 10° off its grasped pose.

- From the take grip the index finger cannot pull the trigger: after the tipped lift (out, 117 mm), flexed at 0.5 rad/s to both joints' limits (1.5 rad), it never touches the blade (0 N). A read by the G1 alone would need a regrasp into the pistol grip (a handover between the hands or a set-down and regrasp), which is not built.

(c) Hold, aim and trigger pull with the G1's own index finger, and a decoded read (seed 2, target slot A03-3-C). ASSISTED setup, labelled in the output: because the take grip of (b) cannot reach the trigger and no regrasp is built, a phantom hold (`probes.fixtures.PhantomHold`, gravity compensated) lifts the scanner out of the cradle and holds it in the air while the right hand closes the probe's pistol grip around the handle; the phantom is then ramped to zero over 0.2 s and stays off. Everything after is unassisted physics driven by joint targets.

Step	Result
grip after closing	thumb 23.5 N, middle 16.4 N, palm 3.3 N on the scanner (rendered run)
hold, 2 s, fingers only	slip 0.009 mm, 0.008° (1.96 mm and 1.41° with the unconverged no-slip pass)
aim (arm alone has no reachable solution for this scan pose; arm + waist)	window 2.4 mm and 0.64° from the planned scan pose (0.15 m from the label, 20° down)
trigger pull: right index flexes further at 0.5 rad/s	switch closes at 3.06 mm with 3.85 N of index force on the blade; releases on the index's return
read (rendered run, Isaac RTX imager, shipped decoder)	**Code128** `A03-3-C` from the first frame of the session, attributed to the printed label `loc_A03-3-C` at 0.201 m, no ghost, no misread; one rollout from tick 0 (phantom setup included, no checkpoint restored)
scanner in the hand from the hold through the pull	slip 0.18 mm, 0.31° (12.0 mm and 15.3° with the unconverged no-slip pass: the Dex3 pinch rolls the scanner)
evaluator (unmodified), after the read	location_recorded, one_location_read, nothing_on_floor, no_unexplained_reads, no_key_pressed, no_misreads: pass; scanner_seated: **fail** (score 0.86)

(The rendered run: 18.0 s simulated, 259 s wall; the physics-only run with the reseat attempt: 29.7 s simulated.) Physics-only run (dark stand-in frames): the same switch closure; the firmware ran a session of 55 frames without a read, as it must with dark frames.

(d) Reseating: not achieved. The pistol grip's middle finger sits in front of the handle under the head's overhang: lowering the held scanner along the socket's axis, the middle finger's links meet the cradle's lip once the handle bottom is 20 mm above the floor (kinematic check, current finger pose), so the lowest clear release is 30 mm up. Let go there, the scanner does not stay in the socket: it falls to the floor (`nothing_on_floor` fails too). The FR3's insertion lets go 10 mm above the floor, upright, with the loop closed on the scanner's pose (section 3.4); the G1's operator lets go at its seated pose without that loop, and neither a lower release nor a closed loop has been tried for it.

Runtime through GAR (section 12; head_rgb observation every action, the render worker kept between operations): 0.1 s actions 24.9× mean (30.7× max), 0.5 s actions 16.2× (17.6×), startup 50–56 s; under the limit (the per-step trace costs the G1's 0.5 ms steps the most). (With a fresh worker per operation, 0.1 s actions run at 217–228×, over it.)

6.5.6 Limitations of the G1 configuration (each is unfinished work, not a property of the job)

1. **No regrasp and no return to the cradle** (6.5.5b, 6.5.5d). The G1 takes the scanner out only with a tipped lift (3 of 7 variants; a straight lift wedges the handle between the lip and the rear wall), in an upper-handle grip whose index finger cannot reach the trigger; a regrasp into the pistol grip is not built, and let go at the lowest height the pistol grip clears (30 mm) the scanner falls out of the socket. Every G1 read starts from the phantom setup.
2. **No balance controller.** The robot stands on PD servos; a hand pushing a fixture can tip it (observed). A real G1 doing this job runs Unitree's locomotion/balance controller under an arm SDK; that controller is not modelled, and the waist gravity feed-forward stands in for its posture holding.
3. **Runtime depends on the kept render worker**: 15.5× at 0.1 s actions with the default; 217–228× with a fresh worker per operation (6.5.5 runtime, section 12).
4. **Seed 4's slot (A03-4-B, 1.43 m) is out of reach standing** (6.5.4); the robot cannot walk.
5. **No Dex3 pad compliance, tactile sensing or grip-force mode**; finger friction is the environment's unsourced 0.8. The operator's touch detection reads contact forces (standing in for the Dex3-1's tactile sensors).
6. **No protective stop** beyond torque saturation and joint limits.
7. The head camera has no lens offset calibration or noise model (as in the whiteboard environment).

7. Parameter ledger

Every constant in `params.py`, with its provenance as written in the code. The table is generated from the file's source, and `params.ledger()` is written into evidence metadata. Solver settings are in `config.py` (section 6).

Group	Parameter	Value	Provenance and reason
scanner body (Zebra DS2278 class)	SCANNER_MASS_KG	0.214	datasheet [S1]
	SCANNER_CORDED_MASS_KG	0.1616	datasheet [S1]: the corded DS2208, 1 cm shorter; the cordless unit's extra 52 g (its battery) sits in the handle and base
	SCANNER_CORDED_HEAD_FRACTION	0.6	estimate: the corded unit's head (imager, window, LEDs) against its handle
	SCANNER_ENVELOPE_M	(0.175, 0.066, 0.090)	H x W x D, datasheet [S1]
	CRADLE_MASS_KG	0.151	datasheet [S1]
	CRADLE_ENVELOPE_M	(0.072, 0.094, 0.122)	H x W x D, datasheet [S1]
imager	WIDTH_PX, HEIGHT_PX	640, 480	datasheet [S1]
	HFOV_DEG, VFOV_DEG	32.8, 24.8	datasheet [S1]
	WAVELENGTH_M	645e-9	illumination LEDs, datasheet [S1]
	AIM_WAVELENGTH_M	624e-9	aim line, datasheet [S1]
	FRAME_RATE_HZ	30.0	estimate: SE4710 default [S3]; DS2208 unpublished
	PUPIL_SETBACK_M	0.050	derived: at the datasheet's 13 mm UPC near limit a UPC-A with its quiet zones (113 modules x 0.33 mm = 37.3 mm) just fills the 32.8 deg field 63 mm from the pupil (37.1 mm); the near limit is a field-of-view limit ([S2] Table 3-3 note)
	PUPIL_DIAMETER_M	0.00041	derived: decoder-independent fit: the least worst blur, in module widths, over every published near and far limit of PRG Table 3-3 [S2]
	FOCUS_DISTANCE_M	0.232	derived (the same fit), from the entrance pupil: 182 mm before the window, against the 194 mm two other Zebra engines publish [S3]
	LED_OFFSET_M	0.010	estimate: LEDs either side of the lens
	AIM_OFFSET_M	0.008	estimate: aim source above the lens
	AIM_CROSSING_M	0.150	estimate: aim/optical axes cross (SE4710: 195 mm [S3])
	LED_FWHM_H_DEG, LED_FWHM_V_DEG	39.3, 28.3	derived: the SE4710 engine's pattern (50.3 x 32.0 deg full widths at half intensity [S3]) scaled by the ratio of the fields of view (DS2208 32.8 x 24.8 deg [S1], SE4710 42 x 28 deg [S3]): the DS2208's engine is not the SE4710 and its pattern is unpublished; elliptical Gaussian IES profile (a hard 25 deg cone leaves the edges unlit, E9)
	AIM_FAN_DEG, AIM_THICK_DEG	32.8, 0.3	estimate: fan = HFOV ("aiming line smaller when closer" [S2])
sensor	SENSOR_WEIGHTS	(0.35, 0.45, 0.20)	estimate: mono CMOS response to the R, G, B render bands
	FULL_WELL_E	6000.0	estimate
	READ_NOISE_E	8.0	estimate
	PRNU	0.01	estimate
	AE_TARGET_DN	110.0	estimate
	AE_HIGHLIGHT_DN	230.0	estimate: hands-free mode holds the 99.5th percentile of the frame's block means at or below this (highlight protection; a clipped label loses its bars with it)
	AE_BLOCK_PX	8	estimate: AE statistics on 8 x 8-pixel block means (a subsampled histogram)
	GAIN_MAX	8.0	estimate
	RENDER_MC_REF	(128, 0.0127)	measured: (samples, relative Monte Carlo noise)
	RENDER_NOISE_FRACTION	1/3	derived: the convergence criterion of section 4.5
	RENDER_SPP_RANGE	(64, 256)	authored: floor and ceiling (cost per sample in the task scenes: section 4.5; the bench scene costs ~4 ms)

Group	Parameter	Value	Provenance and reason
	RENDER_IMAGER_MAX_BOUNCES	3	derived: path depth of the scanner's frames (section 4.5: 1.7-2.5x cheaper per sample than 6; frames with a read within 2.7 % of 6's radiance, empty dark frames up to 8 % darker; the same decodes on all 17 recorded task frames compared)
	T_MIN_S	20e-6	estimate
	T_EXPOSURE_START_S	5e-4	estimate
	T_MAX_HANDHELD_S	0.5*0.33e-3/0.13	derived: half a 13 mil module of blur at the 13 cm/s motion tolerance [S1]
	T_MAX_HANDSFREE_S	0.5*0.33e-3/0.762	derived: 76.2 cm/s swipe speed [S1]
	SENSOR_K	3.1e6	identified (render calibration): white paper under LED+200 lx at 150 mm -> DN 128 at 0.4 ms, gain 1
light levels (renderer units, calibrated in E1)	LED_INTENSITY	2.03e6	derived from an estimate (render calibration): LED:room sensor signal 10:1 on white paper at 150 mm, 200 lx room; the 10:1 is a power-budget estimate (0.3 W to the LEDs at 20-30 %, ratio 6-14), not a device measurement; includes the factors x 2.19/83.4 for the IES profile in place of a hard cone and / 1.163 for the pattern scaled to the field of view (the on-axis gain per unit intensity measured with the normalised IES)
	AIM_INTENSITY	2.0e4	authored (render calibration): aim-line peak ~7x the 200 lx room illuminance at 150 mm (a visible line on white stock; no published luminance)
	ROOM_LUX_STOCKROOM	200.0	search [S17]: 150-300 lx manned rack areas / packing
	ROOM_LUX_COUNTER	500.0	search [S17]: till areas
firmware (DS2208 Product Reference Guide [S2])	DECODE_SESSION_TIMEOUT_S	9.9	
	SAME_SYMBOL_TIMEOUT_S	0.5	
	DIFFERENT_SYMBOL_TIMEOUT_S	0.1	
	HANDSFREE_SESSION_MIN_S, HANDSFREE_SESSION_MAX_S	0.25, 2.5	
	AIM_IDLE_OFF_S	2.0	
	DECODE_LATENCY_S	0.030	estimate
	GOOD_READ_LED_S	0.2	estimate
	MOTION_WAKE_LINEAR_M_S, MOTION_WAKE_ANGULAR_RAD_S	0.02, 0.2	estimate (stand-in for motion sensing)
	PRESENTATION_WAKE_SPEED_M_S, PRESENTATION_ZONE_M	0.005, 0.40	authored stand-in for image-change detection
trigger (estimates; switch bounds from Omron D2F [S12])	TRIGGER_ARM_M	0.025	
	TRIGGER_PRELOAD_N	1.0	
	TRIGGER_RATE_N_PER_M	450.0	
	TRIGGER_CLOSE_M	0.0030	
	TRIGGER_HYSTERESIS_M	0.00019	D2F MD <= 0.12 mm x lever ratio 1.6 (estimate)
	TRIGGER_STOP_M	0.0036	>= 0.25 mm overtravel (D2F)
	TRIGGER_MASS_KG	0.003	
	TRIGGER_DAMPING_RATIO	0.4	estimate: plastic pivot + snap switch (undamped, the lever rang 0.4-3.5 mm)
	SWITCH_DEBOUNCE_S	0.005	
	STOP_SOLIMP	(0.99, 0.999, 0.5, 2.0)	numerical: MuJoCo limit impedance (d0, dmax, mid, power) for the trigger and key stops (MuJoCo's default 0.9-0.95 lets a 35 N squeeze drive the 3 g trigger ~1 mm through its plastic stop; E6)

Group	Parameter	Value	Provenance and reason
keypad (Cherry MX Red class [S13])	KEY_OPERATE_M, KEY_TRAVEL_M	0.0020, 0.0040	
	KEY_FORCE_OPERATE_N, KEY_FORCE_INITIAL_N	0.45, 0.30	
	KEY_RELEASE_M	0.0018	estimate
	KEY_MASS_KG	0.002	estimate: ABS keycap + slider (1u); larger keys scale with area
	KEY_DAMPING_RATIO	0.3	estimate: slider friction + leaf contact
	KEY_PITCH_M	0.01905	
cradle seat sensor (authored stand-in [S2])	SEAT_TOL_M, SEAT_TOL_DEG, SEAT_HOLD_S	0.003, 8.0, 0.2	
hardcover book	BOOK_BOARD_M	0.003	estimate: 2.5-3 mm greyboard + cloth (hardcover)
	BOOK_COVER_MASS_KG	0.08	derived: 0.156 x 0.235 x 0.003 m greyboard at ~700 kg/m3 + cloth (K12)
	BOOK_HINGE_FRICTION_NM	0.02	estimate: an opened cover lies open, a closed one stays closed
	PAPERBACK_COVER_M	0.0003	estimate: 10-12 pt (0.25-0.30 mm) coated cover board
	PAPERBACK_COVER_KG_M2	0.28	estimate: 250-300 g/m2 cover stock
	PAPERBACK_HINGE_FRICTION_NM	0.002	estimate: a creased card flap; its own weight (3 mN m) swings it open past vertical
friction (Coulomb, estimates)	MU	dict(pad_housing=0.8, board_deck=0.40, can_deck=0.30, glass_deck=0.35, film_deck=0.35, item_tote=0.35, board_board=0.45, book_counter=0.30, scanner_cradle=0.30, robot_default=0.8)	One coefficient per SHAPE, named by the material the value was estimated for (the keys read like pairs, but a value is given to each shape of that material). MuJoCo applies the LARGER of the two shapes' coefficients at equal priority (the fingertip pads, priority 1, govern their contacts): the coefficient each contact pair actually gets is listed in section 6.2.
	BOX_AS_HULL_MIN_MASS_KG	5.0	authored: box SKUs at least this heavy (the 12.5 kg water case) collide as 8-vertex convex hulls: MuJoCo's box-box routine threw the case at 31.7 m/s from a bogus 584 mm contact. All free boxes as hulls was measured and rejected (resting creep 24x, held-scanner slip 3.5x; scene.Scene.box); the reset sweep looks for deep box-box contacts (section 11.2)
	TORSIONAL_FRICTION_M	0.001	estimate: spin resistance of a free body's contacts, (2/3) mu a for a contact patch of radius a ~ 4 mm at mu 0.35-0.4 (MuJoCo condim 4; world.py); the fingertip pads too (a ~ 1.9 mm at the pad's mu 0.8)
scene geometry	SHELF_BAY_M	(36*IN, 18*IN, 72*IN)	Uline H-7115 class [S19]
	SHELF_PITCH_M	1.5*IN	[S19]
	DECK_THICK_M	5/8*IN	particle board [S19]
	CART_TOP_M	0.75	authored from reach
	ARM_Y_M	0.25	
	BASE_TO_SHELF_M	0.40	
	COUNTER_TOP_M	0.915	ADA 2010 §904.4 [S18]
	READY_POSE	(0., -1.3, 0., -2.8, 0., 1.6, 0.785, 0.04, 0.04)	robot ready pose (arm joints 1-7, fingers): authored from forward kinematics
physics	RIGID_DT_S	0.001	the Dex3 component probe's default step; the environment's steps are in config.py (1 ms for the FR3/Panda configurations, 0.5 ms for the G1; section 6)
	GRAVITY	9.81	standard gravity
contact detection and the robot's protective stop	REFLEX_RECOVERY_S	1.0	authored: after a collision reflex the arm holds and refuses new targets this long (the real FR3 stays stopped until an error recovery; libfranka does not publish its duration)

Group	Parameter	Value	Provenance and reason
	REFLEX_TORQUE_NM	(10., 20.)	sourced: libfranka examples/examples_common.cpp setDefaultBehavior, upper torque thresholds (nominal, acceleration) on every joint
	CONTACT_GAP_M	0.0	contact detection envelope of boxes and planes (E17: MuJoCo's box-box routine returns bogus deep points for separated boxes inside a positive gap; at 1 kHz a 3 m/s link moves 3 mm per step, less than the thinnest free collider it can meet, the 4 mm basket wall)
	MESH_CONTACT_GAP_M	0.002	convex hulls (GJK/EPA): with no envelope resting hull contacts flicker and a shelf of jars creeps faster and faster (2 mm/s after 10 s, rest check)
numerical-failure guard (world.py)	MAX_OBJECT_SPEED_M_S	20.	authored: 3x the fastest plausible (a 2 m drop reaches 6.3 m/s)
	MAX_OBJECT_SPIN_RAD_S	1000.	authored: 8x the fastest plausible (a 16 mm lip balm rolling at 1 m/s spins at 128 rad/s)
	MAX_FINGER_COUPLING_M	0.010	authored: a Franka Hand's two fingers ride one drive, so their joint positions cannot part; the stiff coupling lets them part <= 2.7 mm under the hardest legal impacts (158 FR3/Panda abuse cases); unconverged solves of a cut-short line search part them over 10 mm in 36 of those cases, up to 232 mm. Robot links are not under the speed bound above (driven at its URDF joint speeds a G1 fingertip can reach 24 m/s)

7.2 Render materials

The appearance constants outside `params.py`: the print and finish constants that set a printed symbol's contrast and glints in the imager's frames, and every base colour. Generated from `render.py` (`COLORS`, `PRINT_WHITE`, `FINISH`, `MATERIAL_NOTES`) and `artwork.INK`. None is identified; the coat roughnesses are not anchored to a dead zone (section 9.4).

Material	Value	Provenance
PRINT_WHITE	0.85	estimate: diffuse reflectance of white label stock / coated board; the substrate of every printed symbol
INK (sRGB)	(63, 63, 63)	estimate (artwork.INK, sRGB 63 = 5 % of the substrate's white): carbon-black thermal-transfer and litho inks reflect 2-5 %
FINISH['matte']	None	estimates: clear-coat roughness per finish; not anchored (E2/E2b: no specular dead zone at any roughness 0.07-0.35, section 9.4)
FINISH['semigloss']	{'coat': 1.0, 'coat_rough': 0.25}	estimates: clear-coat roughness per finish; not anchored (E2/E2b: no specular dead zone at any roughness 0.07-0.35, section 9.4)
FINISH['glossy']	{'coat': 1.0, 'coat_rough': 0.07}	estimates: clear-coat roughness per finish; not anchored (E2/E2b: no specular dead zone at any roughness 0.07-0.35, section 9.4)
FINISH['film']	{'coat': 1.0, 'coat_rough': 0.03}	estimates: clear-coat roughness per finish; not anchored (E2/E2b: no specular dead zone at any roughness 0.07-0.35, section 9.4)
FINISH['metal']	{'coat': 1.0, 'coat_rough': 0.1, 'metal': 0.35}	estimates: clear-coat roughness per finish; not anchored (E2/E2b: no specular dead zone at any roughness 0.07-0.35, section 9.4)
FINISH['glass']	{'coat': 1.0, 'coat_rough': 0.05}	estimates: clear-coat roughness per finish; not anchored (E2/E2b: no specular dead zone at any roughness 0.07-0.35, section 9.4)
COLORS['floor']	(0.3, 0.31, 0.3), rough 0.75, metal 0.0	authored appearance estimate (scene dressing and fixtures; does not enter a decoded symbol's contrast)
COLORS['robot']	(0.86, 0.87, 0.86), rough 0.3, metal 0.0	authored appearance estimate (scene dressing and fixtures; does not enter a decoded symbol's contrast)
COLORS['robot_dark']	(0.07, 0.075, 0.08), rough 0.4, metal 0.0	authored appearance estimate (scene dressing and fixtures; does not enter a decoded symbol's contrast)
COLORS['camera_dark']	(0.03, 0.03, 0.035), rough 0.45, metal 0.0	authored appearance estimate (scene dressing and fixtures; does not enter a decoded symbol's contrast)
COLORS['aluminum']	(0.62, 0.64, 0.66), rough 0.25, metal 0.9	authored appearance estimate (scene dressing and fixtures; does not enter a decoded symbol's contrast)
COLORS['cart_steel']	(0.3, 0.31, 0.33), rough 0.45, metal 0.5	authored appearance estimate (scene dressing and fixtures; does not enter a decoded symbol's contrast)

Material	Value	Provenance
COLORS['cart_frame']	(0.16, 0.17, 0.18), rough 0.5, metal 0.3	authored appearance estimate (scene dressing and fixtures; does not enter a decoded symbol's contrast)
COLORS['tablet_black']	(0.02, 0.02, 0.025), rough 0.3, metal 0.0	authored appearance estimate (scene dressing and fixtures; does not enter a decoded symbol's contrast)
COLORS['shelf_steel']	(0.42, 0.44, 0.46), rough 0.5, metal 0.4	authored appearance estimate (scene dressing and fixtures; does not enter a decoded symbol's contrast)
COLORS['scanner_body']	(0.055, 0.055, 0.06), rough 0.38, metal 0.0	authored appearance (dark grey moulded plastic, product photographs)
COLORS['scanner_grip']	(0.03, 0.03, 0.032), rough 0.75, metal 0.0	authored appearance estimate (scene dressing and fixtures; does not enter a decoded symbol's contrast)
COLORS['trigger_black']	(0.04, 0.04, 0.045), rough 0.35, metal 0.0	authored appearance estimate (scene dressing and fixtures; does not enter a decoded symbol's contrast)
COLORS['cradle_black']	(0.05, 0.05, 0.055), rough 0.45, metal 0.0	authored appearance estimate (scene dressing and fixtures; does not enter a decoded symbol's contrast)
COLORS['keypad_black']	(0.06, 0.06, 0.065), rough 0.5, metal 0.0	authored appearance estimate (scene dressing and fixtures; does not enter a decoded symbol's contrast)
COLORS['keycap']	(0.8, 0.8, 0.79), rough 0.55, metal 0.0	authored appearance estimate (scene dressing and fixtures; does not enter a decoded symbol's contrast)
COLORS['carton']	(0.62, 0.45, 0.28), rough 0.85, metal 0.0	authored appearance estimate (scene dressing and fixtures; does not enter a decoded symbol's contrast)
COLORS['basket_blue']	(0.1, 0.32, 0.72), rough 0.45, metal 0.0	authored appearance estimate (scene dressing and fixtures; does not enter a decoded symbol's contrast)
COLORS['tote_grey']	(0.36, 0.38, 0.4), rough 0.6, metal 0.0	authored appearance estimate (scene dressing and fixtures; does not enter a decoded symbol's contrast)
COLORS['laminate']	(0.78, 0.76, 0.72), rough 0.35, metal 0.0	authored appearance estimate (scene dressing and fixtures; does not enter a decoded symbol's contrast)
COLORS['wood_counter']	(0.55, 0.38, 0.22), rough 0.45, metal 0.0	authored appearance estimate (scene dressing and fixtures; does not enter a decoded symbol's contrast)
COLORS['counter_panel']	(0.3, 0.3, 0.32), rough 0.55, metal 0.0	authored appearance estimate (scene dressing and fixtures; does not enter a decoded symbol's contrast)
COLORS['bezel']	(0.02, 0.02, 0.022), rough 0.4, metal 0.0	authored appearance estimate (scene dressing and fixtures; does not enter a decoded symbol's contrast)
COLORS['led_off']	(0.018, 0.026, 0.02), rough 0.35, metal 0.0	authored appearance estimate (scene dressing and fixtures; does not enter a decoded symbol's contrast)
COLORS['lid_metal']	(0.72, 0.72, 0.7), rough 0.25, metal 0.9	authored appearance estimate (scene dressing and fixtures; does not enter a decoded symbol's contrast)
COLORS['cap_blue']	(0.1, 0.35, 0.75), rough 0.35, metal 0.0	authored appearance estimate (scene dressing and fixtures; does not enter a decoded symbol's contrast)
COLORS['jar_glass']	(0.35, 0.05, 0.05), rough 0.1, metal 0.0	authored: the sauce seen through the jar's glass (opaque, glass coat); the headspace and shoulder are OmniGlass n 1.52 (render.jar_glass)
COLORS['pet_water']	(0.8, 0.88, 0.94), rough 0.06, metal 0.0	authored: the PET bottle's label-free body is OmniGlass n 1.33 (render.water_pet); this colour is the fallback
COLORS['bag_white']	(0.9, 0.9, 0.88), rough 0.55, metal 0.0	authored appearance estimate (scene dressing and fixtures; does not enter a decoded symbol's contrast)
COLORS['caster_rubber']	(0.05, 0.05, 0.055), rough 0.8, metal 0.0	authored appearance estimate (scene dressing and fixtures; does not enter a decoded symbol's contrast)
COLORS['lid_gold']	(0.72, 0.58, 0.3), rough 0.3, metal 0.9	authored appearance estimate (scene dressing and fixtures; does not enter a decoded symbol's contrast)

8. Tasks, host application and evaluation

Preset	Task	Instruction (abridged)	Judged
stockroom	`scan-location`	take the scanner, scan slot {slot}'s location label, return the scanner	location recorded, no read of another slot's location label, scanner seated, no key pressed, nothing on the floor
stockroom	`count-by-scanning`	cycle count {slot} (a slot runs from its location label to the next label on the right): location label, then every unit once; leave stock faced; return the scanner	the counts the host booked **under the target location** equal truth per GTIN; each physical unit's own symbol booked exactly once and nothing else booked (no inner-pack reads, no unit of the same SKU from a neighbouring slot; a read the host refused books nothing); units in slot (±20 mm), resting on the deck (not held above it, not stacked), upright (≤15°), faced (±30°, cans and bottles on their label front), each facing lane fronted by its own product (≤ 40 mm behind its found front); the rest of the shelf as found, its facing lanes fronted by the same rule; no misread the record holds data from; scanner seated and not held. Score: section 8.2
stockroom	`count-key-quantity`	cycle count {slot} (where the slot ends, as above): location label, one unit per item, key the quantity, ENTER; leave stock faced as found; return the scanner	quantities keyed and entered under the target location, each after a read of that item, equal truth; nothing keyed under any other location; no key held at the end; stock as `count-by-scanning` (section 8.3); tool as above
stockroom	`hold-scanner`	take the scanner, hold it steady 5 s, put it back	out of the cradle and held (finger contact > 0.2 N) and steady (< 20 mm/s, < 0.25 rad/s) for 5 s without a break, from the environment's own per-tick history; seated and released at the end; no reads; nothing on the floor
counter	`checkout-present`	present each item to the cradled scanner once, bag it	sale equals the basket; each item read once; each item's volume inside the tote, the tote standing on the counter; nothing on the floor; scanner seated
counter	`checkout-bulky`	as above, plus scan the 12.5 kg water case with the handheld scanner without lifting it	as above; the case's own label read (not an inner bottle's); the case's lowest point never rose more than 10 mm during the episode
counter	`checkout-counter`	ring up the customer's items set out on the counter: present each to the cradled scanner once, bag it	as `checkout-present` (sale equals the items, each read once, each inside the standing tote, nothing on the floor, scanner seated)
library	`library-checkin`	scan each book's library label (not the ISBN; back cover or inside the front cover), put the book in the returns bin	checked-in list equals the books; no ISBN reads; each book's volume inside the bin, the bin standing on the desk; scanner seated

Seeds vary the target slot, the stock mix and poses, pack contents, and which library labels are inside the cover. `checkout-counter` is the group's own checkout row 087, "merchandise resting on counter unscanned": in `checkout-present` and `checkout-bulky` every item lies in a 200 mm-deep hand basket, where a Franka Hand (208 mm across its fingers) has no collision-free grasp (section 6.3), so a phantom customer does every extraction. On the counter the items stand as they stand on a shelf, at least 55 mm apart, and the hand grasps every item narrower than its 80 mm stroke itself (all but the 85 mm sauce jar; section 10.5). The basket tasks are the harder variants. The catalogue holds 8 000 frozen variations (the split is a hash of task and seed). Design choice in `library-checkin`: any read of a book's ISBN fails the task (`no_isbn_reads`). At a real circulation desk an ISBN scanned into the check-in screen usually gives "item not found" and the operator rescans; the task treats it as the workmanship error it is (scanning the wrong symbol), rather than modelling the library system's lookup. Every criterion reads physical state or the host app's record (`task.py`), never commanded targets. Thresholds are authored with physical meaning (a slot is ±20 mm around its facing position; a unit tilted more than 15° is not upright; a faced unit's front is within 30° of the aisle). A ghost read or a non-finite state is a backend fault that stops the episode; a misread (a printed label decoded as another number) is a reported read that fails the `no_misreads` check (in `count-by-scanning` only a misread the record holds data from, section 8.2); a wrong outcome is a valid result. In `hold-scanner` a tick counts as held only while a finger or hand body presses the scanner or its trigger with more than 0.2 N (`world.pair_contact_forces`), so a scanner lying still on the cart top earns nothing.

Every check (all tasks): no free body, the scanner included, has its lowest point below 15 mm above the floor; no read that the environment cannot attribute to a printed label (a ghost read is a backend fault, and a checkpoint restore does not launder one); no misread (`count-by-scanning`: no misread the record holds data from, section 8.2); no key held down at the end.

8.1 Adversarial evaluator tests

An adversarial test suite (79 tests that run without physics, 25 more that need it) attacks each task's evaluator with the cheapest way to fake success, built as a real physical state wherever the state is physical (a unit really lying on the floor, a scanner really held by a phantom hold above the cradle, a tote really set on the basket), plus a correct end state that must pass. Each row is an attack the evaluator refuses:

Attack	Defence
hold-scanner credited without any hold (seated scanner, no reads)	per-tick history in the environment: seconds out of the cradle, held and steady; checkpointed with the state
a unit standing upright on the floor, or a non-SKU body (the scanner, the case) on the floor, not seen as "on the floor"	lowest point of every free body's box/hull corners, not the centre height of SKU units
counts credited whatever location they were recorded under	the host keeps counts per location; the evaluator reads the target location's
a same-SKU unit taken from the neighbouring slot counted	each read is attributed to its physical unit; foreign units are a failed check
tote/bin checks by horizontal footprint only (an item held above the tote)	item volume inside the container's inner box, the container itself on the counter
a tote set on top of the basket counted as bagging	the container must stand on the counter/desk
the water case lifted and put back	the case's maximum rise over the episode, from the per-tick history
a scanner still gripped in its cradle counted as returned	seated **and** no finger contact on the scanner
a ghost read laundered by a checkpoint restore	the fault latch and read history are part of the checkpoint
a refused keyboard chord (or an invalid joint target) raising a plain error that the HTTP service latches as a runtime fault, locking the policy out	every command is planned and validated without side effects and refused with `ActionRejectedError` (the session stays usable)
a scanner set down on the cart top and left still for 5 s, earning the hold	a tick counts only while a finger or hand presses the scanner or its trigger with more than 0.2 N; tests: a robot hold through the skills (credited), the scanner lying still (not credited), a phantom mount holding it (not credited)
a misread of a printed label ending the episode as a backend fault, so that a legal aim could end it unscored	reported and failing `no_misreads`; a decode with no label on its ray still faults; tests for each, and for a misread that spells the target's own number

The same suite checks the keyboard contract (unknown or conflicting keys refused without motion, the 10 s cap, the commanded Cartesian speed, rotation about world axes, grip levels, finger speed, an IK refusal leaving the arm unmoved) and the policy-access contract (an allowlist; diagnostic cameras and privileged routes denied to the policy; policy steps return no reward or evaluation).

8.2 count-by-scanning: description, verifier and score

The instruction says where the counted slot ends, "(a slot runs from its location label to the next label on the right)": each label sits at the left end of its slot, and at seed 0 the counted soup cans stand 9 mm from the next slot's label. The verifier's stock and read rules:

Rule	What it judges
facing	a can's or bottle's label front is judged like a box's front panel (±30°)
resting	a counted unit's lowest point within 5 mm of the deck top (`REST_TOL_M`), so a unit held above its place does not pass
lanes fronted	in each facing lane a unit of the lane's own product has its front face ≤ 40 mm behind the lane's found front (identical units and rows may be swapped)
`neighbouring_stock_undisturbed`	every other shelf unit and shipping case in its own slot (±20 mm), its lowest point within 5 mm of where it was found, upright (≤ 15°), turned ≤ 30° from its found heading **or** ≤ 30° from the aisle; every other slot's facing lanes fronted by the same rule, an unfronted lane charged once in `disturbed`
location	`location_recorded` is not applied: `record_matches` judges the count booked under the target, so a read of another location's label after a correct count changes nothing
booked reads	only reads the host booked count towards "exactly once" (`HostApp.on_scan` sets `rep["booked_location"]`); a read the host refused ("scan location first") books nothing
misreads	only a misread the record holds data from fails: a booked misread, or a misread location label that a later booking was made under (`HostApp.on_scan` sets `rep["sets_location"]`). A misread location label is in the record only if a booking was made under its code before any correct read of that code in the same stretch (consecutive location reads setting one code); such an unbacked stretch costs one error only for the counted slot's own code, because bookings under another code are errors already

Score = 1.0 on success, else 0.9 × progress; progress = max(0, hits − errors − ½ × disturbed) / N, × 0.9 while the scanner is not back (definitions in `task.py`'s docstring). The constants are authored: half a counted unit per disturbed unit; 0.9 for the scanner's return; and a 0.9 presentation factor that puts every failure below every success. The location read, aiming, lifting or turning a unit earn nothing by themselves, and no unit order, hand, trigger pull or hands-free presentation is preferred. E22's count-by-scanning run (seed 5, 10/10, section 10.6) was scored with an earlier version of this verifier; it saved no unit poses, so it cannot be re-judged.

The correctly read location label earns no credit. Every valid completion contains that read (the host books nothing before a location), so crediting it would not disadvantage any route, and aiming and pulling for this read is the default robot's tightest step. Against it: the read alone puts nothing correct in the count, and partial reward does not pay for anticipated intermediate milestones. A faulted state embeds score 0.0 in `get_state()`

and the faulting step's reward; that path is shared by all eight tasks, and `/evaluate` refuses faulted states.

Reading the score along an episode: each unit off its place costs 0.5/N of progress (× 0.9 × 0.9 in the score with the scanner out) until it is back, so a correct trace dips at every pick-up and can sit at 0 with units already booked; `metrics.hits`, `errors` and `disturbed` give the parts. Mid-episode a handheld partial count scores 10 % below the same count presented to the docked scanner (the scanner factor); final scores agree. With the ½ constant, "all units counted, none put back" ties "half counted, shelf intact", and a unit on the floor costs the same as one left in another slot. Along the recorded seed-0 assisted run, the resting test charges a unit while the phantom stocker holds it just above its place (8 mm lifts, releases from 3 mm), in 243 of its 2451 samples; the end state scores 1.0.

8.3 The eight-task set: one score contract for every task

All eight tasks share `count-by-scanning`'s score contract. **Score = 1 if every check passes, else 0.9 × progress**; progress lies in [0, 1], is 0 at an untouched start, and depends only on the end state, the host record, what was read and the episode history, never on the order of the work, the hand, a trigger pull or a hands-free presentation. Every evaluation reports its `contract` (`task.CONTRACTS`). The evaluator scores quoted in sections 10.1–10.7 were computed with an earlier rule, the share of checks passed (under which an untouched `hold-scanner` scores 6/7, 0.86, and an untouched `scan-location` 5/7, 0.71); they carry no `contract` field and stand as recorded.

Checks common to all eight: `nothing_on_floor` (no free body, whether stock, a container, the case or the scanner, with its lowest point less than 15 mm above the floor), `scanner_seated` (the cradle's seat sensor true and no finger or hand body pressing the scanner or its trigger with more than 0.2 N), `no_key_pressed`, `no_unexplained_reads` (a ghost read is a backend fault that removes the evaluation), `no_misreads` (no read whose source label's payload differs from the decode; `count-by-scanning`: only a misread the record holds data from). **Progress factors shared with** `count-by-scanning` (`scoring.py`): × 0.9 unless the scanner is back (seated, released, no key down); each free body on the floor, the scanner included, costs half an item (0.5 / N).

#	Task	Difficulty	Success: the checks above and	Progress (clipped to [0, 1])
1	`hold-scanner`	entry	`held_steady_5s`: the history's longest unbroken run out of the cradle with a finger pressing the scanner > 0.2 N at < 20 mm/s and < 0.25 rad/s is ≥ 5 s; `no_reads`	min(1, hold / 5 s) × (0.7 + 0.3 × scanner back) × (0.5 if anything was read) − 0.5 × floor bodies
2	`scan-location`	entry	`location_recorded` (the host's location is the slot's code); `no_wrong_location_read` (no read of another slot's location label, none that made another slot's code the location)	hit × (0.7 + 0.3 × scanner back) − 0.25 × wrong-location reads − 0.5 × floor bodies; hit = the slot's own label read correctly and its code still the recorded location
3	`checkout-counter`	medium	`sale_matches` (the sale equals the items' GTINs); `each_item_once` (each item's own label read exactly once, no inner unit read); `items_bagged` (each item inside the tote: within its walls, lowest point above the floor's underside, centre ≤ 50 mm above the rim; the tote tilted ≤ 20° and standing within 20 mm of the counter top)	(Σ items rung × (0.6 + 0.4 × bagged) − 1 per other read (a repeat, an inner unit, a misread, a label that is no item's) − 0.5 per item bagged without being rung − 0.5 × floor bodies) / N × 0.9 unless the scanner is back
4	`library-checkin`	medium	`checkin_matches` (the checked-in list equals the books' library item numbers); `no_isbn_reads` (no EAN-13 or UPC-A read); `books_in_bin` (as `items_bagged`, the returns bin on the desk)	(Σ books checked in × (0.6 + 0.4 × in the bin) − 1 per other read (an ISBN, a misread, a repeat check-in, a label that is no book's library label) − 0.5 × floor bodies) / N × 0.9 unless the scanner is back
5	`checkout-present`	medium (harder variant of 3)	as 3, the items starting in the 200 mm-deep basket	as 3
6	`count-key-quantity`	hard	`record_matches` (the quantities under the slot's location equal the truth per GTIN); `no_counts_under_other_locations`; `stock_faced_in_slot`, `no_foreign_units_in_slot`, `neighbouring_stock_undisturbed` exactly as `count-by-scanning` (section 8.2: slot ±20 mm, resting within 5 mm of the deck, tilt ≤ 15°, front or label front within 30° of the aisle, lanes fronted within 40 mm; the rest of the shelf as found)	(Σ products 0.3 × a unit of it read correctly under the slot + 0.7 × max(0, 1 − abs(keyed − truth) / truth) − 1 per saved quantity under another location, for a product not in the slot or keyed for a misread − 0.5 × disturbed units, cases, lanes and floor bodies, as 8.2) / products × 0.9 unless the scanner is back. ENTER overwrites, so a corrected quantity earns full credit
7	`checkout-bulky`	hard	as 3 for the basket (the case is not bagged); the case's own label read once (in `sale_matches`, `each_item_once`); `case_not_lifted`: the case's highest rise over the episode < 10 mm	as 3 over the basket items plus the case, whose credit is read × (1 up to a 10 mm rise, falling linearly to 0 at 50 mm), divided by N + 1
8	`count-by-scanning`	hardest	as section 8.2	max(0, hits − errors − 0.5 × disturbed) / N × 0.9 unless the scanner is back

`count-key-quantity`'s instruction says where the slot ends and that the stock is left "as you found it", as `count-by-scanning`'s does: "Cycle count slot {slot} (a slot runs from its location label to the next label on the right): scan the slot's location label, then for each different item scan one unit and key the number of units of that item in the slot on the keypad, then ENTER. Leave the stock faced on the shelf as you found it and return the scanner to its cradle."

Two checks differ from a plain reading of the instructions. `scan-location` checks `no_wrong_location_read`, not exactly one read: a second read of the right label changes nothing in the record. `count-key-quantity` uses `count-by-scanning`'s stock rules, which judge the facing of cylinders, a unit held above its place and the stock outside the slot, and it has no `location_recorded` check: a stray read of another label after a correct count changes no quantity, while quantities saved under another location are wrong data that `no_counts_under_other_locations` catches.

Solvability. Three tasks have a hard step; none is impossible as built:

- `checkout-present`, `checkout-bulky` (basket): the Franka Hand has no collision-free grasp of any item inside the basket at the start (section 6.3; of 10 physical attempts 9 stopped on the basket, 1 on the cradled scanner) and every extraction on record is the phantom customer's (10.5, success 8/8 and 9/9). But the basket is a free 0.6 kg body whose 4 mm wall the hand reaches, nothing in the checks forbids tilting it to bring items to the rim or tipping them onto the counter, and from the counter the robot took every item itself (10.5). Hard and unshown, not physically blocked.
- `checkout-counter`'s sauce jar (K04, 85 mm body over the 80 mm stroke): its 66 mm lug cap is graspable and the planner found collision-free grasps of every counter item, the jar included (10.5); the seed-0 run failed on the swinging jar's protective stops, not on a missing grasp.
- `library-checkin` (a book lying flat has no side under the stroke): sliding a book over the desk edge or off the stack and pinching it across its 25-30 mm thickness is untried; the phantom patron stands in, as the phantom stocker does in `count-by-scanning`.

The Franka Hand has not pulled the trigger of a robot-held scanner: the other hand's approach has no IK solution from the turn poses tried and a search over turn and approach together is not built (9.1b), and a squeeze around the handle does not move the trigger (10.2); the G1's Dex3 index finger pulls it (6.5). `scan-location`, `count-key-quantity` and `checkout-bulky` share this step with `count-by-scanning`, where it is shown as assisted evidence.

Open risks. The eight-task score contract has not been run in simulation: the scoring arithmetic and the count-key-quantity replay are unit tested without physics (the replay against `device.HostApp` on random key and scan sequences), and the evaluator's minimal pairs for it have not been run. `count-key-quantity`'s real scenes (slots three or more deep) have not been judged by the adopted stock rules; its cans spawn within about 26° (3σ) of the aisle against the 30° facing limit, so a rare seed could start with a can the rules call unfaced (the same holds for `count-by-scanning`). Its disturbed cost is per product (0.5 / S), so a lifted unit dips the trace more than in `count-by-scanning` (0.5 / N). `hold-scanner` charges the scanner on the floor like any free body, as `count-by-scanning` does. A disagreement between the count-key-quantity replay and the host's record raises an error instead of scoring.

9. Component validation

Scope: each mechanism and each stage of the imaging chain tested apart, against a prediction written before the run. Assisted means phantom forces or a phantom mount; every such run says so in its output.

9.1 Trigger and keys (E6, E12)

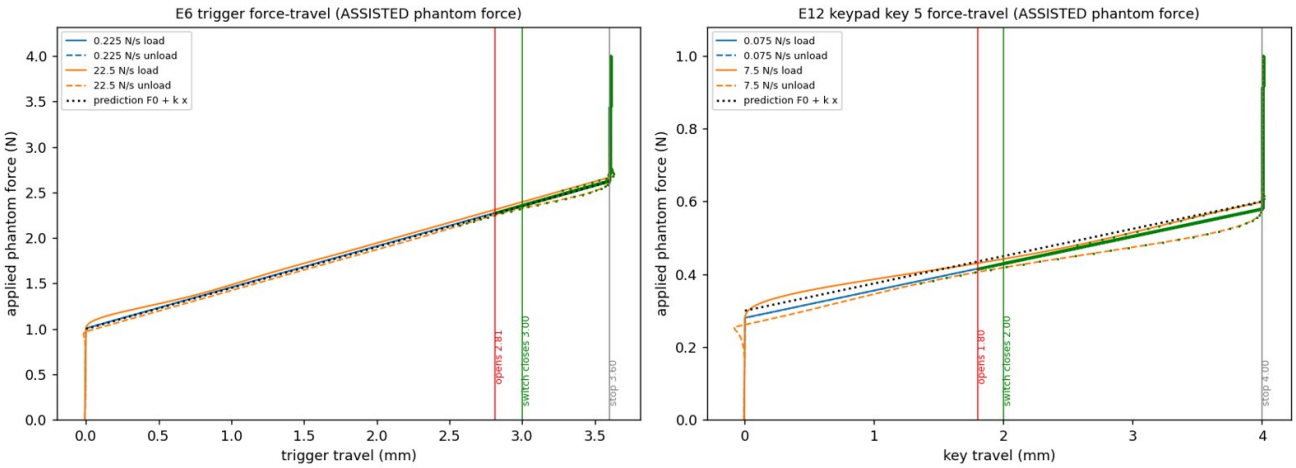


Figure 4. E6: trigger and key force against travel under prescribed phantom fingertip forces.

ASSISTED: the scanner on a phantom six-axis mount, prescribed phantom fingertip forces on the trigger blade and on the "5" key (a force gauge driven by displacement is itself unstable).

Mechanism	Prediction	Measured, slow ramp	Fast ramp	Stop overtravel
trigger	F = 1.0 N + 0.45 N/mm; switch 3.0 / 2.81 mm; stop 3.6 mm	F = 1.006 N + 0.451 N/mm (rms 0.009 N); closes at 3.003 mm (2.36 N), opens at 2.807 mm; max 3.609 mm; no chatter	F0 1.042 N (lever inertia and damping at 22.5 N/s); switch reported at 3.28 / 2.53 mm (5 ms debounce)	0.009 / 0.017 / 0.026 / 0.054 mm at 4 / 10 / 35 / 70 N
key	F = 0.30 N + 0.075 N/mm; switch 2.0 / 1.8 mm; stop 4.0 mm	F = 0.281 N + 0.075 N/mm; closes at 2.006 mm (0.43 N), opens at 1.795 mm; max 4.007 mm	F0 0.316 N; switch at 2.57 / 1.26 mm (debounce)	0.007 / 0.019 / 0.039 mm at 1 / 5 / 20 N

Figure 4 shows the force-travel loops against the prediction.

9.1b A robot fingertip on the trigger of a robot-held scanner (E6r)

ASSISTED only in the choice of motion (joint targets through IK, Franka grasp; the script reads the switch to know when to stop, as a policy would read the firmware's beep or the screen); no phantom force. The idle left arm parks; the right hand takes the scanner by its head, lifts it and turns it so the trigger faces the left hand, in open space in front of the cart; the left hand, fingers closed, brings its fingertips 30 mm in front of the blade and advances in 0.5 mm steps along the blade's travel.

The attempt fails before the pull, in both runs: the right hand's turn ends 5.6 mm from its target (one collision reflex, its wrist-camera bracket against its own link 6 at 10.1 N m), and the left hand's approach to the turned scanner's blade has no IK solution within tolerance (1.3 mm and 3.0° short); the script stops with nothing executed for the left hand, and the trigger never moves. The blade of the head on the raked handle sits where the left hand cannot square up to it from this turn pose. Five other turn positions 4–5 cm around the first, and extra turns of −45° to +45° about the vertical, leave the left hand's approach 3–37 mm or 1–34° out of reach.

So no robot hand has closed the Franka-held scanner's trigger; the G1's Dex3 index finger does (6.5.5c). A turn chosen for this geometry (or a two-arm search over turn and approach together) is not built.

9.2 Grasp, carry, hold and reseat of the scanner by the robot (E7)

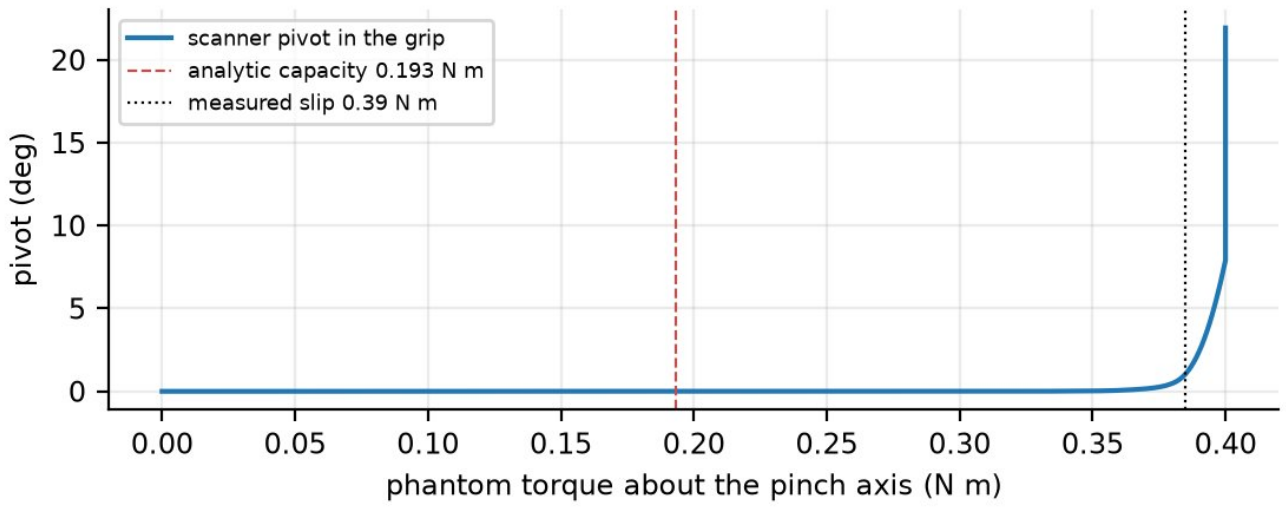


Figure 7. E7b: a phantom torque ramp about the pinch axis of the held scanner; the scanner's pivot in the grip.

ASSISTED only in the choice of motion: the scripted operator sends joint targets (IK waypoints) and Franka Hand grasp commands, exactly what a policy can send; no phantom force touches the scanner. The right hand takes the scanner by its head (17.5 N per finger), lifts it 70 mm along the socket's axis, carries it 0.17 m back and up and returns, holds 10 s, and puts it back (upright into the socket, released 10 mm above its floor; section 3.4).

Quantity	Result
finger opening on the head	65.7 mm (the head is 66 mm)
seat sensor after the lift / at the end	false / **true**
scanner in the hand, whole run	slip ≤ 0.94 mm, pivot ≤ 0.71°, peak scanner speed 0.26 m/s
10 s still hold	slip 0.56 → 0.51 mm, pivot 0.24 → 0.08° (no creep: −0.0002 mm/s), pad force 35.0 N
120 s still hold	slip 0.56 → 0.50 mm, pivot 0.24 → 0.07°, no creep (−0.00005 mm/s), pad force 35.0 N; reseated at the end, no reflex
collision or velocity reflexes	none

With the unconverged no-slip pass (10 iterations at MuJoCo's tolerance) the same 10 s hold creeps 0.107 mm/s (0.59 → 1.66 mm, 0.26 → 1.15°), a 30 s hold reaches 3.9 mm and 3.2° (0.117 mm/s at `impratio` 1, 0.096 at 5, 0.048 at 10 with a 0.5 ms step), and a 120 s hold turns the scanner 7° and slips it 8 mm in the first minute before it stops.

Pinch torsion (E7b). ASSISTED: a phantom torque ramped about the pinch axis on the held scanner. The pinch breaks away at 0.385 N m (21.9° of pivot at 0.40 N m, with the pads' 1 mm torsional term; 0.38–0.40 N m without it). Two flat 17.5 N pads with μ = 0.8 hold 2μF r_eff: 0.19 N m for uniform pressure (mean lever 6.9 mm), 0.36 N m for pressure at the patch corners (12.7 mm). The simulated pinch holds to the corner bound and breaks away there: MuJoCo's box-box contact points lie at the patch corners; the pads' torsional term (condim 4, 1 mm) adds about 2 x 17.5 N x 1 mm = 0.035 N m at most, within the breakaway's spread. With the unconverged no-slip pass it creeps from well below that (0.62° at 0.20 N m, 1° at 0.26 N m). A condim-4 variant with a torsional coefficient of 5.5 mm (which alone would give the uniform-pressure 0.19 N m: 0.0055 m × 35 N) holds the whole 0.40 N m ramp with 0.02° of pivot (0.78° with the unconverged pass): the torsional term adds to the corner points' sliding friction instead of replacing it, so it would overstate the capacity further, and it is not used. A trigger pull of 3 N at ~0.1 m below a head grasp is 0.3 N m: close to the pinch's capacity, which is why a two-hand trigger pull needs care (section 10.2).

9.3 Decode range (E1)

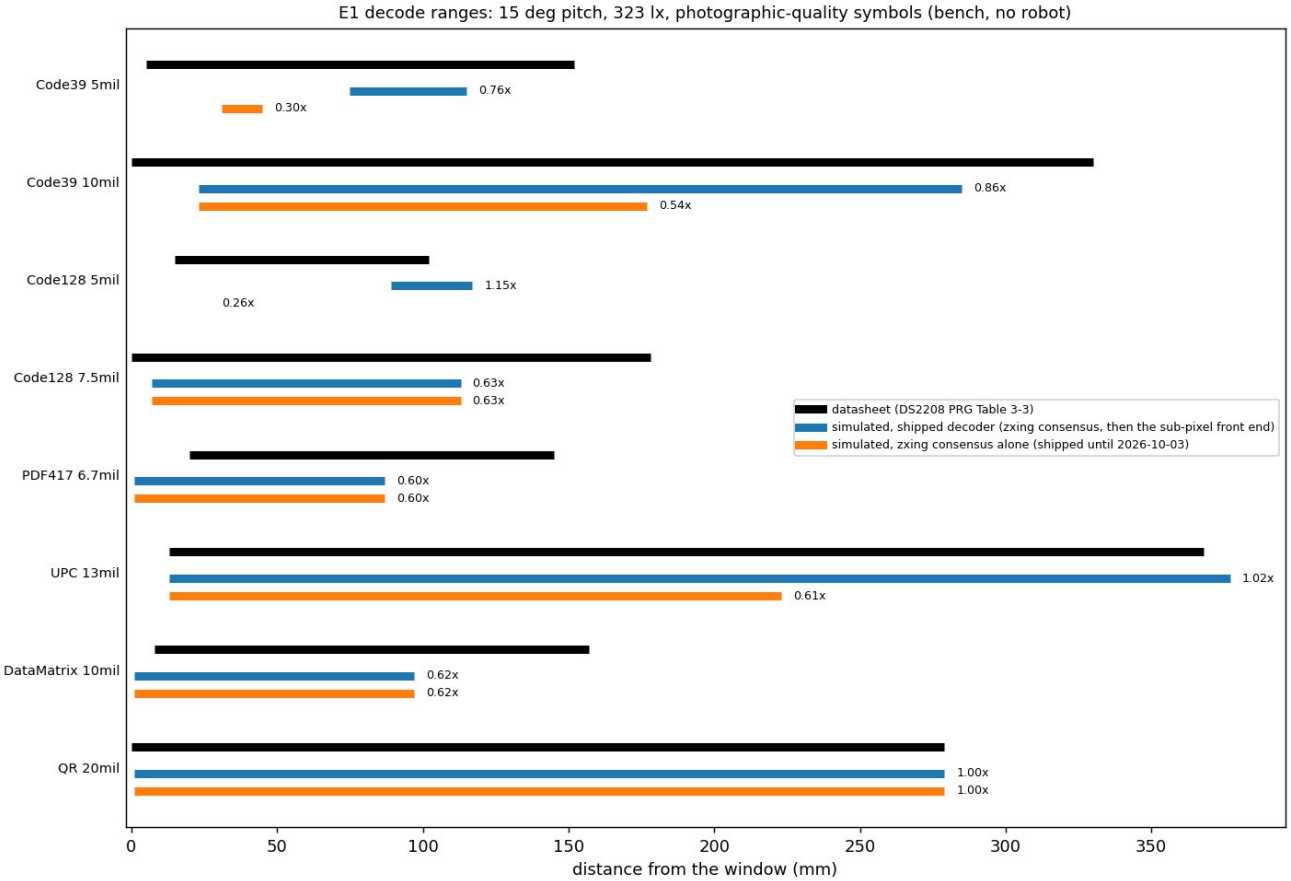


Figure 5. E1: decode range per symbology against the DS2208 PRG table.

No robot, no physics. The environment's own imager chain (thin lens, LEDs, aim off, sensor, decoder) on photographic-quality test cards pitched 15° under 323 lx, moved along the optical axis on a 2 mm grid; at each distance five independent trigger sessions (sensor noise and PRNU) of up to six frames; a distance reads when the majority of sessions read. The manufacturer's test conditions are those of the PRG Table 3-3 [S2].

Symbol	Datasheet near–far (mm)	Shipped decoder: working range (mm)	Far ratio	Other reading bands (mm)	Misreads	zxing consensus alone
Code 39 5 mil	5–152	75–115	0.76	31–39, 43–45	0	31–45 (0.30), 1 misread
Code 39 10 mil	0–330	23–285	0.86		0	23–177 (0.54)
Code 128 5 mil	15–102	89–117	1.15	27	0	27 (0.26)
Code 128 7.5 mil	0–178	7–113	0.63	155–205	0	7–113 (0.63)
PDF417 6.7 mil	20–145	1–87	0.60		0	1–87
UPC-A 13 mil	13–368	13–377	1.02		0	13–223 (0.61)
Data Matrix 10 mil	8–157	1–97	0.62		0	1–97
QR 20 mil	0–279	1–279	1.00		0	1–279

The working range is the longest run of reading distances (single 2 mm gaps allowed); a row's other bands are shorter runs. The bench radiance uses the decoder-independent lens of section 4.1 (A = 0.41 mm, focused 182 mm before the window), the LED pattern scaled to the field (4.2) and 256 samples per pixel at path depth 3. It is decoded by the shipped decoder of section 5.1 (the scan-line separation rule and the front end as the last 1-D stage), five sessions per distance, and by the zxing consensus alone for comparison.

With the front end, UPC-A reads to the datasheet's far limit (377 of 368 mm) and the 1-D far limits are 0.63–1.15 of the DS2208's. The lens is the thin lens whose worst blur over all of PRG Table 3-3's published near and far limits is least, 2.55 module widths at 0.68–0.91 pixels per module at the 1-D far limits; zxing reads at about one module of blur and 1.35 pixels per module, the front end at about 0.7. Where zxing reads, the shipped decoder reads the same (the front end only runs on frames where the consensus reported no 1-D symbol), so the near limits are zxing's. The 5 mil codes read in two bands with a gap: zxing near the window (Code 39 31–45 mm), then nothing while the symbol is too blurred for zxing and too coarse for the front end's hypotheses (it refuses blur of a module or more), then the front end farther out (75–115 mm). Code 128 7.5 mil reads to 113 mm and again at 155–205 mm, past the datasheet's 178 mm; Code 128 5 mil to 117

mm against 102. A real imager reads without such gaps; the simulated one can read a little farther than the device for these two Code 128 rows (lenient). **No misread** by the shipped decoder in the five-session sweep (the zxing consensus alone misread Code 39 5 mil once; decoded without the rules of section 5.1, the same radiance gives a Code 128 7.5 mil misread at 97 mm, two passes on one scan line).

The task symbols read at the task distances (UPC/EAN 13 mil at the counter's 120–180 mm, the location and library labels at 150–200 mm; sections 10.1–10.7). **Misreads over the task frames**: every imager frame the task runs saved, decoded again and each read classed against the symbols printed in its scene: 2 497 frames of the task runs (E9, E11, E14, E21, E22, E23, E24 passes, the G1), 618 reads by the firmware at record time and 617 on re-decoding (one frame decodes differently between runs of zxing), **one wrong read**: the UPC-E of the second counter checkout (10.5). None of the task reads came from the front end (the task distances are inside zxing's range). One wrong read in 618 is 0.16 % per read, with a 95 % upper bound of 0.8 % (Poisson); the AIM/Ohio University worst case for UPC-A (5.1) is about 0.003 % per symbol. The simulated scanner misreads far more often than the real one is measured to; the benches add none (E1, the optics bench, the synthetic bench) except the rolled UPC-A of E1c (9.3c).

9.3b Auto-exposure rule (E24)

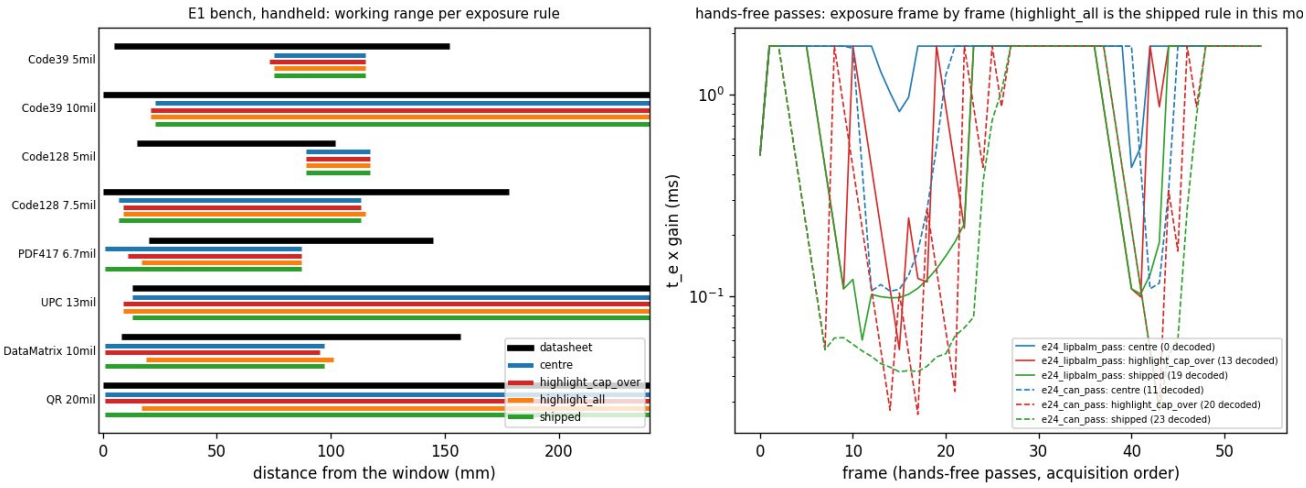


Figure 5b. E24, left: E1 working range per exposure rule (handheld). Right: t_e × gain frame by frame over the

recorded hands-free passes of a lip balm (solid) and a can (dashed).*

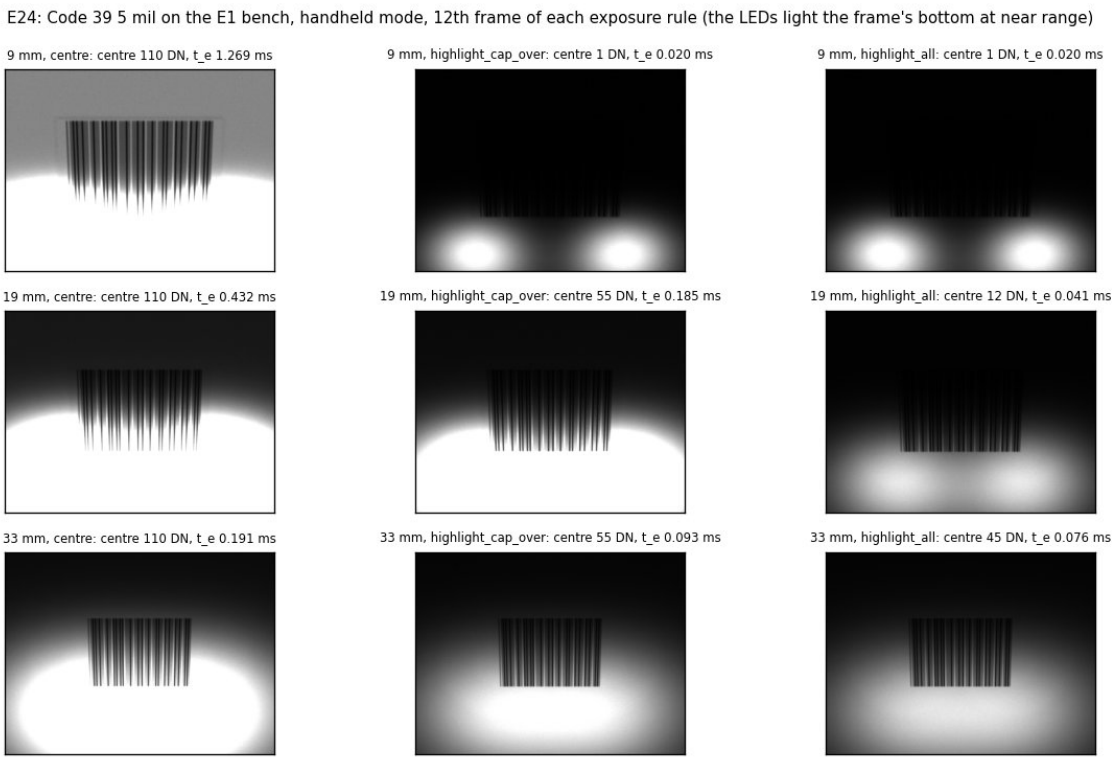


Figure 5c. What the imager sees at 9, 19 and 33 mm (Code 39 5 mil, handheld, 12th frame) under centre metering

and the two highlight caps. At near range the LEDs light the bottom of the frame; a highlight cap exposes for that band and leaves the code dark.*

No robot, no rendering in the comparison: recorded radiance is re-exposed under four rules: `centre` (the central third's mean held at 110 counts), `highlight_cap_over` (plus the frame's per-pixel 99.5th percentile capped at 230 counts once over it, halved if clipped, in both modes), `highlight_all` (the shipped hands-free cap, predictive on 8 × 8 block means, applied in both modes) and `shipped` (handheld = `centre`, hands-free = `highlight_all`). The radiance is identical across rules, so every difference is the rule's. Three cases:

1. the E1 bench (handheld): the same saved radiance, seeds and majority rule as E1, consensus decoder, three sessions per distance (E1 uses five);
2. two recorded hands-free passes: the lip balm (the E21 failure) and a can, carried past the docked scanner at 0.2 and 0.5 m/s on the phantom mount, re-exposed frame by frame in the recorded order;
3. a still frame (dim centre, bright band, 12 % per-pixel noise like the render's Monte Carlo noise; hands-free): the largest over the smallest t_e × gain in the last 10 of 30 frames.

Symbol (handheld)	**Datasheet**	`centre` = `shipped`	`highlight_cap_over`	`highlight_all`
Code 39 5 mil	5–152	27–45	29–37 (1 misread)	none
Code 39 10 mil	0–330	23–177	21–177	21–175
Code 128 5 mil	15–102	27–29	29–33	none
Code 128 7.5 mil	0–178	7–115 (1 misread)	9–113	9–115
PDF417 6.7 mil	20–145	1–87	11–87	17–87
UPC-A 13 mil	13–368	13–223	9–223	9–223
Data Matrix 10 mil	8–157	1–97	1–95	19–101
QR 20 mil	0–279	1–279	1–279	17–277

Near and far limits in mm, three sessions per distance (decoded without the symbol-length rule, which none of these reads is affected by). The misreads: under `centre` the Code 128 7.5 mil card as `SK&4471#5382` at 97 mm (E1's own misread, 9.3), under `highlight_cap_over` the Code 39 5 mil card as " 4" at 19 mm. Handheld, the shipped rule is centre metering and reproduces E1's limits (its 5 mil near limit 27 against E1's 29 mm: three sessions against five). A highlight cap moves the near limits out (10–18 mm for PDF417, Data Matrix and QR), and the predictive cap of all blocks loses the 5 mil band entirely. Figure 5c is the cause: at 9–19 mm the LEDs light a band at the bottom of the frame, and a cap on the highlights exposes for that band.

Hands-free pass (recorded, re-exposed)	`centre`	`highlight_cap_over`	`shipped`
10 g lip balm, 0.2 and 0.5 m/s	0 frames decoded	15	**20**
6 oz can, 0.2 and 0.5 m/s	11	20	**23**

Hands-free, centre metering never reads the lip balm (the dark room drives the gain up and the label clips white) and reads the can in fewer frames; the shipped cap reads both in the most frames. The still frame (dim centre, bright band, render-like noise): t_e × gain varies by ×8.0 over the last 10 frames with `highlight_cap_over` (a four-frame limit cycle: 0.21 → 1.67 → 0.83 → 0.42 ms), by ×1.004 with the shipped one, and by ×1.002 with centre metering.

The decision follows the cases, not a sweep. Handheld mode keeps centre metering: it is the only rule whose near limits stay where E1 found them, and a highlight cap exposes for the LEDs' near-range hot band instead of the aimed code. Hands-free mode keeps the cap: without it the lip balm is never read. The cap is predictive because capping only a highlight already over the limit makes the exposure hunt. `AE_HIGHLIGHT_DN` (230) and `AE_BLOCK_PX` (8) are estimates. Nothing here was tuned to a result: every rule was replayed once on the same recorded radiance.

9.3c Roll, skew and pitch (E1c)

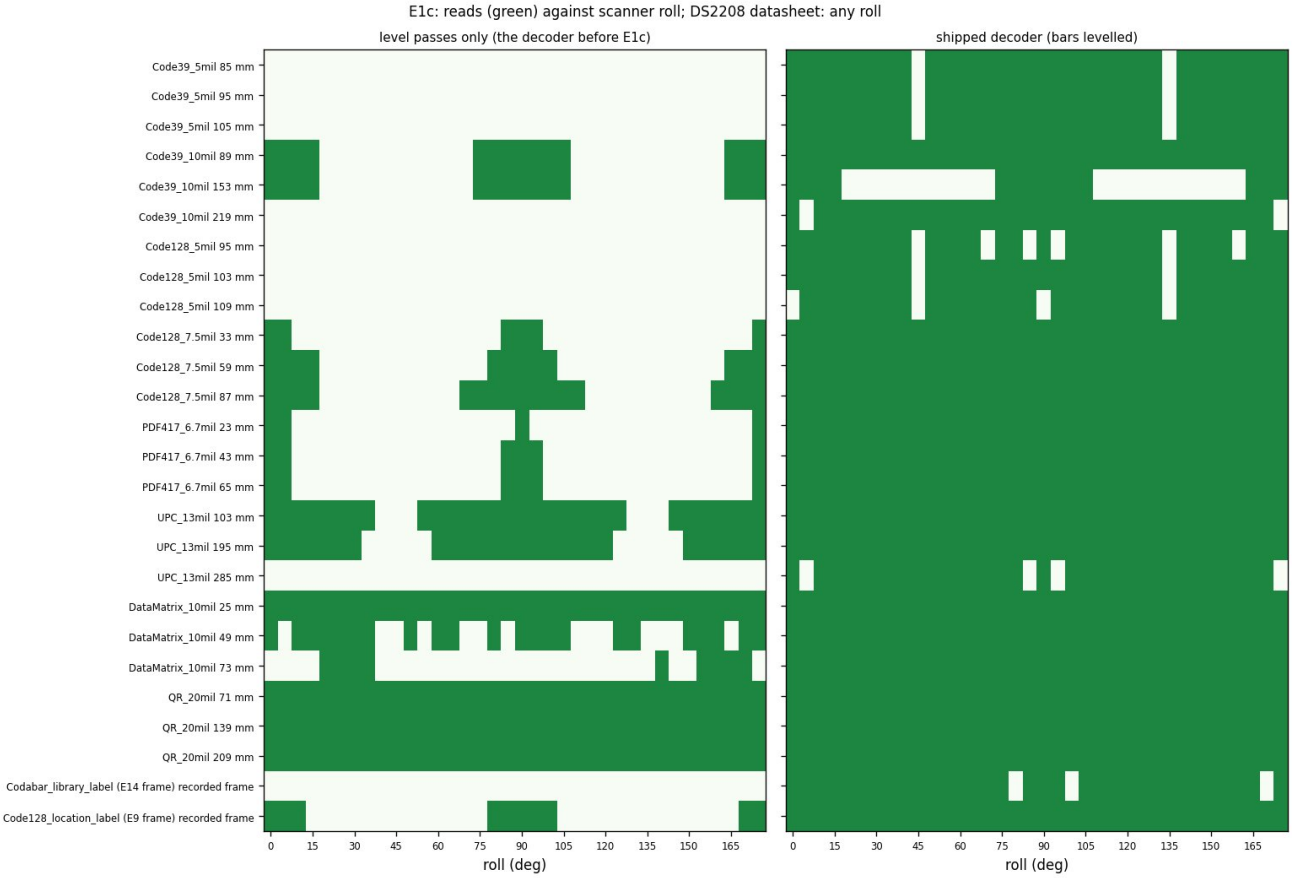


Figure 5d. E1c: reads against scanner roll, 0–175° in 5° steps (a symbol turned 180° decodes as itself), at a

quarter, half and three quarters of each E1 row's working range, and on two recorded task frames. Left: zxing's passes on the frame as captured, without levelling. Right: the shipped decoder.*

Scope: rolling the scanner about its optical axis turns the image of the scene on the sensor, LED pattern included, so the saved E1 radiance is turned about the frame centre before it is exposed (the auto-exposure meters the turned scene; sensor noise and PRNU stay on the sensor); 3 handheld sessions of up to 6 frames per roll, majority rule. Two recorded 8-bit task frames (the E14 library label and the E9 location label) are turned as frames.

Symbol, distance	Rolls read, level passes only	Rolls read, shipped	Misreads (shipped)
Code 39 5 mil, 85 / 95 / 105 mm (the front end's band)	0 / 0 / 0 of 36	34 / 34 / 34 (not at 45° and 135°)	0
Code 39 10 mil, 89 / 153 / 219 mm	14 / 14 / 0	36 / **14** / 34	0
Code 128 5 mil, 95 / 103 / 109 mm (the front end's band)	0 / 0 / 0	30 / 34 / 32	0
Code 128 7.5 mil, 33 / 59 / 87 mm	6 / 12 / 17	36 / 36 / 36	0
PDF417 6.7 mil, 23 / 43 / 65 mm	4 / 6 / 6	36 / 36 / 36	0
UPC-A 13 mil, 103 / 195 / 285 mm	30 / 26 / 0	36 / 36 / 32	**2** (195 mm, 40°)
Data Matrix 10 mil, 25 / 49 / 73 mm	36 / 22 / 9	36 / 36 / 36	0
QR 20 mil, 71 / 139 / 209 mm	36 / 36 / 36	36 / 36 / 36	0
library Codabar (E14 frame)	0	33 (not at 80°, 100°, 170°)	0
location Code 128 (E9 frame)	10	36	0

(The shipped decoder on the E1 radiance and the E9 and E14 frames; the distances are a quarter, a half and three quarters of each row's working range as decoded by the shipped decoder, 9.3.) **Not every roll reads, and the sweep found two misreads.** The front end reads a rolled symbol where zxing cannot (the 5 mil bands), except within about 5° of 45° and 135°; Code 39 10 mil at 153 mm, near zxing's limit and short of the front end's, reads only near level (14 of 36: the levelled passes fail on the bicubically turned frame at 1.4 px per module); UPC-A at 285 mm, the front end's range, misses four rolls near level and 90°. At 195 mm rolled 40° one session's levelled zxing passes read the UPC-A 036000291452 as the checksum-valid EAN-13 0036000201451 (two passes agreeing, its scan lines 17 and 31 px apart, so the line rule does not apply) in two frames, and in one of them the true number too; the front end, on the whole frame, read the true number in another session. The DS2208 reads at any roll [S1]; the simulated imager falls short at some rolls of some distances, and the levelled passes can misread near zxing's range limit. The library label (recorded frame)

misses 3 rolls of 36 (cause not separated). Noise-free synthetic symbols (every 3°, UPC-A, EAN-13, Code 128, Codabar and Code 39 at 3 and 5 px per module): every angle reads except four for EAN-13 at 3 px per module, 0 misreads.

Skew and pitch. E1's bench with a card 127 mm (5 in) in front of the window, turned about the symbol's horizontal axis (pitch) or its vertical axis (skew), −70° to +70° in 5° steps, 3 sessions per angle, 323 lx ambient:

Card	Pitch read	Skew read	Datasheet	Misreads
Code 39 20 mil (the SE4710 guide's test symbol)	−70° to +70°	−60° to +55°	±65° (DS2208), ±60° at 5 in (SE4710)	0
UPC-A 13 mil	−70° to +70°	−50° to +50°	±65°	0

(Decoded without the symbol-length rule, which drops none of its reads; without the flat-field stage the same sweep reads skew only to ±25° and ±30°.) Pitch meets the datasheet; **skew comes close but falls short**: 5–10° for Code 39 and 15° for UPC-A. Not for lack of pixels: at 40° and 60° the symbol is sharp and resolved (the saved frames), and the decoder alone reads a projected synthetic card to 50–60° skew. A skewed card is lit unevenly by the scanner's own LEDs, its far side about 1.7 times dimmer, with the dark room beyond its edge in the same rows, and zxing thresholds each 1-D row as a whole; the flat-field stage of section 5.1 divides the frame by the local white level and recovers most of the band. The remaining 5–15° were not investigated further. Section 14, item 8.

9.4 Optics bench (E2–E5)

No robot. One path-traced radiance image per configuration, then the sensor (up to six frames of auto-exposure) and the decoder, as in a task.

Decoded by the shipped decoder (5.1); 923 configurations; **no misread in any of them**. The front end reads many configurations zxing does not (E2 364 of 372 against 294; the roughness sweep 444 of 465 against 335).

- **E2 specular (UPC-A on matte, semigloss and glossy stock, pitch −30…+30° in 2° steps).** With the LEDs on, every finish reads at every pitch at 100 and 150 mm; with the LEDs off all read at 150 mm. **At 250 mm a dead zone around normal incidence appears**: matte reads at every pitch, semigloss misses −4…+4° (26 of 31) and glossy 0…+4° (28 of 31); the clear-coat roughness sweep (0.07, 0.12, 0.18, 0.25, 0.35) reads at every pitch at 100 and 150 mm and misses 3–5 pitches within −4…+6° at 250 mm at every roughness. At 250 mm only the front end reads this symbol (zxing's limit is 223 mm, 9.3), and the LEDs' mirror reflection on the symbol near normal incidence spoils the edge model's fit where zxing at 100–150 mm still found a clean row beside it. So the simulation has a dead zone, 4–8° wide, where the read depends on the front end; the SE4710 guide gives up to 20° with illumination on at working distances [S3], and at 100–150 mm there is still none. See section 14, item 7.
- **E2b: what would make glare matter.** UPC-A on matte, semigloss and glossy stock at 100 and 150 mm, pitch −30…+30° in 2° steps, rendered once with the shipped 1 mm-radius LEDs and once with 2.5 mm-radius emitters (intensity × $1/2.5^2$, the on-axis signal unchanged: LED-to-room 10.80 against 10.80), each radiance then exposed with lens veiling glare η = 0, 0.5, 1, 2 and 4 % (a fraction η of the light spread over the frame by a $1/(r^2 + 2^2)^{\wedge}1.5$ kernel; the thin-lens render has none) and decoded with the flat-field stage on and off: 120 rows. With the flat-field stage every row reads at every pitch; without it, semigloss at 150 mm fails at +2° only, at both radii and every η but one (`saturated_max` 5 %). Neither the emitter's size nor veiling glare up to 4 % makes a dead zone: the mirror images grow slightly and the decoder still finds a clean row. Neither change is part of the shipped model; section 14, item 7.
- **Glare on unconverged frames.** At coat roughness 0.18, 150 mm, pitch +2°, unconverged frames decode the UPC-A 036000291452 as the checksum-valid 7046606291452 (the LEDs' two mirror reflections saturate the symbol's left half and zxing reads the row just above them). On the converged frames the same configuration reads the true number, and the bench has no misread anywhere. A check over scan lines is as unstable as the read itself, so none is added for it.
- **E3 ink colour.** UPC-A in black, blue (0.05, 0.10, 0.60) and red (0.70, 0.06, 0.06) reads to 200 mm in all three inks, with the LEDs on and off; at 300 mm, where zxing reads none, the front end reads black and blue with the LEDs on and off and red with the LEDs on, not off. The chosen red keeps a 15–18 % reflectance difference under the 645 nm LEDs. That is below the 25 % minimum reflective difference the DS2200 spec sheet and the DS2278 guide specify [S1, S2]. The simulated scanner therefore reads a red symbol below the device's specified minimum: the figure is a guarantee, not a cliff, so a real unit may read it too, but here the simulation is lenient and unvalidated. A red whose reflectance at 645 nm equals the paper's would not read. The environment's own packaging prints symbols in dark ink.
- **E4 curvature.** UPC-A wrapped on cylinders of 30–100 mm diameter, presented 8° off square (then square and 8° the other way if that fails): every diameter reads in both orientations except the 30 mm cylinder with the bars along the axis (picket fence), which reads at none of the three; GS1's curvature rule for 100 % UPC [S9] excludes picket-fence symbols on cylinders this small.
- **E5 clear film.** A Code 128 bin label behind a clear PVC holder reads at every tilt from −30 to +30° (42 of 42) with about 10 % less light than bare (exposure 0.47–0.55 ms against 0.43–0.48 ms), and the two LED

glints appear on the film at normal incidence.

9.5 Illumination pattern

On white paper at 100 / 150 / 250 mm, the frame edges receive 0.50–0.57 of the centre's illumination and the corners 0.13 / 0.20 / 0.27; the region above half the centre value spans 623 / 640 / 640 of the 640 columns and 374 / 405 / 442 of the 480 rows (the optics bench): the published 50.3° × 32.0° half-intensity pattern [S3] seen through the 32.8° × 24.8° field.

10. Assisted end-to-end demonstrations

Scope: complete interactions from tick zero through the normal action interface (joint targets, grasp commands), scored by the unmodified evaluator, with the assistance named. A policy has none of the phantom forces or mounts.

Who does what in the robot demonstrations (E21–E23). The scripted operator (`skills.py`) sends only what a policy can send, joint targets and Franka Hand commands, every 10 ms; everything it plans is planned with the robot's own geometry. Where the Franka Hand physically cannot do a step of the human job, a labelled phantom does it:

Step	Done by	How it is planned or bounded
take an item that has a collision-free grasp where it lies	robot (in `checkout-counter`, every item where it stands on the counter)	`grasping.plan`: the Franka Hand's collision boxes against the scene's shapes (separating-axis test), then the whole-arm path checked by MuJoCo (`collide.py`)
take an item with no such grasp (a hand basket is 200 mm deep and the hand's housing is 208 mm across its fingers; shelf facings stand touching, fingers are 26 mm thick; a book lying flat has no side under the 80 mm stroke; at the counter, an item the left hand set down where the right hand finds no grasp)	**phantom customer / patron / stocker** lifts it straight out, turns it upright up high and holds it at a handover point; the robot grasps it from the phantom's hold and the phantom lets go once the fingers have closed. Each hand-over is recorded (`handed_over`, `handed_over_from`)	the hold is a six-axis PD spring with gravity compensation, each axis's stiffness capped at 0.25 I/dt² (explicit-spring stability). **Never from the floor** (below 0.30 m): an item the robot drops stays on the floor and the evaluator sees it
hold a book's front cover shut while handing it over, or open flat (180°) to show a label inside the cover	**phantom fingertip** (torque on the spine hinge, capped at 0.15 N m)	the patron's thumb
sweep a barcode across the cradled scanner's window	robot	the free rotation about the barcode normal is chosen for the most downward hand whose turn, approach and sweep (the item carried in the hand) the collision test clears and that keep the item's other barcodes (a book's ISBN) out of the imager's view; afterwards the item is lifted above the window's field of view and carried behind the window's plane before it is turned for the set-down (not for books, which go straight to the bin)
set an item down, bag it in the tote, drop a book in the returns bin	robot	set down upright with its footprint ≥ 15 mm inside the counter's front edge; lowered to the lowest of four release heights with a clear path (the item's lowest point 20, 60 or 100 mm above the container floor, or 20 mm above its rim), then released; each item's height is in its record (`release_height_above_floor_mm`: sections 10.5 and 10.7 give them); an item that cannot be set down goes back into the basket
take, aim and reseat the handheld scanner	robot (right hand, 17.5 N per finger on the head)	E7 grasp; aim by IK on the held scanner's measured pose
pull the handheld scanner's trigger	**phantom fingertip** (ramped to 3 N on the blade in 0.1 s)	a Franka Hand holding the scanner cannot also pull its trigger (a second Franka fingertip has not closed it, 9.1b; the G1's Dex3 index finger does it, 6.5)
key a quantity and ENTER	robot (one fingertip corner of the closed hand)	`skills.plan_press`: per key, a hand orientation whose leading fingertip corner is ≥ 4 mm lower than any other finger point and whose hand, camera and link 7 clear every other keycap, the display and the shelf; the whole-arm path and the pressed pose ±5 mm checked by MuJoCo; closed loop on the measured fingertip; the switch, not a target, decides when the key is down

10.1 Handheld read of a shelf location (E9, task `scan-location`)

ASSISTED: the scripted operator moves the arms (IK waypoints turned into joint targets; Franka grasp), and a **phantom fingertip** pulls the trigger (its force ramped from 0 to 3 N over 0.1 s, as a finger squeezes, then held for 0.5 s; reaction on the scanner). Applied in one step, the 3 N would drive the blade into its stop at 0.85 m/s and overrun it by 0.3 mm; ramped, it reaches 3.65 mm against the 3.6 mm stop. Sequence: the idle left arm parks aside; the right hand takes the scanner by its head, lifts it out of the cradle, aims the window 150 mm from the A03-3-B location label pitched 20° down (aim error 1.2 mm); the phantom finger pulls the trigger (switch closed, firmware SESSION); the first frame (path depth 3, decoded) reads **Code 128 "A03-3-B"**, attributed to the physical label `loc_A03-3-B` at 0.201 m (not a ghost, not a misread); the trigger is released; the right hand reseats the scanner (seat sensor true) and the hands are withdrawn for a 2 s hands-off hold.

Evaluator (unmodified): **success, score 1.0, 7 of 7 checks**: location recorded, exactly one location read, scanner seated, no key pressed, nothing on the floor, no misread; 1 frame, 0 ghost reads, 0 reflexes; 21.4 s

simulated, 250 s wall.

The same task by the two-arm Panda and by the one-arm FR3. The same script, the one-arm FR3 taking the scanner with its only (left) arm: aim errors 1.0 and 0.8 mm; the first frame reads Code 128 "A03-3-B" attributed to `loc_A03-3-B` at 0.201 m; reseated; **success, 7 of 7** for both, 0 reflexes; 21.4 and 19.8 s simulated, 244 and 156 s wall.

Negative control: the same script aimed at the next label on the beam, A03-3-C. The read is correct and attributed to `loc_A03-3-C`; the evaluator returns **failure, score 0.71** (5 of 7: location not recorded, no read of the requested location).

10.2 Robot-hand trigger pull (attempt)

The same rollout with the trigger pulled by the left hand instead of a phantom force: the left gripper closes front-to-back around the handle just under the head (one pad on the blade, one on the back of the handle) with 3 N per finger. **It fails** (score 0.43, 3 of 7): the left hand reaches the handle, but its squeeze does not move the trigger (0 frames); on the way the two hands touch (one stop of the right arm, joint 4) and both touch the scanner's head (one stop each), and after the scripted reseat the scanner is not in its socket and rests on a key (`no_key_pressed` fails too). A second fingertip on the trigger of the scanner turned towards the other hand in open space fails before the pull (9.1b). A read with the trigger pulled by a Franka Hand while the scanner is aimed at a label has not been shown. See section 14, item 5.

10.3 Presentation scanning at the counter (E11)

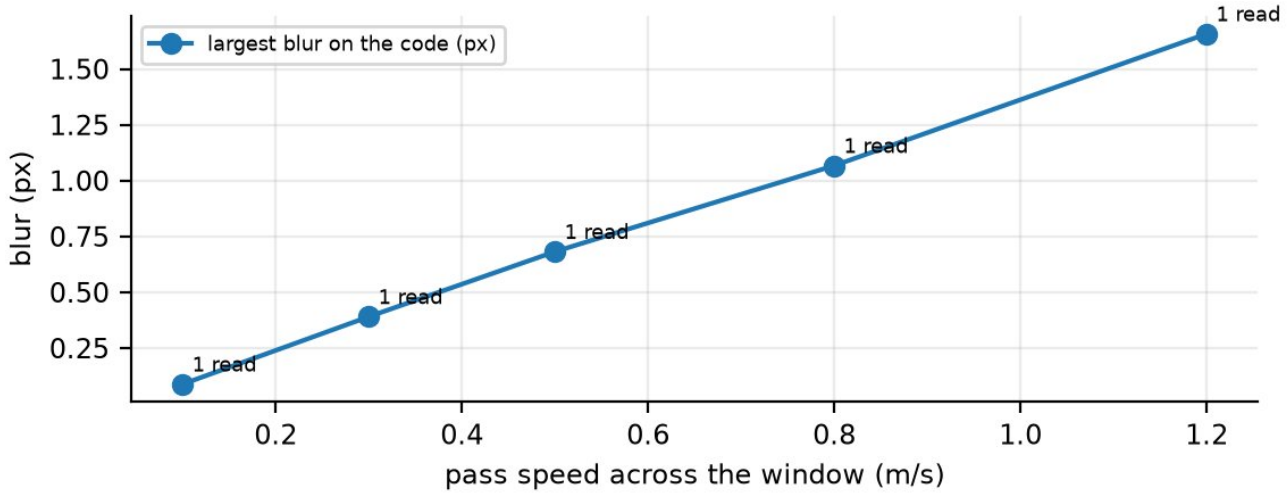


Figure 8. E11: motion blur on the barcode against pass speed, with the reads per pass.

ASSISTED: the item (a 6 oz can from the basket) rides a **phantom six-axis mount**; the arms park aside through the normal joint targets. The scanner stands in its cradle (DOCKED, hands-free exposure cap 0.22 ms, brighter LED drive).

Pass across the window, 0.12 m in front	Frames	Reads	Attributed to	Max blur	Sub-exposures
0.1 m/s	64	1	`basket_K02_0_code`	0.09 px	1
0.3 m/s	26	1	same	0.39 px	1
0.5 m/s	19	1	same	0.68 px	2
0.8 m/s	15	1	same	1.07 px	3
1.2 m/s	13	1	same	1.66 px	4

Held still in view for 3 s: exactly **one** read. Moved out for 0.3 s and back: **refused** (same-symbol timeout 0.5 s). Out for 0.8 s and back: **one** read. The scanner stayed seated throughout; no ghost reads, no misread. The datasheet's 76.2 cm/s swipe rating [S1] is met with margin at the hands-free exposure cap (0.22 ms; the predictive highlight cap shortens it to 0.09–0.19 ms in some frames).

The robot presenting the basket items itself is E21 (10.5).

10.4 Library book with the label inside the front cover (E14)

ASSISTED: a **phantom fingertip** lifts the cover's free edge (1.5x the cover's gravity torque plus spine friction) until the hinge passes 100°; a **phantom mount** carries the scanner (entrance pupil 150 mm from the label, pitched 15°, handle up) and a **phantom finger** pulls the trigger. The top book of the returns stack (seed 1) is the paperback.

Step	Result
hands-off 1 s, cover closed	hinge −0.03° (stays closed)

Step	Result
cover lifted to 100° and released	falls open, comes to rest at **217°** draped over the stack's side
book block displacement	≤ 0.43 mm (with a 180° joint stop instead: 27 mm)
scanner at the aim pose	0.001 mm, 0.03°
read	**Codabar "A31234000051126B"** from the first frame, by the flat-field stage, attributed to `book_3_library` (the inside-cover label) at 0.150 m, host records `31234000051126`

(6.6 s simulated, 91 s wall. The phantom mount's hold peaked at 2.4 N.)

10.5 Checkout: the robot scans and bags; from the basket a labelled phantom customer takes every item out, from the counter the robot takes them itself (E21; tasks `checkout-present`, `checkout-bulky`, `checkout-counter`)

One rollout from tick zero per run, scored by the unmodified evaluator. Per item: the left hand takes it (where it lies if it has a collision-free grasp, else from the phantom customer's hold), turns its barcode to the cradled scanner and sweeps it across the window at 0.2 m/s and 12 cm, lifts it out of the scanner's view, sets it down upright on the counter's front strip; the right hand takes it there and lowers it into the bagging tote (or, where it has no collision-free grasp of the item as it stands, takes it from the phantom customer's hold: recorded per item). For `checkout-bulky` the right hand then takes the handheld scanner, aims it at the 24-pack's label where the pack stands, a phantom fingertip pulls the trigger, and the hand reseats the scanner. Negative controls: the same run with the sweep commanded at 3 m/s (the arm caps it near 1 m/s, still beyond the rated 76.2 cm/s), and with each barcode turned away from the window.

Rendered (Isaac RTX imager frames decoded by the firmware). Basket runs: seed 0 (tomato paste, spring water, marinara jar, lip balm, AA battery pack). Counter runs: seed 0 (the same five items standing on the counter) and seed 3, the first counter layout without the jar (AA pack, lip balm, tomato paste, soup, water), run after seed 0 failed on the jar.

Run	Who took each item	Reads (sale)	Bagged	Stops	Evaluator
`checkout-present` (items pass the window at 0.22–0.25 m/s)	from the basket: the phantom customer (5 of 5; peak 9.9 N, tracking ≤ 36 mm); from the counter: the robot (5 of 5)	5, each item once, no misread (the jar read correctly)	5 of 5	2	**success, 8/8**; 206.7 s simulated, 4 695 s wall
control: sweep commanded at 3 m/s (items pass at 1.06–1.21 m/s), tomato paste and lip balm	as above	0	the two	0	fails `sale_matches`, `each_item_once`, `items_bagged` (the three items not in the run stay in the basket); 83.6 s simulated
control: barcodes turned away, the same two items	as above	0 (86 frames)	the two	0	fails the same three; 86.1 s simulated
`checkout-bulky` (then the handheld scanner, aimed at the 12.5 kg case where it stands, aim error 1.4 mm; phantom trigger; reseated)	as `checkout-present`	6: each item once, the case's own label once, no misread	5 of 5; the case never rose	2	**success, 9/9**; 230.7 s simulated, 6 429 s wall
`checkout-counter`, seed 0	**the robot, all five, where they stand** (4–13 collision-free grasps each); the right hand takes four from the counter strip, the phantom hands over the battery pack there	4: the jar is never read	5 of 5	12 (11 the left hand on the jar)	fails `sale_matches`, `each_item_once` (6/8); 197.2 s simulated, 3 242 s wall
`checkout-counter`, seed 3	**the robot, all five, where they stand**; the phantom hands over the battery pack on the counter strip	6: each item once, and the tomato paste a second time **as the wrong UPC-E 0067100000040**, 0.13 s before its correct read (5.1, section 14 item 2)	5 of 5	2	fails `no_misreads`, `sale_matches`, `each_item_once` (5/8); 191.9 s simulated, 3 506 s wall

On the counter the robot takes every item itself (in the basket tasks no Franka Hand grasp exists, section 6.3, so the phantom customer does every extraction). Neither counter run succeeded. Seed 0: the 85 mm jar is wider than the hand's stroke, so the planner grasps its 66 mm lid, tilted 20°; carried that way the jar's body swung against the hand and the collision reflex stopped the left arm 11 times in 10 s, and in what was left of the sweep its barcode never came into a decodable view. Seed 3: the misread of section 5.1 (an EAN-13 entering the field of view at the frame's edge read as a UPC-E from two scan lines 5 px apart through the ragged ends of its bars). One run was made per layout; no rerun was made to look for a passing one.

Three details of the operator matter for these runs. `skills.take_item` lifts an item 10 cm after its grasp (lower if needed) and 22 cm only out of a container: a 22 cm lift is out of reach for a tall item grasped at its top. The set-down tries a second row of spots 4 cm further in when the first row has none (with one row the battery pack finds no spot beside the scanner among 72). The water case is scanned in place, never taken. Every item was carried out of the scanner's view after its sweep (`cleared_view`), so none was read twice by re-presentation.

10.6 Cycle count: the robot scans and keys, a labelled phantom stocker lifts every unit off the facing (E22; tasks `count-key-quantity`, `count-by-scanning`)

Seed 5. The right hand takes the scanner and aims it at the slot's location label (phantom trigger); for each unit (one per SKU when keying) the phantom stocker lifts it from the facing and turns its barcode to the held scanner (phantom trigger), then puts it back where it stood. For `count-key-quantity` the right hand then reseats the scanner, swings aside, and the keys of the quantity and ENTER are pressed one fingertip corner at a time by whichever hand has a collision-free plan for that key (the left first; a hand that cannot is retried from the ready pose, then tucked aside for the other); the keypad's switches, not the script, register the keys. Each press is planned with the closed hand's own geometry (orientation, path and pressed pose checked with MuJoCo's collision test, preferring plans with every joint at least 0.15 rad from its limits: the keypad is close to the arm bases and the right-column keys need an arm folded near its limits) and closed-loop on the planned fingertip corner, which descends 0.5 mm at a time until the key's switch closes. Negative control: a wrong quantity keyed.

Rendered (Isaac RTX imager frames decoded by the firmware), seed 5 (slot A03-3-B, six cans of one SKU):

Run	Reads (host record)	Keys pressed by the robot (switch closed; fingertip centred on the keycap)	Protective stops	Evaluator
`count-key-quantity`	the location label A03-3-B, then the unit's EAN-13 (one frame each)	6, ENTER: left hand, closed at 2.7 and 2.2 mm below the keycap top, fingertip within 0.2 mm of the keycap centre	0	**success, 9/9**; 49.9 s simulated, 877 s wall
negative control: 5 keyed instead of 6	same	5, ENTER: same hand, 2.6 mm below the keycap tops	0	fails `record_matches` only (8/9); 49.6 s simulated, 667 s wall
`count-by-scanning`	the location label, then each of the slot's four units once (EAN-13, each held up to the scanner by the phantom stocker and put back), one frame each	none	0	**success, 10/10** (the scanner returned to its socket); 59.1 s simulated, 485 s wall

The phantom trigger fingertip ramps to 3.0 N (blade to 3.65 mm, the stop at 3.6 mm).

10.7 Library check-in: the robot scans and bins, a labelled phantom patron hands every book over (E23; task `library-checkin`)

Seed 0 (three books; the second is labelled inside its front cover). The right hand takes each book from the phantom patron's hold (the cover held shut, or opened flat for an inside label), sweeps the library label across the desk scanner in its cradle, and lowers the book into the returns bin.

Rendered, seed 0: **success, 8 of 8 checks**, 0 protective stops, 0 misreads, 396 scanner frames, 84.9 s simulated (4 621 s wall on a shared GPU). The two back-cover books also carry an ISBN: every hand rotation whose approach, turn or sweep would bring it into the imager's view is rejected (15 and 22 rotations; `other_barcodes_kept_out_of_view`). For the third book none of the 36 most downward hand rotations has a reachable, collision-free presentation at the scanner, so the presentation tries the other 36 rotations, then 3 cm farther and 2 cm nearer, before giving up (`skills.present_to_cradle`); the third book was presented from the second group, hand 91° from down.

Book (label)	Taken from	Presented	Read (Codabar library label)	Into the bin
The Tidewater Almanac (back cover)	the patron's hold (cover held shut), handed over 1.28 m up	69° from hand-down, swept across the window	A31234000050748B, frame at 16.74 s, rolled 21.6° in the image (levelled passes)	let go 20 mm above the bin floor
Notes on Salt and Wind (inside the front cover)	the patron's hold, cover opened flat (179.7°) by the phantom fingertip, handed over 1.24 m up	12.0°, the cover held open	A31234000050375B, frame at 45.30 s, flat-field stage	let go 176 mm above the bin floor (the opened cover limits how low the hand can go)
The Tidewater Almanac (back cover, second copy)	as the first, 1.28 m up	91° (fallback rotations)	A31234000050003B, frame at 70.34 s, flat-field stage	20 mm above the floor

No ISBN (EAN-13 on the back covers) was read. How hard the labelled phantoms pushed: the patron's hold peaked at 7.1, 2.8 and 7.4 N with tracking errors of 59, 37 and 59 mm, and the cover fingertip at 0.05, 0.15 (its cap, holding the inside cover open) and 0.05 N m. The patron carries each book over whatever stands between,

at the lowest clear crossing height plus 3 cm (`collide.Checker.free_body_contacts`); carried straight, a book sweeps through the cradled scanner (phantom peaks 48–80 N, the scanner knocked off its seat).

11. Stability at rest and under legal abuse

11.1 Rest (E0)

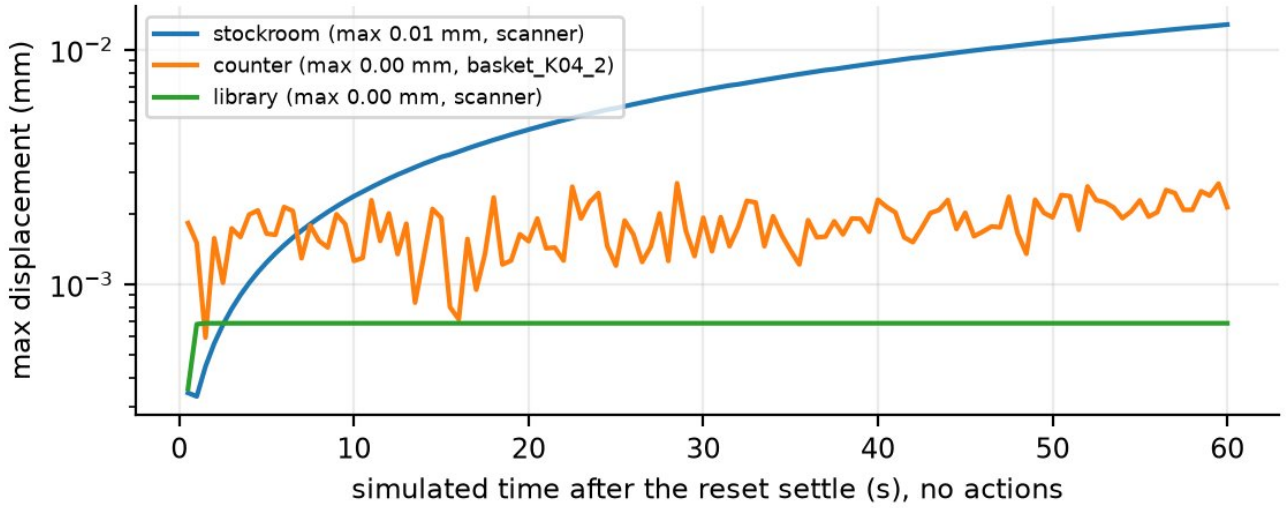


Figure 6. E0: the largest displacement of any free body from its settled pose, 60 s hands off (log scale).

UNASSISTED: after the environment's own settle (until every free body has stayed below 2 mm/s through a whole 0.1 s: 1.0 s in every preset), 60 s with no actions (the robots hold their ready pose). Every 0.5 s: each dynamic body's displacement from its post-settle pose and its speed; the trigger, keys and seat sensor.

Preset	Bodies	Largest displacement	Grows in the second 30 s by	Speed at the end	Switches, seat, reads, reflexes	Physics wall/sim
stockroom	54	0.0127 mm (the cradled scanner, settling into its socket; bounded, E20)	0.0060 mm	≤ 0.062 mm/s (solver jitter)	none closed; seated throughout; 0 frames, 0 reads, 0 reflexes	4.6x
counter	8	0.0020 mm (the sauce jar in the basket)	0.0008 mm	≤ 0.50 mm/s (jitter of a body in the basket)	same	4.0x
library	8	0.0006 mm (the scanner)	0.0003 mm	≤ 0.08 mm/s	same	2.5x

Nothing is non-finite, no sensor changes. The "speed at the end" is the solver's micrometre contact jitter (a body vibrating within a few micrometres), not motion. The stockroom's scanner settles 12.7 µm further into its socket in the minute, and the 300 s soak shows it bounded (E20, 11.5).

Rest on more seeds and layouts (120 s hands off after the reset's settle, four more seeds of every task layout):

Layout (seeds 1–4)	Largest displacement in 120 s	Fastest steady rate (least squares over the run)	Seated at the end; faults; reflexes
stockroom `count-by-scanning`, `scan-location`	0.0206 mm (the scanner, every seed)	≤ 0.14 µm/s	all seated; none; 0
counter `checkout-present`	0.0056–**0.975** mm	≤ 1.0 µm/s (seed 2: the cereal box leaning in the basket)	all seated; none; 0
counter `checkout-bulky`	0.0066–0.120 mm	≤ 1.0 µm/s (the same basket at seed 2)	all seated; none; 0
counter `checkout-counter`	0.0004–0.0006 mm	≤ 0.001 µm/s	all seated; none; 0
library `library-checkin`	0.0034–0.012 mm	≤ 0.10 µm/s	all seated; none; 0

One slide: at counter seed 4 the cereal box in the basket slides 0.97 mm in the first 40 s after the reset's settle (about 25 µm/s, below the settle's 2 mm/s criterion) and then stops (its rate over the run's second half is zero). The seed-2 cereal box creeps steadily (1.0 µm/s, linear over the whole run). The items standing on the counter (`checkout-counter`) do not move. Section 14, item 10.

11.2 Legal-action stress (E17)

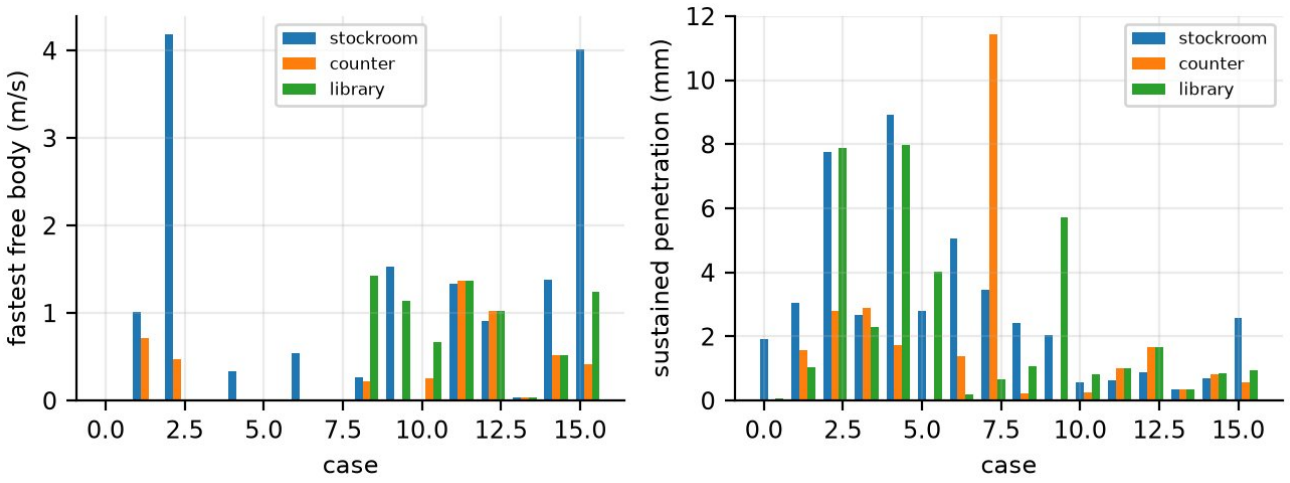


Figure 9. E17: fastest free body and sustained penetration per legal-action case.

Only legal actions (joint targets within limits, grip force within the hand's limit, 35 N per finger on the FR3, the finger joint's effort limit; Cartesian targets through the same IK as the API); no phantom forces. Per preset (a different random stream per preset): 10 cases of random targets for every arm and finger joint with random grip forces, redrawn every 0.5 s for 4 s (an arm in a protective stop keeps being sent targets, which it refuses for its 1 s hold; the two fingers of a hand get one target); a **push** (the left hand driven at full speed 15 cm sideways through the nearest stock item and held there 2 s with its target inside it); four directed abuses of a held scanner (whipped ±0.15 m six times at full joint speed in the hand's full 35 N grip; slammed down at full speed to 5 cm below the surface; grip toggled 0/35 N every 20 ms while rising; dragged through the cradle's cup); and a fingertip driven 4 cm into the keypad then onto the nearest item and squeezed at 35 N. **Every case ends with a judged settle**: 3 s hands off, grips released; the held cases **15 s** with the hand keeping its grip and its last target (3 s is too short to judge a held scanner's slide), the scanner's pose in the hand measured after 3 s and at the end, and the slide rate over the last 5 s. Judged per case: no non-finite state, no non-robot body faster than 5 m/s or 200 rad/s, no penetration held for 5 ticks deeper than 5 mm, no joint more than 10 mrad (prismatic: 10 mm) past its limit, no fault, no body whose positions in the last second of the settle span more than 0.1 mm without shrinking, and (held cases) no slide faster than 0.01 mm/s in the last 5 s. The table was recorded with a line-search budget of 20 evaluations (section 6.1); the re-run with the shipped budget follows it.

Preset	Cases	All finite	Fastest non-robot body (single tick)	Deepest sustained overlap (≥ 5 ticks)	Joint past limit	Largest settle span of the passing cases	Flags
stockroom	16	yes	4.19 m/s (a 5 kg shipping case knocked off a deck by a link at about 2 m/s, then free fall to the floor)	8.9 mm (a left fingertip into the shelving, random case 4)	18 mm (a finger, random 7; 13 mm random 5), else ≤ 0.2 mrad	0.002 mm	5: three overlaps over 5 mm (random 2, 4, 6: fingertips and link 7 into the shelving, 5.0–8.9 mm); two finger excursions past the 40 mm stop (random 5 and 7)
counter	16	yes	1.37 m/s (the scanner, whipped)	11.4 mm (link 7 into the counter, random case 7)	≤ 0.1 mrad	0.046 mm	1: the counter overlap
library	16	yes	1.43 m/s (the trigger, random 8)	8.0 mm (a fingertip into the other arm's link 5, random case 4)	≤ 2.5 mrad	0.004 mm	3: overlaps over 5 mm (random 2, 4, 9: a fingertip into the other arm, 5.7–8.0 mm)

Held scanners settle: in every held case the scanner stops sliding (≤ 0.001 mm/s over the last 5 s of the 15 s hold; it moved 1.4–11.2 mm and turned 0.4–6.9° in the hand during the abuse itself), and the stockroom cradle jam does not knock it out of the hand (moved 44 mm and turned 68° in the grip, still held). **No case leaves a body moving** after the settle (largest span 0.002–0.046 mm). **The counter runs clean** with the counter top as a hull; with a box top, the held-whip settle threw the bagging tote at 26.9 m/s twice, at different ticks, from one 753 mm box-box contact between its 4 mm floor and the counter top (section 6.2). **What fails** (judged problems, not fixed):

- Seven random cases (of 30 per preset) hold a fingertip or link 7 5.0–11.4 mm inside a shelf, the counter or the other arm for at least 5 ms: impacts at arm speed (the 4 ms contact time constant; 2 ms goes unstable, section 6.2).

- In the table, two stockroom random cases drive one finger 13 and 18 mm past its 40 mm stop for about 10 ms. Traced (random case 5): the hand strikes the shelving and the impact drives the two fingers apart in opposite directions (one to 52.8 mm, the other to 3.8 mm) against the coupling that makes them one mechanism (section 3.3) and against the stop. The cause is the line search cut short at the impact (section 6.1), not the coupling.

With the full line search (50 evaluations, section 6.1), E17 re-run with the same seeds and cases: stockroom random 5 and 7 do not pass the finger stop (0 against 12.7 and 18.4 mm; fastest robot link 1.8 and 1.7 m/s against 15.4 and 18.7 m/s), so the stockroom has 3 flags instead of 5, all three the overlaps of random 2, 4 and 6 (7.8, 8.9, 5.1 mm). The library's 3 flags and the counter's overlap are the same (to 0.1 mm). Most cases are the same to the printed digits; where an impact differs, the change is within the judged bounds: the stockroom keypad case's deepest sustained overlap 2.6 and 4.8 mm, its fastest body 4.0 and 4.6 m/s, 3 and 8 reflexes (budget 20 and 50). One new flag: at the counter, random 1 leaves the 4-pack of batteries spanning 0.10 mm in the last second of the 3 s settle (the judged limit is 0.1 mm); the same case re-run with a 15 s settle ends with every body still to within 0.001 mm, so it is a slow settle, not a body that keeps moving.

In-hand: whip 2.9–7.8 mm and 0.4–6.6°, slam 1.4–11.2 mm and 5.3–6.9°, grip toggle 4.4–4.5 mm and 2.6–2.8°, jam 8.4–43.8 mm and 1.7–68°. The collision reflex stopped the arms in 15 of the 16 cases per preset (up to 46 stops in one case). At the counter and the library, arms moving in front of the docked scanner woke its hands-free mode (up to 58 frames); no case closed the trigger switch. This probe renders dark stub frames (physics only), so it produces no reads by construction. **Box-box contacts at reset**: every task's reset and 2 s of rest for 25 seeds of each of the eight tasks (200 resets), the deepest contact recorded at every physics step. The deepest anywhere is 10.0 mm; the 43 resets with a contact deeper than 5 mm are all neighbouring boxes of the stockroom's facings and shipping cases overlapping at spawn (stock is faced with a little yaw noise), within the first 0.1 s of the settle and for at most 6 ticks, with no body faster than 0.74 m/s; at the counter and the library the deepest is 2.1 mm. The bogus contacts of section 6.2 were 584 and 753 mm deep. A sweep cannot prove the routine never misfires on the remaining box-box pairs; it shows it did not in any of these resets, and E17's dynamic cases count single-tick spikes over 5 mm (none).

11.3 Timestep refinement (E18)

Scope: the same fixtures at 1 ms (shipped), 0.5 ms and 0.25 ms, and 2 ms for context, nothing else changed, gate $|q(1\text{ ms}) - q(0.5\text{ ms})| / |q(0.5\text{ ms})| \le 2\ \%$; 0.25 ms shows where the values converge; settings as shipped (`impratio`* 3, section 6.1).*

26 of 50 quantities pass. Every force-travel law passes: the trigger's fitted F0 and k (1.006 N, 0.4512 N/mm), switch make and break travel and force, the key's law and switch points, at slow and at fast ramps. The grasp's pad force (35.0 N) and finger opening (65.7 mm) pass.

9 fail by construction, as predicted: the trigger's and the key's travel at rest and stop overtravel. The stops are MuJoCo joint limits with a 2·dt time constant, so the overtravel scales with the step (under 2 % of the 3.6 mm travel at every step), and the rest travel is the preload sinking into the stop.

15 fail and are not by construction; every one is a micrometre quantity. The held scanner's slip over 5 s is 30 µm and 0.017° at 1 ms, against 15 and 7 µm (0.007°, 0.003°) at 0.5 and 0.25 ms; the grasp-to-hold transient 52 against 39 and 39 µm. The stockroom's rest is 2.2–2.4 µm at every step (passes the spirit, fails the 2 % gate by 4 %); its peak speeds (0.13, 0.08, 0.06 mm/s) are contact jitter scaling with the step. At the counter the finer steps let a basket item settle 41 and 37 µm in the first ten seconds, against 2.4 µm at 1 ms; the last second's speeds are jitter in place. The library's rest is 0.6, 0.9 and 1.1 µm. Relative differences of micrometre quantities fail a 2 % gate without consequence at any scale a camera or the evaluator resolves.

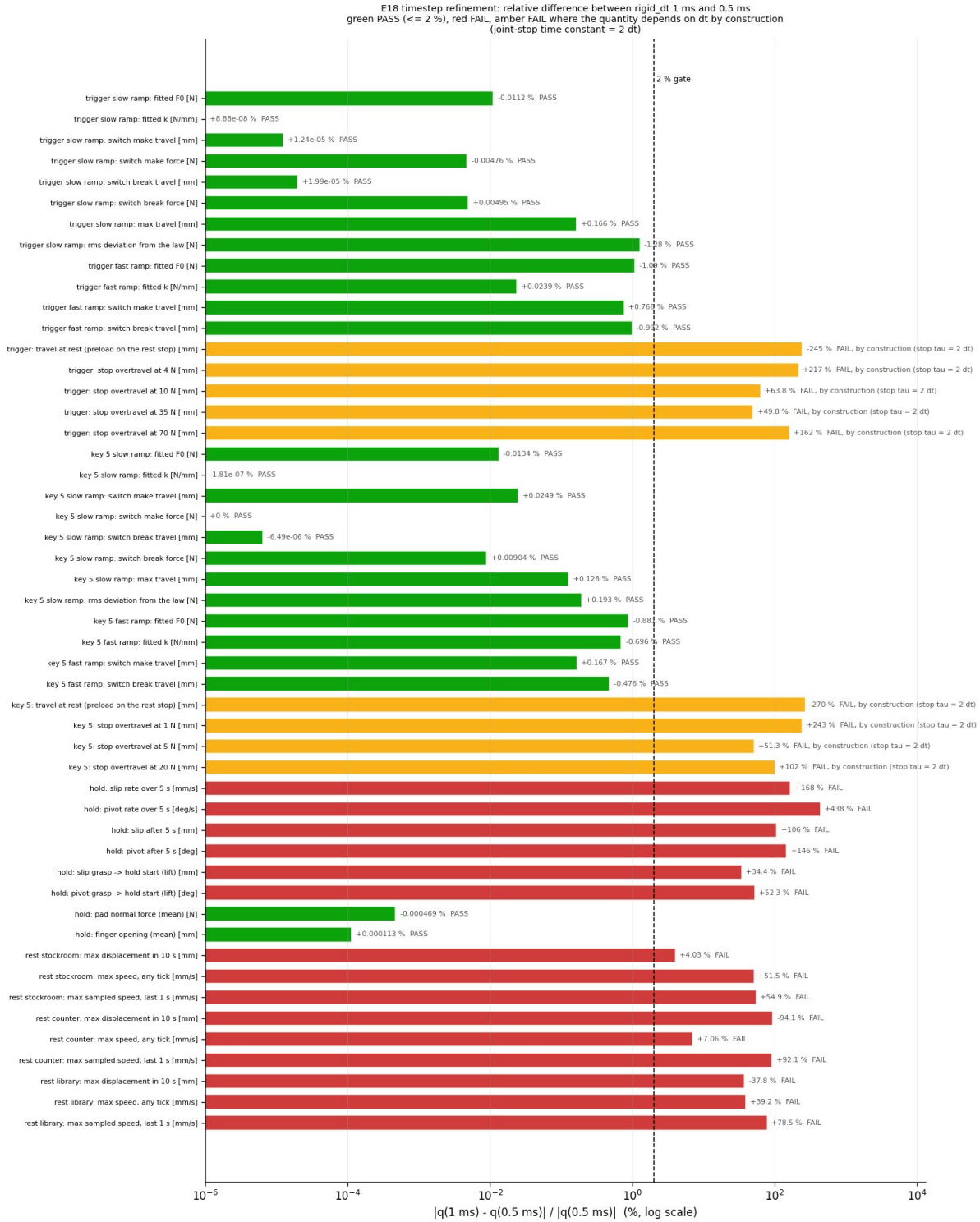


Figure 13. E18: relative difference of each measured quantity between 1 ms and 0.5 ms (the 2 % gate dashed); quantities that depend on the time step by construction marked.

11.4 Drop impacts and energy (E19)

Scope: a cereal box flat, a soup can upright and the scanner on its side dropped from 100 mm onto the counter at 1, 0.5 and 0.25 ms, against analytic free fall (contact after 0.1428 s at 1.4007 m/s); the total energy (kinetic, potential, the trigger spring's) logged every tick; `python -m barcode_scanning.probes.e19_drop`.

Case	Impact speed error (1 / 0.5 / 0.25 ms)	Peak penetration (mm)	Rebound (mm)	Energy ever above its start
cereal box, flat	+0.15 % / +0.15 % / −0.02 %	1.85 / 2.07 / 1.91	0.30 / 0.46 / 0.28	never (0 ticks)
soup can, upright	+0.15 % / +0.15 % / −0.02 %	1.85 / 2.07 / 1.91	0.30 / 0.46 / 0.27	never
scanner, on its side	+0.15 % / +0.15 % / −0.02 %	2.09 / 2.15 / 1.95	0.25 / 0.73 / 0.52	never

(The scanner penetrates 2.0–2.2 mm and slides 8.5–11.1 mm after tipping 8.4–8.7°.) Free fall and the moment of contact agree with the analytic values to 0.15 % (one step's discretisation); no contact ever added energy

(the largest excess over the initial energy is -10^{-6} to -2×10^{-5} J, i.e. none). Every material has the same contact compliance here (MuJoCo's mass-normalised contact law: the box, the can and the scanner sink the same 1.85 mm at 1 ms), a model property, not a measured one. The fast corner impact of E17 is E19b (section 6.1): with the torsional friction neither `impratio` 3 nor 10 gains energy at any step (rebound 27.6 mm at 1 ms, 29.7–30.2 mm at 0.5 and 0.25 ms; without it, `impratio` 10 at 1 ms gains 3.2×), and the two values give the same rebound.

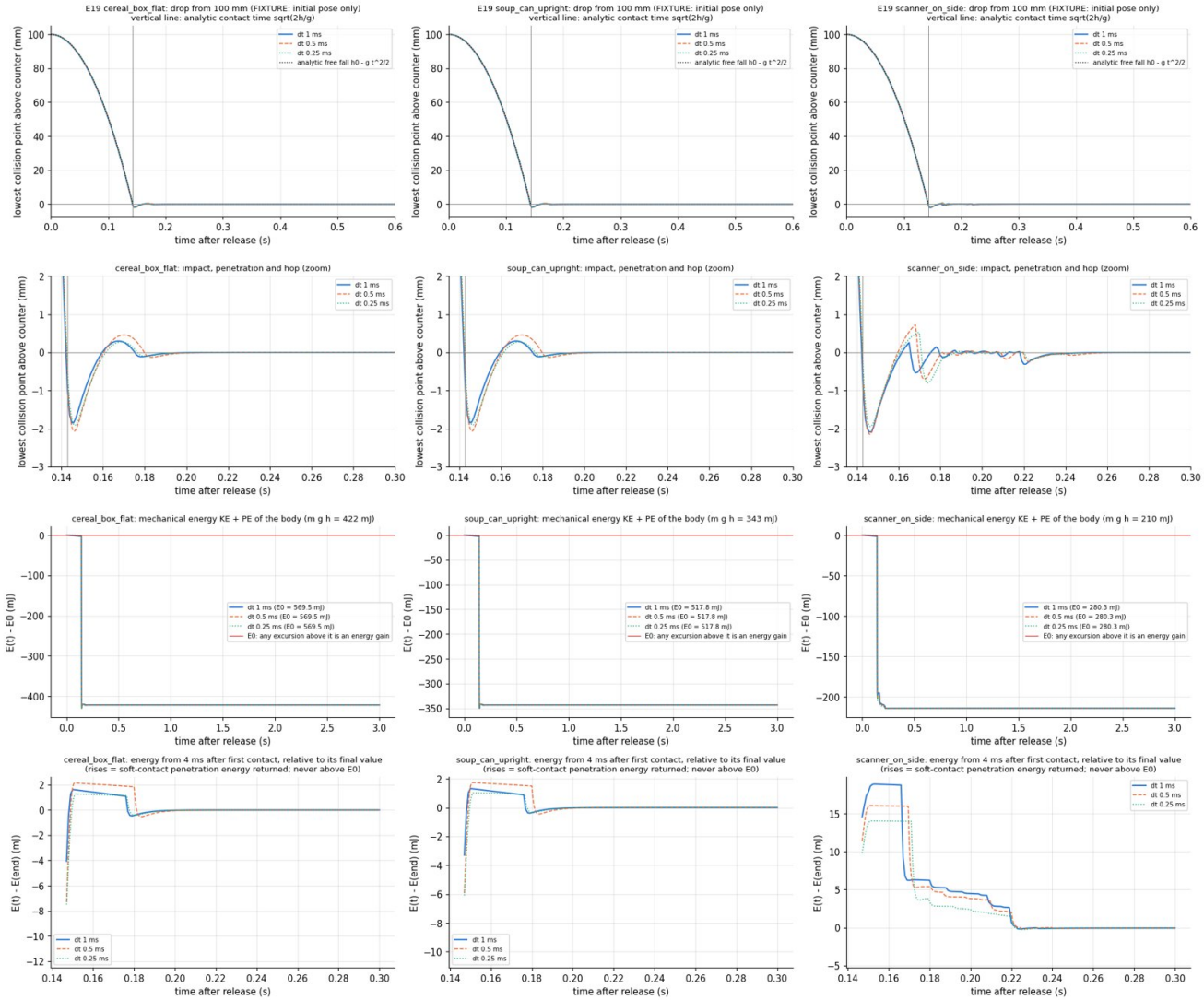


Figure 14. E19: 100 mm drops onto the counter at 1, 0.5 and 0.25 ms: contact time and impact speed against the analytic free fall, and the total energy against its initial value.

11.5 Long soak (E20)

Scope: 300 s hands off per preset after the reset settle, every free body's displacement from its settled pose sampled every 0.5 s and classified bounded / linear / other; settings as shipped (MuJoCo keeping its own state and the converged no-slip pass, section 6.1).

Preset	Bodies	Largest displacement at 300 s	Bodies growing linearly (rate over the last 100 s)	Fault / non-finite
stockroom	55	0.021 mm (the scanner, settling into its socket; bounded)	0	none
library	9	0.0038 mm (a book's cover)	0	none
counter	9	0.0094 mm (the tomato-paste can in the basket)	1: the tomato-paste can 0.032 µm/s	none

The largest single-tick speed after the settle was 0.13 mm/s in the stockroom, 0.17 mm/s in the library and 0.88 mm/s at the counter (contact jitter; nothing moves). With an unconverged no-slip pass the counter's lip balm reaches 0.95 mm in 300 s, with the basket walking across the counter at 0.8 µm/s. With Newton's default state round trip (section 6.1) the stockroom has 21 bodies walking linearly (worst 1.6 mm in 300 s).

E20c: the small cylinder under a sideways load (the counter at seed 0, 120 s hands off). In this layout the battery pack can lie partly on the 15.6 mm lip balm, so that the stick carries a steady sideways load. In the run shown nothing creeps (whether the pack rests on the stick after the settle was not checked):

Variant	Lip balm creep (second half of 120 s)	displacement at 120 s	Largest single-tick speed	Basket creep
shipped (32-facet hulls, 1 ms)	0.0 µm/s	0.0001 mm	0.18 mm/s	0.0 µm/s
64 facets	0.0 µm/s	0.0075 mm	0.33 mm/s	0.0 µm/s
16 facets	0.0 µm/s	0.0 mm	0.0 mm/s	0.0 µm/s
shipped at 0.5 ms	0.0 µm/s	0.0004 mm	0.02 mm/s	0.0 µm/s

In a run where the pack does rest on the stick the creep is 0.17–0.23 µm/s at every hull resolution and both steps (0.01 mm per minute), and the basket does not move. With the unconverged no-slip pass (10 iterations at MuJoCo's tolerance) the same four variants creep 2.6, 9.2, −3.6 and 1.5 µm/s (0.50, 2.56, 0.81 and 0.32 mm at 120 s) and the basket 0.6 µm/s; at `impratio` 10 the shipped hull creeps 1.65 µm/s, and only a 0.25 ms step stops it (section 6.1). A steady rate under a steady tangential load is MuJoCo's regularised ("soft") static friction: a contact under a tangential force below the friction limit still slides at a small velocity proportional to that force, and an unconverged no-slip pass stops at its tolerance long before the slip is gone (section 6.1). The remaining creep stays in the environment as a property of the contact model (section 14).

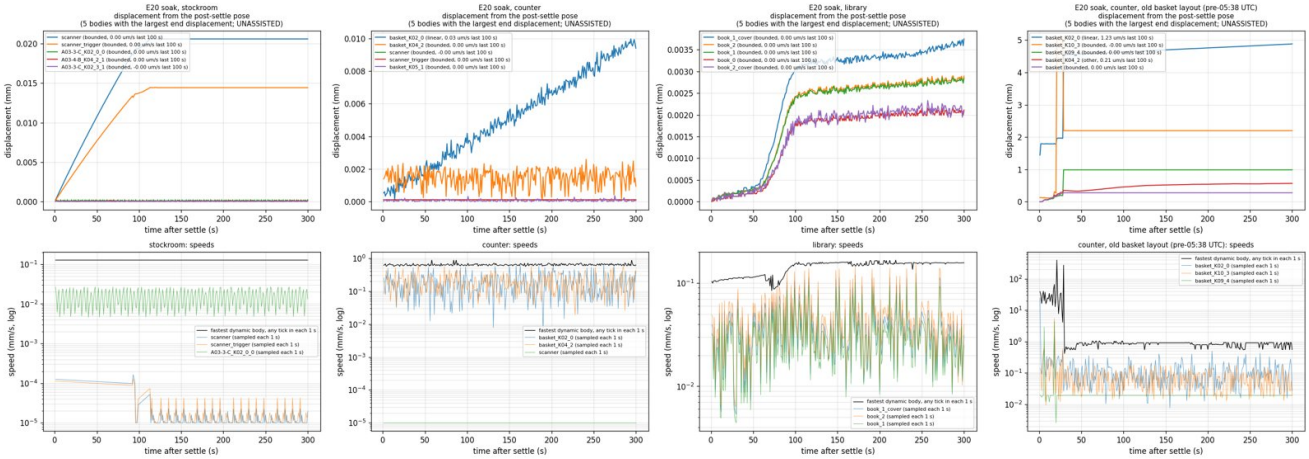


Figure 15. E20: displacement of every free body over 300 s hands off, per preset.

12. Runtime

Scope: wall-clock cost per simulated second through the public API (`gar.create`*, then reset, then policy actions of 0.5 s or 0.1 s, each returning the 1280 × 720* `cart_rgb` *observation,* `head_rgb` *for the G1, decoded to check it is a real image, with the action trace recorded), against the 200× limit; then the same through* `gar run` *and HTTP with actions that wake the scanner, and the shared benchmark runner (below). No other job of this environment was running and the GPU was idle before the first action; the host's 1-minute load, from other jobs, is in the last column.*

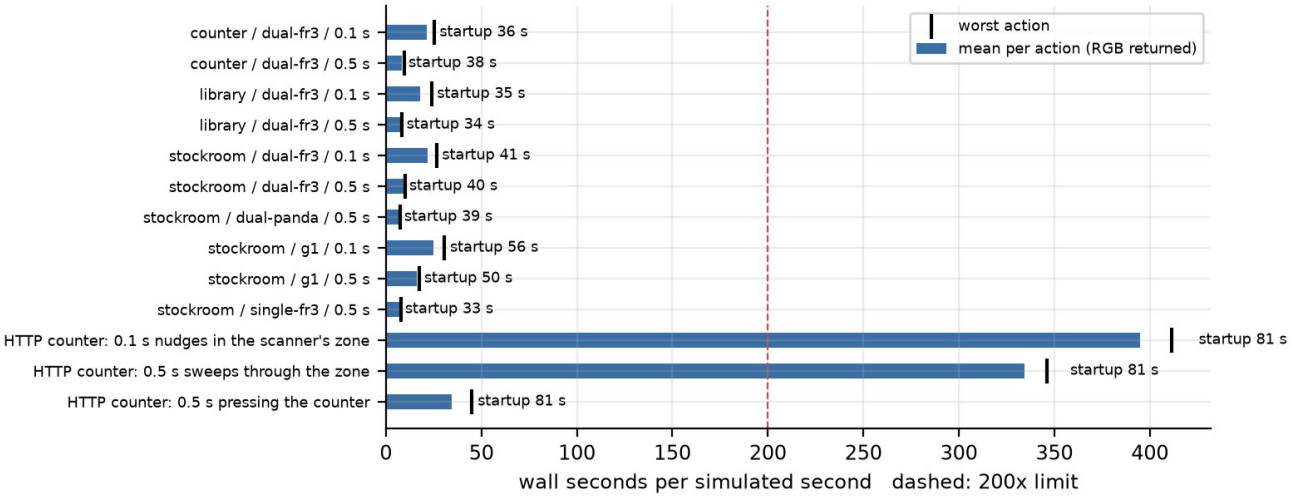


Figure 10. E16: wall seconds per simulated second with the RGB observation, in-process (default worker retention) and through HTTP on the counter for actions that wake the scanner and for pressing.

Render workers: kept for the next operation, then released. An operation (reset, step, observation) starts a render worker while its physics runs, or takes the one kept from the previous operation if that was opened on the same exported scene; after the operation the worker is kept 10 s (`renderer_options` `idle_release_s`, default 10) and then released by a timer, and at once when the session closes. The project's GPU rule forbids keeping a worker indefinitely after an observation; a bounded idle window released by a timer honours it. `idle_release_s = 0` restores release after every operation (the last three rows).

Preset, robot	Action	Worker kept	Startup (create + reset)	Wall per action, mean	Wall / sim, mean (max)	Physics and scanner frames	Waiting for / starting a worker	Observation frame	Actions on a kept worker	1-min load
stockroom, dual-fr3	0.5 s	10 s	40 s	4.8 s	**9.7×** (9.9×)	3.2 s	0.0 s	1.6 s	5 of 5	13–16
counter, dual-fr3	0.5 s	10 s	38 s	4.2 s	**8.4×** (9.6×)	2.9 s	0.0 s	1.3 s	5 of 5	8–8
library, dual-fr3	0.5 s	10 s	34 s	3.8 s	**7.6×** (8.5×)	2.6 s	0.0 s	1.2 s	5 of 5	12–13
stockroom, dual-panda	0.5 s	10 s	39 s	3.4 s	**6.9×** (7.4×)	2.5 s	0.0 s	1.0 s	5 of 5	6–6
stockroom, single-fr3	0.5 s	10 s	33 s	3.5 s	**7.0×** (7.8×)	2.6 s	0.0 s	0.9 s	5 of 5	5–6
stockroom, g1	0.5 s	10 s	50 s	8.1 s	**16.2×** (17.6×)	7.0 s	0.0 s	1.2 s	5 of 5	7–8
stockroom, dual-fr3	0.1 s	10 s	41 s	2.2 s	**22.0×** (26.6×)	0.7 s	0.0 s	1.5 s	10 of 10	12–15
counter, dual-fr3	0.1 s	10 s	36 s	2.1 s	**21.5×** (25.3×)	0.7 s	0.0 s	1.4 s	10 of 10	7–8
library, dual-fr3	0.1 s	10 s	35 s	1.8 s	**17.9×** (24.1×)	0.5 s	0.0 s	1.2 s	10 of 10	9–10
stockroom, g1	0.1 s	10 s	56 s	2.5 s	**24.9×** (30.7×)	1.4 s	0.0 s	1.1 s	10 of 10	7–8
counter, dual-fr3	0.1 s	0 s	49 s	13.6 s	**135.8×** (153.2×)	0.5 s	6.3 s	6.8 s	0 of 10	9–27
library, dual-fr3	0.1 s	0 s	43 s	18.3 s	**182.6×** (223.5×; 4 of 10 over 200×)	0.8 s	6.7 s	10.7 s	0 of 10	10–16

Preset, robot	Action	Worker kept	Startup (create + reset)	Wall per action, mean	Wall / sim, mean (max)	Physics and scanner frames	Waiting for / starting a worker	Observation frame	Actions on a kept worker	1-min load
stockroom, dual-fr3	0.1 s	0 s	54 s	15.6 s	**156.0×** (163.3×)	0.5 s	6.3 s	8.8 s	0 of 10	7–23

Means are over the steady actions (the first after reset excluded); in every action the simulated time from the world's ticks and from the observation agreed with the request, and every observation decoded as a 1280 × 720 image.

Against the 200× limit. With the default retention every preset and robot runs at 6.9–16.2× at 0.5 s actions and at 18–25× at 0.1 s actions. Released after every operation, each 0.1 s action pays a fresh worker and its first frame (6.3–6.9 s waiting for the worker, 6.8–10.7 s for the frame): 136–183× mean, and in the library 4 of 10 actions over 200× (224× at most). The physics of a 0.5 s action costs 2.5–3.2 s (the G1 7.0 s): the per-action trace goes through the shared `PhysicsTraceStore` with one sample per physics step, about 1.2–2.4 ms per step (section 13), and the pads' torsional rows add to every grasp contact. The observation frame costs 0.9–1.6 s.

Where the time goes (two workers each on the preset's exported scene): exporting the scene 4.9–10.1 s (once per session); starting a worker (process, Kit, stage load) 6.7–7.2 s; its first `cart_rgb` frame 7.2–9.4 s (texture upload, BVH build, shader warm-up); the next frame 0.09–0.11 s; all three policy cameras together 1.2–1.8 s; the first scanner-imager frame 1.8–2.4 s at path depth 3 (11.3–12.5 s at depth 6 on a busy GPU), later frames of an unchanged scene 0.05–0.06 s; closing 2.0–2.2 s (on a background thread, outside the action).

Through `gar run` and HTTP, with actions that wake the scanner. Each session is started with `gar run --policy-key` and driven under the policy key the way a policy is: POST the action, then GET the returned camera image; the owner key reads the scanner's frame counter after each action (untimed). Every action also records the host's 1-minute load and the GPU's utilisation and memory in use when it returned. Workloads: holding still; the right hand moved into the docked scanner's presentation zone and nudged 2 cm (0.1 s) or swept across it (0.5 s), every such action rendering scanner frames; the left hand pressed 5 mm into the counter top; for the G1, 1 cm hand nudges beside the cradle.

Both measurements ran on a shared GPU. The run marked `_shared` ran beside this environment's own video renders (3 render workers and one task render); the other sessions met other jobs on the GPU: in every one of their actions the GPU was at 99–100 % with up to 14 other CUDA contexts (recorded per action). The G1's counter and library rows were measured during the video renders.

Session (gar run, policy key)	Startup to first observation	Workload	Actions	Wall / sim, mean (max)	Scanner frames	Every observation returned
counter_dual-fr3	81 s	hold still, 0.1 s	11	**60.2×** (64.4×)	0	yes
counter_dual-fr3	81 s	hold still, 0.5 s	6	**35.4×** (46.5×)	0	yes
counter_dual-fr3	81 s	hand in the scanner's zone, 0.1 s nudges	11	**394.7×** (411.2×)	33	yes
counter_dual-fr3	81 s	hand sweeping the zone, 0.5 s	4	**334.2×** (345.9×)	60	yes
counter_dual-fr3	81 s	hand pressed on the counter, 0.5 s	4	**34.5×** (44.7×)	0	yes
	startup plus 60 s of 0.1 s idle actions: 61.5×	all_ok True				
counter_dual-fr3_release0	97 s	hold still, 0.1 s	11	**223.4×** (226.5×)	0	yes
counter_dual-fr3_release0	97 s	hold still, 0.5 s	6	**44.4×** (45.4×)	0	yes
	startup plus 60 s of 0.1 s idle actions: 225.0×	all_ok True				
counter_dual-fr3_shared	96 s	hold still, 0.1 s	11	**51.3×** (54.0×)	0	yes
counter_dual-fr3_shared	96 s	hand in the scanner's zone, 0.1 s nudges	11	**242.7×** (279.0×)	33	yes
counter_dual-fr3_shared	96 s	hand sweeping the zone, 0.5 s	4	**208.9×** (239.2×)	60	yes

Session (gar run, policy key)	Startup to first observation	Workload	Actions	Wall / sim, mean (max)	Scanner frames	Every observation returned
	startup plus 60 s of 0.1 s idle actions: 52.9×	all_ok True				
counter_g1	135 s	hold still, 0.1 s	11	**65.5×** (81.3×)	0	yes
counter_g1	135 s	hold still, 0.5 s	6	**68.7×** (72.3×)	0	yes
	startup plus 60 s of 0.1 s idle actions: 67.8×	all_ok True				
library_dual-panda	56 s	hold still, 0.1 s	11	**52.6×** (57.8×)	0	yes
library_dual-panda	56 s	hold still, 0.5 s	6	**32.7×** (43.9×)	0	yes
	startup plus 60 s of 0.1 s idle actions: 53.5×	all_ok True				
library_g1	65 s	hold still, 0.1 s	11	**223.5×** (289.6×)	29	yes
library_g1	65 s	hold still, 0.5 s	6	**72.8×** (77.5×)	0	yes
	startup plus 60 s of 0.1 s idle actions: 224.6×	all_ok True				
stockroom_g1	107 s	hold still, 0.1 s	11	**66.6×** (73.8×)	0	yes
stockroom_g1	107 s	hold still, 0.5 s	6	**78.1×** (80.0×)	0	yes
stockroom_g1	107 s	hand nudges near the cradle, 0.1 s	11	**66.5×** (73.4×)	0	yes
	startup plus 60 s of 0.1 s idle actions: 68.4×	all_ok True				
stockroom_g1_release0	131 s	hold still, 0.1 s	11	**250.7×** (266.2×)	0	yes
	startup plus 60 s of 0.1 s idle actions: 252.9×	all_ok True				
stockroom_single-fr3	58 s	hold still, 0.1 s	11	**59.3×** (63.4×)	0	yes
stockroom_single-fr3	58 s	hold still, 0.5 s	6	**38.4×** (53.1×)	0	yes
	startup plus 60 s of 0.1 s idle actions: 60.3×	all_ok True				

Actions that wake the scanner run over the 200× limit on a shared GPU: 209–243× beside this environment's video renders, 334–395× (411× at most) beside other jobs, against 75–77× on a quiet host; idle actions 35–68×, against 3–18× on a quiet host. Every action returned its observation. The difference is the GPU's other work, which this environment cannot control: on a quiet GPU scanning runs near 75×; shared as here it does not. The capabilities' `wall_time` note tells clients to allow 600 s per request. **The G1's wake**: at the counter the G1's idle actions render no scanner frame (0 of 17 actions); at the library desk its shoulder's sway still wakes it: 29 frames in the first eleven 0.1 s actions of one session and none in the six 0.5 s actions after them, and in the integration session 52 frames in 11 of its 12 actions (section 14, item 3).

The shared benchmark runner (`gar benchmark` with a presenting agent and at most six actions: six 1 s moves into, across and out of the docked scanner's zone; the summary records each action's wall time and its margin to the client timeout). On the shared GPU, with the runner's **default 120 s timeout the run failed on its second action** (the first took 84 s); with `--timeout 900` it completed all six actions, each taking 76–197 s, 703–825 s inside the timeout, 15 min in all. On a quiet GPU all six completed at the default timeout in 3 min 34 s, with 4 s to spare on the slowest action.

The decoder's share. Where frames are decoded, the levelled decoding, the flat-field stage and the misread check of section 5.1 add to the six level passes. On the recorded frames of the checkout (291): 21 ms per frame on average with the level passes alone, 103 ms with levelling, 171 ms shipped (95th percentile 0.44 s, at most 0.72 s); of the library check-in (446): 25, 202 and 302 ms (1.48 s at the 95th percentile, 3.79 s at most: the check's edge-model fit of a 16-character Codabar label is slow). Single process beside other jobs, 1-minute

load 12. The library check-in took 5 139 s of wall time for its 446 scanner frames; the shipped decoder adds (302 − 25) ms × 446 ≈ 124 s to it, 2.4 %. The rendering of each converged frame, not the decoder, is the cost (section 4.5).

Where a worker's first frame goes (E16b, one worker at a time on the stockroom's export; GPU at 100 % utilisation from other jobs, 1-minute load 15–46, so all times are longer than in the table above):

Variant of the same exported scene	Worker start (s)	First `cart_rgb` frame (s)	Next frame (s)
as exported (6 workers)	8.1–10.3	16.1–21.7	0.11–0.17
every texture input replaced by one 4 × 4 image (114 inputs)	7.9, 9.6	10.8, 11.2	0.11, 0.15
room dressing deactivated (its lights kept)	8.6, 9.4	10.3, 10.7	0.10
the dressing's PNG/JPG maps at half size (48 files)	7.9, 9.1	14.3, 15.0	0.11, 0.12
only the dressing models' OpenEXR maps replaced (46 files)	8.0, 8.7	15.8, 15.8	0.12, 0.16

The dressing's textures (94 files, 50 megapixels) are 6–9 s of every worker's first frame; the geometry, the packaging art and the shelf labels are not measurable against the noise, and what remains (~10.5 s here) is the renderer's own start-up of a frame. Shrinking the dressing's maps below what the cameras resolve would buy a few seconds of the fixed cost; it is not done, because the worker start and the renderer's own first frame would still cost about 15 s per action.

Worker retention at 30 s (library, dual-fr3, 0.1 s actions, 1-minute load 29–33): 34.5× mean (40× max) against 217× released per operation, with 1.0–1.4 s of physics and scanner frames and 2.1–2.5 s for the observation per action. The warm frame costs 2.1 s rather than 2.7 s because the host display's texture is named by its content (a new file per tick would be reloaded every frame) and the worker skips attribute writes whose value is unchanged. On one warm worker (E16c) a frame of an unchanged scene costs 0.13 s, and one after moving a single book by 2 mm, a single link, the whole robot, or after changing the LEDs' intensity costs 2.2–2.5 s every time: a per-change cost inside the renderer, not proportional to what changed. A kept worker holds about 4 GB of the shared GPU for up to 10 s after the last operation.

13. Interfaces

Scope: the delivered entry points, exercised as a user would: the GAR CLI, a live HTTP session under both keys, the session dashboard in a browser, and the package installed from a wheel outside the checkout.

CLI. `gar tasks barcode-scanning` lists the eight tasks and resolves any selection to its instruction; `gar inspect barcode-scanning` shows the robot choices (dual-fr3 default, dual-panda, single-fr3, g1); `gar doctor barcode-scanning` checks zxing-cpp, trimesh, Pillow, SciPy, the Isaac render runtime (`ISAAC_RENDER_PYTHON`) and the fonts; `gar run barcode-scanning --select preset=… --select task=… --select seed=… --policy-key` starts a session and prints its address, dashboard and keys.

Policy view (policy key): `/v1/capabilities`, `/v1/task` (the instruction), `/v1/observation` (joint proprioception; the robot's own status, that is each arm's remaining protective-stop time and the count of joint targets it refused; and RGB from `cart_rgb` and the wrist cameras, 1280 × 720), `/v1/action` with `named_joint_target`, `gripper` (position, or force mode by `inward_effort_n`), `cartesian_pose`, `cartesian_delta` and `keyboard` (the shared GAR chord action: W/S A/D R/F translate the hand, I/K J/L U/O rotate it about world axes, Q/E close/open the fingers, H a force-mode grip of 35 N with G for 10 N and B for the hand's maximum (70 N on the FR3, 40 N on the Panda), Alt/Shift ×0.2/×6 speed; the key list is in the policy's `keyboard_keys`, the hardware notes say what the gripper and the protective stop do); each owner step's info reports reflexes, the joint targets refused because their arm was in its post-reflex hold, the hold time left, and scanner frames and reads. Checked through the public API: the right shoulder driven forward strikes the cradled scanner and the reflex stops the arm after 0.5 s (34.0 N m at joint 4); the next command's joint target is refused (target unchanged, `refused_targets` 1, 0.86 s left) while its finger target is taken; after the hold the same command is accepted (the shoulder strikes with a 1.3 rad stroke). **Grip limits**: the backend checks the grip against the selected hand's limit while planning (twice the finger joint's effort limit: 70 N FR3, 40 N Panda) and refuses with `ActionRejectedError` ("nothing executed"), so a grip over the limit never starts executing and never latches a runtime fault; B is the hand's own maximum, and the capabilities state the range. The policy key is refused the state, checkpoints, diagnostic cameras and the evaluator (403). **Owner view** (owner key): `/v1/state` (scanner firmware, trigger travel, seat, reads with their attribution, host record, object poses), `/v1/checkpoints/save|restore` (restore returns the saved tick), diagnostic cameras, `/v1/evaluation`.

A policy that scans, through HTTP. The render worker switches between the policy camera and the scanner without disabling a render product (a product disabled and re-enabled never returns an image). Measured through `gar run` and HTTP under the policy key (section 12): every one of the 18 scanner-waking actions on the counter returned its observation (140 scanner frames), as did every action of the five other sessions, the G1's included; on a quiet GPU a scanner-waking action costs 75–77× wall/sim. The shared runner with a presenting agent completed all six actions with the runner's default 120 s client timeout on a quiet GPU (3 min 34 s) and with `--timeout 900` (4 min 0 s). On a GPU busy with other jobs the default timeout ends the run on its second action, which is why the capabilities say to allow 600 s per request.

Per-action physics trace through the shared routes. The trace is written through the shared `PhysicsTraceStore` (`barcode_scanning/trace.py`): one sample per physics step (body poses, joint positions, trigger travel and switch, seat, keys, grip force, the fastest body and the step's contact count) and events (protective stops, trigger and seat changes, reads), with the action's request id. Checked in-process (a 0.2 s gripper action): complete, 200 of 200 samples, no missing tick ranges; the trace adds about 1.2 ms per physics step (0.9 ms of it the store's per-sample commit), about 20 % of a manipulation action's physics. Through `gar run` and HTTP, an integration script (the stockroom with dual-fr3 and the manual-control proxy, the library with the G1) checks: `/v1/history/{id}/trace` and `/trace/events` return 200 with the trace available, `gar export --include-traces` completes (exit 0) as does the export without traces, every owner route returns 403 under the policy key and 200 under the owner key, the dashboard and the manual-control proxy's page, script, styles, session and health routes return 200 (401 without its token) and a keyboard chord through the proxy is accepted (200), an action while paused is refused before execution, and `gar verify` exits 1 (task not successful, as expected for an unsolved session). In the G1 library session the docked scanner woke on 11 of 12 actions (section 12).

Live HTTP transaction (the keys are read from the launcher's JSON in memory and never written): `gar run barcode-scanning --select preset=stockroom --select task=scan-location --select seed=0 --policy-key --port 0 --json` started a session in 38 s. Under the policy key: `/v1/capabilities`, `/v1/task`, `/v1/observation` and `/v1/cameras/cart_rgb` (a PNG) returned 200; four actions returned 200 (a 0.1 s named joint target in 1.4 s wall, 14×; 0.5 s joint target 6.2×, gripper 6.2×, Cartesian delta 5.9×); `/v1/checkpoints/save` and `/v1/state` returned **403**. Under the owner key: checkpoint save, state, an action, restore (the restore returned the saved tick) and `/v1/evaluation` returned 200. The session dashboard's page, script and calls returned 200 under both keys (the policy view reads "Restricted policy connected", the owner view "Owner · ready", each showing the task's instruction). The session was stopped with `gar stop`; all ten of the transaction's checks pass.

Manual control (`manual.py`): a local web page drives a live session's robot from the keyboard through the backend's `keyboard` action only (the GAR key stays in the proxy; the page gets a one-page token). Checked with real key events in headless Chromium, effects measured through the owner's state history, on an earlier build

of the environment (not re-run on the final code): all 14 checks pass (Shift+D 1.03 s commanded +31.0 mm, measured +30.80 mm with the other arm unmoved; U 1 s at 10°/s gives +10.01°; Q/E at 10 mm/s per finger take the fingers from 79.99 to 59.99 and back to 79.99 mm; H gives a closed force grip, 0.04 mm; two queued chords cost one render); the wrist-camera subset passes as well. Latency from key release to new image with the default worker retention: 5.4–9.5 s under light load, 37–112 s on a heavily loaded machine. Behaviour of the API that the page works around: every command renders; a later chord refused inside a multi-chord request latches the session's partial-execution fault (the shared GAR contract: earlier commands already ran), so the page pre-checks every chord with the backend's own IK; step feedback is owner-only.

Variation catalogue and splits (`variations.py`, `variations.json`): 8 000 variations (seeds 0–999 × 8 tasks), each assigned to train / validation / test by a hash of (task, seed), 6 378 / 845 / 777, frozen with a sha256 of the list; `gar tasks` resolves any selection to its split. A seed varies what the task varies (the counted slot, its stock mix and yaws; the basket's contents and their places; the returned books, their order and label side); dimensions, masses and materials never. `python -m barcode_scanning.variations --validate 12` builds and settles a stratified sample of every task and checks each reset (finite, scanner seated, nothing on the floor, nothing thrown by the settle, ground truth non-empty): **96 of 96** resets (12 per task) pass. In 18 counter resets items dropped into the basket settled by rolling or sliding off a neighbour (up to 66 mm), all at rest inside the basket; the record lists them.

Session dashboard: the page and its script load (200); in headless Chromium the key is typed into the key field and Connect pressed, once per principal. Opened and looked at: the policy view reads "Restricted policy connected", shows the `cart_rgb` observation (the two arms, the cereal shelf, the scanner in its cradle, the keypad and the host display) at simulation 1.6 s, the task instruction and the action form; the owner view reads "Owner · ready" and adds the session id and the pause/resume controls. Connecting did not reset or step the environment (the page says so and the simulation time is unchanged).

Installed wheel: the wheel built from the repository, installed into a clean venv layered only on the shared runtime's site-packages and run from a directory outside the checkout, with the checkout removed from the import path (imports resolved to the venv): for dual-fr3, dual-panda and single-fr3, create through the public registry, reset, a joint action, a checkpoint, a force-mode grasp and a restore back to the saved tick all succeed; single-fr3 offers `cart_rgb` and `left_wrist_rgb` only; the packaged artwork and decoder round-trip a UPC-A, a Code 128, a Codabar and a Data Matrix. The Panda meshes are not packaged: `dual-panda` needs `TITRATE_PANDA_URDF` (or the checkout's pinned copy), as for the other environments that use the Panda. Nor are the G1's: the shared catalog resolves them from `$GAR_ROBOT_ASSETS/g1` when that is set, else from the checkout's asset cache. From the installed wheel without it, `g1` fails at creation ("g1 asset missing or outside package"); with `GAR_ROBOT_ASSETS` pointing at a directory whose `g1/` is the locked unitree_ros package, the G1 creates, resets, acts, checkpoints, refuses the Franka gripper action and restores. `gar doctor barcode-scanning` with the G1 checks that the assets resolve and names `GAR_ROBOT_ASSETS` when they do not. The manual-control page (`manual_static/`) and the variation catalogue (`variations.json`) are package data (`pyproject.toml`).

Tests. A unit suite of 36 tests covers the catalogue round-trip of every printed symbol through the decoder, any roll, no UPC-E misread near the decoder's limit, firmware timing rules including the stationary-symbol case and the power-on state, switch hysteresis and debounce, imager geometry, both auto-exposure rules, the sensor's shot noise at any gain, the parameter table, the render worker's release, the policy interface, the grip limits and the one-target fingers, the setup-object validator, rest, the phantom press and exact restore, the reflex after a restore, the rigs' own controller profiles, the G1's start pose and policy observation. The evaluator suite holds 110 adversarial evaluator, keyboard-contract and policy-access tests (section 8.1), many on real physics scenes. With the physics tests enabled: **144 passed, 1 failed, 139 subtests passed**; the failure was a stale copy of the parameter table, which passes after regenerating it.

14. Limitations

Each item is a measured gap between the simulation and the real job, or work not done, with its evidence. None is hidden in a parameter. Items 1–4 are work the environment still needs.

1. **Decode range has gaps for the 5 mil codes** (9.3, E1): with the front end (5.1) the 1-D far limits are 0.63–1.15 of the DS2208's published ones (UPC-A 377 of 368 mm) and 2-D 0.60–1.00, but the 5 mil codes read in two bands with nothing between them (Code 39 31–45 and 75–115 mm; Code 128 at 27 and 89–117 mm): too blurred for zxing and too coarse for the front end's hypotheses in between. Code 128 5 and 7.5 mil read a little past the datasheet (117 against 102 mm; a band at 155–205 against 178 mm): lenient. PDF417 and Data Matrix stay at about 0.6. The task symbols read at the task distances (sections 10.1–10.7).
2. **Misreads remain possible** (5.1, 9.3, 10.5): in the task runs one wrong read in 618 (0.16 %, 9.3), and two frames of E1c's roll sweep (9.3c). The one misread found: in the second counter checkout (seed 3) a tomato-paste can turning into the field of view at the frame's left edge was decoded as the checksum-valid UPC-E 0067100000040 from two scan lines 5 px apart through the ragged ends of its EAN-13's bars, four frames before the true EAN-13 was read; both went to the host, and the run fails `no_misreads`. It is the mechanism the scan-line rule guards against (5.1: lines grazing bar ends), at a line separation the 4 px rule lets through. A quiet-zone check on the scan line does not reject this read (the 7 modules beyond the false guard fall on a space). Against the AIM/Ohio University reference (UPC-A worst case about 1 in 33 000 symbols) the simulated rate is far higher; a misread reaches the host application and fails the task's `no_misreads` check; it does not end the episode (a decode that no printed label on its ray can explain still does).
3. **The G1 cannot do a handheld task end to end** (6.5): it takes the scanner out of the cradle only with a tipped lift (3 of 7 variants), in a grip whose index finger cannot reach the trigger (no regrasp is built); it cannot put the scanner back (let go 30 mm above the socket, it falls to the floor); it has no balance controller; it cannot reach the top shelf row (seed 4) or the counter's water case standing. From an installed wheel it needs the locked assets through `GAR_ROBOT_ASSETS` (13). At the library desk its right shoulder stands inside the docked scanner's presentation zone and its standing sway (5–8.5 mm/s by the wake rule's measure, which adds 50 mm of rotation to the linear speed) wakes the scanner on most actions (11 of 12 in one HTTP session, the first 11 of 17 in another; none at the counter): the reset's settle waits for the robot's links, but by their linear speed only, and the sway outlasts it.
4. **The scanner housing is modelled from the envelope, not a drawing** (2.1): the colliders are two boxes inside the DS2278's 175 × 66 × 90 mm with the handle raked 15° back; the moulded neck, tunnel and grip taper are visual. No CAD model or dimensioned drawing was obtained.
5. **With the Franka Hands, every handheld read uses a phantom fingertip for the trigger** (10.1, 10.6, E21 bulky). A second robot fingertip on the trigger fails before the pull (the left hand cannot reach the blade from the scripted turn, 9.1b), and while the scanner is aimed at a label the scripted squeeze does not move the trigger (10.2). A parallel gripper holding a pistol grip has no finger for the trigger; the G1's Dex3 hand has one and pulls it itself (6.5.5c).
6. **Items the Franka Hand cannot grasp are handed over by a labelled phantom** (10, 6.3): no collision-free grasp in the 200 mm basket, on a tight facing or for a book lying flat; the robot's 15 physical attempts at the planner's rejected grasps and at side approaches all fail on the basket's rim, the neighbouring stock or the desk. Not tried: pushing an item to an edge first, tilting it against a wall, two-hand grasps. Where the items stand on the counter (`checkout-counter`) the hand takes every item itself; there the 85 mm jar, held by its 66 mm lid, swung against the hand, the collision reflex stopped the arm 11 times and its barcode was not read (10.5).
7. **The specular dead zone appears only where the front end reads** (9.4, E2): at 250 mm glossy and semigloss UPC-A miss 3–5 pitches within −4…+6° of normal incidence; at 100 and 150 mm, where zxing reads beside the LEDs' glint, there is none, where the SE4710 guide gives up to 20° with illumination on [S3]. E2b (9.4): neither a 2.5 mm emitter nor lens veiling glare up to 4 % makes one at 100–150 mm. Candidates not modelled: sensor blooming, lens ghosts, the decoder searching near the aim line, the exit window (whose housing rules, SE4710 guide pp. 2-7 to 2-12, keep its reflections out of the imager). The LED placement (10 mm either side, 12 mm below the axis) is an estimate. The simulation is lenient at working distances: a policy is not taught to tilt away from glare there.
8. **Skew tolerance is ±50–60°, not the DS2208's ±65°, and not every roll reads** (E1c, section 9.3c; pitch reaches ±70°): with the shipped decoder some rolls fail near 45°/135° in the front end's bands and near zxing's range limit (Code 39 10 mil at 153 mm reads 14 of 36 rolls), and the levelled passes misread a UPC-A rolled 40° at 195 mm. Code 39 20 mil reads to −60/+55°, UPC-A to ±50° at 127 mm with the flat-field stage (±25–30° without it).
9. **Pinch torsion capacity is at the corner-loaded bound** (breakaway 0.385 N m against 0.19 N m for uniform pad pressure; 9.2): box-box contact points sit at the pad corners; the pads' torsional term (1 mm, 6.2) adds spin resistance at a point contact, not a pressure distribution.
10. **Soft-constraint residue.** Fast impacts hold overlaps of 5–11 mm for a few ticks (E17: seven random cases at link speeds up to ~4 m/s; 11.2), and at full-speed impacts the fingers' stops and coupling yield by

up to 3.5 and 2.7 mm (the worst of 158 FR3/Panda impact cases, 6.1); a line search cut short drives fingers far past their stop (up to 104 mm at 73 m/s), which the shipped budget prevents (6.1). MuJoCo's regularised static friction leaves a creep under a steady tangential load: at counter seed 2 a cereal box leaning in the basket walks at 1.0 µm/s, everything else much less (11.1). MuJoCo's box-box routine can return a bogus deep contact: it threw the 12.5 kg water case and the bagging tote from the counter top, which is therefore a convex hull, as are box SKUs of 5 kg or more; the remaining box-box pairs met no deep contact in 200 resets (11.2), which does not prove they never will. The converged no-slip pass costs 1.55 times the physics of MuJoCo's default.

11. **The collision reflex's recovery time (1 s) is authored**; libfranka does not publish one. The reflex thresholds are the libfranka examples' reference setting, not a measured factory default.
12. **Presentation wake-up and seat sensing are authored stand-ins** for image-change detection and the stand's hall sensor (5.3, 3.4); their thresholds (5 mm/s within 0.4 m; 3 mm / 8° / 0.2 s) are not published.
13. **Estimated constants**: trigger and key force laws (bounded by switch datasheets), sensor noise and full well (the OV9281 datasheet lists them as TBD [S33]), frictions (the fingertip pad $\mu = 0.8$ is unsourced and the G1 probe shows it matters), book covers, basket and tote masses (section 7, labelled). No sensitivity study of the tasks' outcomes to them was run.
14. **The paperback's cover is a rigid 0.3 mm board on a hinge** (E14): it falls open and rests flat rather than flexing.
15. **Scope**: RFID, OCR of non-barcode text, keyboard-wedge timing, battery and cable are not modelled; the scanner is cordless (DS2278 class) so there is no cable to simulate. Shipping and parcel labels, the largest cluster of the action group not built, are not among the presets.
16. **Runtime depends on the shared GPU, and on a shared GPU scanning exceeds 200×** (section 12). The HTTP measurements both ran beside other GPU work: actions that keep a hand in the docked scanner's zone ran at 209–243× beside this environment's own video renders and 334–411× beside other jobs (75–77× on a quiet host); idle actions 35–68×. In-process on a quieter host, idle and manipulation actions run at 6.9–25× with the default worker retention and 136–224× when the worker is released after every operation (four library actions of ten over 200×). The per-step trace through the shared store adds about 1–2 ms per physics step (section 13). The shared benchmark runner failed at its default 120 s timeout on the shared GPU and completed at 900 s. Every render worker needs about 4 GB of GPU memory; parallel rendering must be sized against the measured free memory.
17. **Red ink** (9.4, E3): the E3 red keeps 15–18 % contrast under the red LEDs and reads, below the datasheet's 25 % minimum reflective difference: lenient where the device's behaviour is not specified.
18. **Robot selection is a closed list** of four configurations (6.3): the four rigs are accepted as aliases or as setup objects; any other setup is refused with what the environment needs (`robot_setup.validate_setup`), not built. The two-arm FR3 runs every task; the two-arm Panda and the one-arm FR3 ran the scan-location task (10.1); the one-arm FR3 cannot reach the tote or the bin, and those pairs (and the G1 with the bulky checkout) are refused at launch. The G1 has no assisted run at the counter or the library desk (its timing there: section 12).
19. **The** `scan_zone` **video camera is placed for the G1 layout** (a diagnostic view, not a policy camera). With the FR3s its eye is 5 cm from the right arm's shoulder link and inside it as the first joint turns (black frames).
20. **Both** `checkout-counter` **runs failed** (10.5): seed 0 because the 85 mm jar, held by its lid, swung against the hand (11 protective stops) and was never read; seed 3 on the UPC-E misread of item 2. The basket checkouts succeeded (8/8 and 9/9). One run per layout; no rerun was made to look for a passing one.
21. **The rooms are dimmer than designed in the stockroom** (2.2): 109–136 lx rendered at the shelf face where the troffers were sized for 200 lx; the counter and the desk are within 10 % of 500 lx.
22. **The phantom trigger finger ramps to 3 N in 0.1 s** (10.1): an estimate of a finger's squeeze; the blade still reaches its 3.6 mm stop (3.65 mm).
23. **Render materials are estimates** (7.2): every base colour, print and finish constant is listed with its provenance; none is identified against a measured reflectance.

15. Reproduction

The environment runs through the GAR CLI from the release (section 13):

```
gar tasks barcode-scanning                    # the eight tasks and their instructions
gar inspect barcode-scanning                  # robot choices and notes
gar doctor barcode-scanning                   # zxing-cpp, Isaac render runtime, fonts, G1 assets
gar run barcode-scanning --select preset=stockroom --select task=scan-location --select seed=0 --policy-key
gar run barcode-scanning --robot g1 ...         # the G1 configuration
```

Rendering needs the Isaac Sim 5.1 render runtime (`ISAAC_RENDER_PYTHON`). Physics probes that ship with the release:

```
python -m barcode_scanning.probes.e7_hold --grip head --force 17.5 --out OUT     # E7: take, carry, hold, reseat
python -m barcode_scanning.probes.e17_stress --out OUT                           # E17: legal-action stress
python -m barcode_scanning.probes.e19_drop --dt 0.001 --out OUT                  # E19: drops (also 0.0005, 0.00025)
python -m barcode_scanning.probes.e19_drop --report --out OUT
python -m barcode_scanning.variations --validate 12                              # sampled resets of every task
python -m barcode_scanning.manual --session SESSION_ID --role policy             # manual control page
```

Line-search budget and the recorded evidence. The evidence in this report was recorded with a line-search budget of 20 evaluations; the shipped budget of 50 (section 6.1) shifts every trajectory slightly. Re-run with the shipped budget, these keep their verdicts: both test suites with the physics tests; E0 (three presets); E17 (three presets, section 11.2); E7, E7 torsion and the 120 s hold; E6r (two finger axes); the robot configurations; the reflex hold; the G1 rest (three presets); the stock grasp attempts (three presets); and, with the render worker, the scan-location reads (two-arm FR3, two-arm Panda, one-arm FR3), E14, E11, the two cycle counts and the `checkout-present` run (the checkout's five reads and bagging as before). `checkout-bulky`, `checkout-counter` (seeds 0 and 3) and the library check-in were re-run physics-only (dark imager frames) with the same physical outcomes, but not re-rendered. Other records were not re-made.

Decoder replay. Every run saves each imager frame exactly as its decoder received it and logs what was found in it. Every frame of the task runs (E9, E9's control, E11, the checkout runs, the E22 runs, E23, the E24 captures and the G1 run; 1 641 frames) decodes exactly as recorded when decoded again with the shipped decoder. E14's one frame, saved without its decode log, is read only by the flat-field stage, as section 10.4 says.

Videos are rendered from the recordings (no re-simulation), captioned per phase with ASSISTED / UNASSISTED / LEGAL ACTIONS, sim time, playback speed, trigger travel, switch, seat and firmware state.

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